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Prophylaxis and Empirical Therapy of Infection in Cancer Patients

Małgorzata Mikulska
Elio Castagnola

Short View Summary

RISK FACTORS FOR INFECTIONS IN CANCER PATIENTS

• Neutropenia is the most important risk, particularly if severe (<100 polymorphonuclear neutrophils) and prolonged (≥7–10 days).
• Other risks include mucositis, underlying disease and its status, intensity of chemotherapy, and the use of certain targeted and biologic therapies.

EPIDEMIOLOGY AND ETIOLOGY

• Epidemiology of bloodstream infections in neutropenia is constantly changing, with increased infections caused by gram-negative rods observed in many centers.
• This shift has been accompanied by an increasing rate of resistant pathogens, including gram-negative organisms with extended-spectrum β-lactamases and carbapenem resistance, and gram-positive organisms with methicillin and vancomycin resistance.
• Aspergillosis has become more common than candidiasis in hematology patients, due to increased use of fluconazole prophylaxis. Fluconazole resistance, and infections caused by echinocandin-resistant Candida glabrata and Candida auris, occurs in patients treated for both hematologic malignancies and solid tumors.

PROPHYLAXIS (see Table 311.4)

• The role of fluoroquinolone prophylaxis is controversial, with increased fluoroquinolone resistance.
• Antifungal prophylaxis is indicated when the anticipated incidence of IFD is >8%. This includes, but is not limited to, patients receiving induction chemotherapy for acute myelogenous leukemia (AML), therapy for high-risk lymphoblastic leukemia, therapy for relapsing leukemia of any origin, high-intensity treatment of myelodysplastic syndromes, and in select groups of cancer patients, if the incidence of IFD is approximately 8% to 10% or higher.
• Posaconazole as mold-active prophylaxis is recommended for patients receiving induction chemotherapy for AML or myelodysplastic syndrome. Drug-drug interactions can be problematic between posaconazole (strong CYP3A4 inhibitor) and novel (e.g., venetoclax, midostaurin) or old (e.g., vinca alkaloids) antileukemia drugs, potentially requiring reduced dosing of antileukemia drugs or alternative strategies.
• Fluconazole is recommended for patients at lower risks for mold infections, including those receiving high-intensity chemotherapy for lymphoma.
• Prophylaxis against Pneumocystis jirovecii is beneficial in patients with deficiencies in T-cell immunity, particularly with chronic lymphocytic leukemia, or in those receiving high-dose corticosteroids, fludarabine, or alemtuzumab. Anti-Pneumocystis prophylaxis is also warranted in people who receive certain targeted therapies, such as idelalisib or bendamustine plus rituximab.
• Antiherpes prophylaxis with acyclovir or valacyclovir is recommended in patients with acute leukemia and in those receiving drugs such as alemtuzumab, bortezomib, or fludarabine, and in some patients with lymphoma or chronic lymphocytic leukemia (CLL).
• Varicella postexposure prophylaxis with varicella-zoster immunoglobulin, varicella vaccine, and/or acyclovir is recommended for high-risk patients who lack evidence of immunity.
• Antiviral prophylaxis is recommended to prevent hepatitis B reactivation in cancer patients with chronic inactive hepatitis B (hepatitis B surface antigen [HBsAg] positive, hepatitis B virus [HBV] DNA <2000 IU/mL, normal aminotransferases) and for select patients with a resolved HBV infection (HBsAg negative/hepatitis B core antibody [HBcAb] positive), particularly if receiving B-cell–depleting therapies (see Chapter 152). Screening with HBV DNA and HBsAg testing should be performed after stopping antiviral prophylaxis and can be used as an alternative to prophylaxis in people with lower risk for HBV reactivation.

Immunization

• Influenza and pneumococcal vaccination is recommended.
• Influenza and varicella vaccination of household contacts and health care workers is recommended.
• Antizoster vaccination has shown to be effective in patients with hematologic malignancies.
• Anti–SARS-CoV-2 vaccination should be provided to patients and their household contacts.

MANAGEMENT OF FEBRILE NEUTROPENIA (see Fig. 311.2 and TABLE 311.7)

Initial Approach

• Blood cultures, physical examination, and studies with initial therapy tailored to risk of severe infection and drug resistance (see Table 311.6)
Choice of Appropriate Therapy: Antibacterial
• Oral versus intravenous depending on inpatient or outpatient setting and the risk of complications (see Table 311.6).
• Escalation versus de-escalation strategy:
• Escalation strategies typically start with anti-Pseudomonas β-lactam monotherapy.
• De-escalation strategies typically start with a carbapenem or a combination (e.g., combination of anti-Pseudomonas β-lactam with an aminoglycoside or vancomycin) and de-escalate based on clinical outcome and microbiologic results. In patients colonized with MDR gram-negative organisms, start with appropriate agents targeting the colonizing pathogen (e.g., novel anti–gram-negative β-lactam + β-lactamase inhibitor or combinations of antibiotics) and then de-escalate as noted previously if no MDR pathogen is isolated.
Choice of Appropriate Therapy: Antifungal
• Diagnostic-driven antifungal strategies may decrease need for empirical antifungal therapy (for treatment of persistent fever during neutropenia), depending on availability of highly sensitive in vitro diagnostics, CT scanning, and ability to perform rapid procedures (especially bronchoalveolar lavage).
OTHER THERAPEUTIC RECOMMENDATIONS
• Antimicrobial stewardship in cancer centers is mandatory and should include attention to:
• Infection-control practices
• Local surveillance of antibiotic resistance, antibiotic consumption, and patient outcomes
• Promoting appropriate antibiotic use (e.g., timely de-escalation or discontinuation, appropriate dosing)
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Cancer patients probably represent the best example of how both a disease and its treatment can impair the complex immunologic network aimed at maintaining the integrity of our body and defending it against infections from both the external and the internal environment. For decades we have known that a granulocyte count of less than 500 cells/mm3 (and especially <100 cells/mm3) is associated with an increased risk of severe bacterial and fungal infections.1,2 Patients with granulocyte counts between 500 and 1000 cells/mm3 can present with severe infectious complications (e.g., bacteremia or invasive mycoses), especially if counts are rapidly decreasing, suggesting the presence of a “gray zone” that requires careful monitoring.3 Other factors that affect the risk of infectious complications in these patients include damage to anatomic barriers, such as skin and mucosal membranes; alterations of microbiota diversity; and the presence of indwelling devices. Indeed, mucositis itself might result in severe infections due to microbial translocation, even in the absence of neutropenia, and a central venous catheter (CVC) may facilitate the entrance of endogenous and exogenous bacteria and fungi into the bloodstream or subcutaneous tissues. Alteration of microbiota diversity has been shown to affect the risk of infection and other important outcomes in people with hematologic malignancies, with potential predictive value but with no consensus on potential therapeutic interventions.4 In patients with solid tumors, cancer-related obstructions, surgical procedures, and prolonged hospital admission are additional risks for infection. The increasing use of the new antineoplastic drugs and treatment modalities (e.g., protein kinase inhibitors, monoclonal antibodies, immunotherapy like CAR-T) poses new risks and therapeutic considerations.

Contemporary recommendations for prevention and early treatment have to be mindful of growing antibiotic resistance worldwide. Antibiotic-resistant organisms of growing concern include gram-negatives that produce extended-spectrum β-lactamases (ESBLs) or carbapenemases, which are frequently associated with other antibiotic resistance (e.g., to aminoglycosides, fluoroquinolones [FQs]), and gram-positive organisms with methicillin-resistance (Staphylococcus aureus [MRSA] and methicillin-resistant Staphylococcus epidermidis) and vancomycin resistance (vancomycin-intermediate S. aureus and vancomycin-resistant enterococci [VRE]).5 This increase in multidrug-resistant (MDR) strains is radically challenging our ability to prevent and cure infections in immunocompromised patients.

As shown in Table 311.1, the clinical approach to a cancer patient with signs and symptoms of infection is multipronged. Before planning a rational management strategy, physicians should answer several crucial questions about the type and stage of the underlying disease and the clinical presentation. In addition, factors potentially associated with the presence of antibiotic-resistant bacteria, such as local epidemiology and the patient's colonization or previous infectious history, should be very carefully considered.

In the following sections, the epidemiology and management principles of infections in cancer patients are described. Risk factors and clinical presentations of specific infections, along with their treatment, are not discussed here but are dealt with in chapters focused on individual infectious agents. Similarly, infections in recipients of allogeneic hematopoietic stem cell transplants (HSCTs) and risks associated with biologic immunomodulators are discussed elsewhere (see Chapters 52 and 312).

Epidemiology and Risk Factors for Infections in Cancer Patients

An understanding of underlying malignant disease and therapeutic regimens is critical for the implementation of safe and effective strategies to prevent and treat infection. Epidemiology and risks today vary from those in years past, as hosts, oncologic treatments, and supportive care strategies have changed. Also, outcomes observed in treatment trials (such as those evaluating empirical antibiotic therapies) may not apply to the real world, given inclusion and treatment biases. Incidence calculations that account for duration (days at risk) provide a better measure of infection risk, especially for people who undergo chronic and/or recurrent immunosuppressive therapies.

In general, the incidence and risks for infections are tightly correlated to the intensity of antineoplastic therapies, which varies for different underlying disease and stage (relapse vs. remission). In general, risks for both bacterial and fungal infections are lower in children compared to adults.[3,4,6–16] Patients with acute myeloid leukemia, both adults and children, have the highest incidence of fever, bacteremia, and invasive fungal diseases, especially during the first induction of remission and with relapsing leukemia, when the intensity of chemotherapy is higher. A lower incidence of infection is observed in people with low-risk lymphoblastic leukemia, chronic lymphatic disorders, multiple myeloma, and non-Hodgkin lymphomas, whereas the lowest rates are observed in people with solid tumors, reflecting the intensity of antineoplastic treatment strategies. Concurrent underlying conditions, including chronic obstructive pulmonary disease and other pulmonary complications, such as lung metastases and receipt of thoracic radiotherapy, can increase risk of infections in people with solid tumors.[12]

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remission). In general, risks for both bacterial and fungal infections are lower in children compared to adults.3,4,6–16 Patients with acute myeloid leukemia, both adults and children, have the highest incidence of fever, bacteremia, and invasive fungal diseases, especially during the first induction of remission and with relapsing leukemia, when the intensity of chemotherapy is higher. A lower incidence of infection is observed in people with low-risk lymphoblastic leukemia, chronic

TABLE 311.1

What Should a Clinician Wonder About and Look for When Approaching a Cancer Patient With a Suspected Infection?

QUESTIONS	RATIONALE FOR THE QUESTION
The underlying disease: 1. Type (e.g., Acute leukemia? Solid tumor?) 2. Phase of the UNDERLYING disease (Active? In remission?)	The incidence of infectious complications is different according to the underlying disease and consequent intensity of chemotherapy, being usually the highest in case of acute myeloid leukemia, followed by aggressive lymphomas, and lower in most solid tumors. The stage of disease may influence the type, risk, and outcome of infection, with the risk of infection the highest in case of relapsed/refractory disease.
Recent or ongoing treatments: 1. Recent (within 1 month) or ongoing chemotherapy? 2. Which drugs, which schedule, and when? Is neutropenia expected? 3. Did the patient receive autologous or allogeneic HSCT? 4. If allogeneic HSCT, what donor type; is GvHD present? 5. Did the patient receive monoclonal antibodies (e.g., anti-CD20, anti-CD52) in the past 6–12 months?	Different drugs may give different types of immunosuppression and favor different infectious complications. For example, the immunosuppressive effect of certain monoclonal antibodies might be prolonged (over 12 months for the risk of HBV reactivation after anti-CD20 treatment). Previous transplantation might result in long-term immunodeficiency, particularly if immunosuppressive treatment is continued. Immune reconstitution depends on the type of donor and conditioning regimens used in allogeneic HSCT.
White blood cell count: 1. Is the patient neutropenic (PMNs $<500/mm^3$ or $<1000/mm^3$ but rapidly decreasing)? 2. Was the patient neutropenic in the previous 30 days?	The presence of neutropenia increases significantly the risk of infection. The knowledge of local epidemiologic data on antimicrobial susceptibility is mandatory for a correct choice of empirical antibacterial therapy.
Risk of infection caused by resistant bacteria: 1. Is the patient colonized with resistant bacteria, particularly gram-negatives? 2. Is the hospital or the country endemic for resistant organisms? 3. Any previous infections caused by resistant pathogens?	In patients colonized by resistant bacteria, particularly if neutropenic, initial empirical therapy should cover these pathogens. Also if not colonized but cared for in a setting where resistance is an issue, consider the possibility of de-escalation strategy.
Central venous catheter (CVC): 1. Yes or no? 2. Has the catheter been manipulated (including infusions) within a few hours before the onset of fever?	The central venous access may be an important source of infection. The type of CVC will influence the risk of infection and the management strategy (facility of removal).
Past history of infections (both before and after the diagnosis of cancer)	It may suggest the etiology and drive the therapeutic choice (e.g., tuberculosis, toxoplasmosis, multidrug-resistant bacteria, or opportunistic fungal infections).
Country of origin	Specific endemic infections can reactivate (Chagas disease, strongyloidiasis, tuberculosis, endemic mycoses). Epidemiology of antibacterial resistance varies worldwide; thus patients coming from areas endemic for resistant bacteria should be treated accordingly.
The clinical picture: 1. Presence of (severe) mucositis? 2. New onset of pain (perianal, chest, everywhere)?	It may suggest the etiology and drive the therapeutic choice. The presence of mucositis is suggestive of infection with pathogens from oral flora or the gastrointestinal tract. Pain may help to locate the site of infection, such as formation of an abscess, or indicate the presence of a locally invasive process, such as pulmonary aspergillosis.
Administration of prophylaxis (no, yes, which drugs): 1. Antibacterial? 2. Antifungal, including against *Pneumocystis jirovecii*? 3. Antiviral? 4. Was the patient compliant? 5. Is there the possibility of inadequate blood levels due to lack of absorption, PK/PD problems, or suboptimal compliance?	Breakthrough infections are possible, and fever during prophylaxis should be considered a failure of prophylaxis, unless proven otherwise. The occurrence of a bacterial/fungal/viral infection during specific prophylaxis may influence the choice of empirical therapy, depending on the drug used for prophylaxis. A resistant pathogen should be suspected in every case, unless the patient was clearly noncompliant or there is the possibility of low drug levels caused by poor absorption (e.g., posaconazole, itraconazole), increased metabolism (e.g., voriconazole), or drug interactions (e.g., triazoles and phenytoin or barbiturates). Knowledge of local epidemiology, including susceptibility pattern, is mandatory for correct diagnostic and therapeutic management.

HSCT, Hematopoietic stem cell transplantation; *PK/PD*, pharmacokinetic/pharmacodynamic; *PMNs*, polymorphonuclear neutrophils.

lymphatic disorders, multiple myeloma, and non-Hodgkin lymphomas, whereas the lowest rates are observed in people with solid tumors, reflecting the intensity of antineoplastic treatment strategies. Concurrent underlying conditions, including chronic obstructive pulmonary disease and other pulmonary complications, such as lung metastases and receipt of thoracic radiotherapy, can increase risk of infections in people with solid tumors.12

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Infection rates vary from a mean of 22.5 (95% confidence interval [CI], 21.2–23.7) per 100 devices for Hickman and Broviac catheters, 3.5 to 4 (95% CI, 2.4–5.6) for port-a-cath, and 3.1 (95% CI, 2.6–3.7) for PICC, with mean rates per 1000 catheter-days of 1.6 (95% CI, 1.5–1.7), 0.1 (95% CI, 0.01–0.2), and 1.1 (95% CI, 0.9–1.3), respectively.41 The rates of CVC-related infections vary across centers, with lower rates of infection reported in people with PICCs compared with centrally inserted central catheters (respectively, 3.29 vs. 5.11 per 1000 catheter days).46 Patients with hematologic malignancies, especially those with acute leukemia, have a higher incidence of CVC-related infections compared to patients with lymphoma, myeloma, or solid tumors, probably due to different types of catheters and frequency of catheter use.41

The role of CVCs in bacteremia might be overestimated, because CVCs can be the site of a secondary localization of pathogens actually coming from the intestinal flora.47 Recent studies suggest that 40% to 50% of BSIs in oncologic settings are actually endogenous and associated with mucosal barrier injury.41,48–50

Gram-positive bacteria are most frequently implicated with CVC-related infections, but gram-negatives are reported with increasing frequency, especially in nonbacteremic episodes.41 Finally, some age-related differences in the proportions of pathogens causing CVC-related bacteremia have been reported, with gram-negatives being Page 3648observed more frequently in adults and gram-positives and fungi more frequently in children.48–51

Genetic Factors
Genetic factors that increase or decrease the susceptibility to infection in immunocompromised patients are increasingly appreciated; all cancer patients are not the same, and every single patient might deserve an individualized approach. For example, in nonleukemic patients receiving less intensive chemotherapy, decreased levels of mannose-binding protein were associated with an increased risk of infection (49.9 vs. 29.6 per 1000 days at risk, $P = .01$).4 Pediatric studies observed that mannose-binding lectin deficiency influenced both the incidence and the severity of febrile neutropenia.52 In breast cancer patients, genetic factors also were found to influence the risk of febrile neutropenia.53 Polymorphisms in pattern recognition receptor pathways such as Toll-like receptors and other components of innate immunity, such as CLEC7A, have been associated with an increased risk of invasive aspergillosis, both in cancer patients (including recipients of HSCT) and in other immunocompromised patients.54,55 In the future, increased understanding and methods to measure genetic risks will help in designing individualized prophylactic or therapeutic approaches.

New Therapies
Biologic Agents and Other New Drugs

Biologic and targeted therapies with monoclonal antibodies and various agents targeting specific biologic pathways (e.g., small molecules, kinase inhibitors) are currently the standard of care for most cancer patients. As these agents were developed to provide targeted activity, the rate of infectious complications is predicted to be lower than with conventional chemotherapy, as confirmed in the case of multiple myeloma.56 However, nonspecific effects and/or secondary complications can increase the risk of specific infections.

Several issues impair our ability to precisely determine the exact role of these new compounds on the risk of infection in cancer patients. In randomized, placebo-controlled trials, the reporting of infectious complications is not uniform, and these trials are powered to establish a drug's efficacy, typically with underpowered observations of infectious complications. According to a recent systematic review, only 45% of clinical trials reported the type or source of infections, and none provided the definitions used to diagnose or classify the reported infections.57 There are also many confounding factors, as new drugs are usually used together or in sequence with old therapies and initially not as first line, making it difficult to establish their role separate from concomitant or previous therapies and from underlying disease. Finally, the same new agents might be used to treat people with different diseases, with variability in accompanying drugs and baseline infectious risks. For example, venetoclax can be used for different diseases (acute myelogenous leukemia [AML] and in chronic lymphocytic leukemia [CLL]) and at different doses, particularly when given in combination regimens for different diseases, to include noncancer conditions (multiple sclerosis).

In some cases, infection risks can be anticipated by mechanism of action. For example, eculizumab, a drug initially developed for paroxysmal nocturnal hemoglobinuria, targets the complement component C5, which is well known to increase risks for invasive meningococcal disease in the setting of inherited deficiencies. In such settings, risks can be anticipated and considered to guide appropriate preventative strategies early in drug development. Multicenter efforts focused on infectious complications help in determining infectious risks to standardize early preventive or diagnostic measures.58

The field of biologic agents and anticancer molecules is rapidly changing, with a constant influx of new drugs and new indications or discontinuation of the older ones. Therefore data on the safety profiles, and appropriate risk evaluation and mitigation strategies, are continuously updated.

For the compounds that have been in use for many years, more data are available. For instance, anti-CD20 agents, such as rituximab, are known to cause a prolonged (2–6 months in median, but sometimes longer) B-cell depletion, with consequent suppression of immunoglobulin production in the case of repeated administration,59 and reactivation of hepatitis B virus (HBVr). In people with lymphoma, up to 85% of hepatitis B surface antigen (HBsAg)–positive and 41.5% of HBsAg-negative/hepatitis B core antibody (HBcAb)–positive patients treated with chemotherapy, including rituximab, developed HBVr.60 More rarely, other severe viral infections can occur after rituximab treatment, including those caused by enteroviruses, parvovirus, and polyomaviruses such as JCV causing progressive multifocal leukoencephalopathy (PML), particularly in people with hypogammaglobulinemia.59,61,62

Brentuximab, another monoclonal antibody targeting lymphocytes through CD30, has also been associated with severe infections, such as Pneumocystis jirovecii pneumonia (PCP) and PML. PML after brentuximab therapy was found to develop earlier than with rituximab or natalizumab (2–3 months vs. 63 weeks vs. 26 months, respectively).58,59,53 Therefore a low threshold for brain magnetic resonance imaging should be applied to patients treated with monoclonal antibodies who develop neurologic symptoms.

Bortezomib and ruxolitinib are associated with varicella-zoster virus (VZV) reactivation. Ruxolitinib additionally might increase the risk of reaction of granulomatous infections such tuberculosis (TB) due to its impact on tumor necrosis factor (TNF)-alfa. Whereas idelalisib was found to be associated with herpes and cytomegalovirus (CMV) reactivations and PCP, ibrutinib was initially associated with a high rate of invasive fungal diseases (IFDs), in particular cerebral aspergillosis, when used with high doses of steroids. In subsequent observational studies of ibrutinib use in patients with CLL, the risk for IFD was significantly lower (<3%) and limited mainly to patients with relapsed/refractory disease.64

Novel agents used in the treatment of AML in combination with standard induction chemotherapy (e.g., midostaurin, venetoclax, gilteritinib) seem not to be associated with particularly higher risk of infectious complications, provided that adequate prophylaxis is administered during sequential and prolonged periods of neutropenia. However, their concomitant use with posaconazole prophylaxis, which is a standard of care in this setting, has been challenging because strong 3A4 inhibitors increase drug levels, resulting in potential severe toxicity, which includes prolonged neutropenia, prolonged QT interval, and pulmonary toxicity. Although data are still limited, experts recommend either dose reduction (e.g., venetoclax) or administration of an alternative antifungal drug, in addition to strict monitoring for toxicity (midostaurin, gilteritinib).65,66

Checkpoint inhibitors such as nivolumab or pembrolizumab do not increase the risk of infections, but they are associated with severe immune-related adverse events (irAEs), such as diarrhea, pneumonitis, or hepatitis. In these cases, the differential diagnosis with infectious complications is complex, and if an irAE is confirmed, it is treated with high-dose steroids, subsequently increasing the risks for infections, including PCP.

Finally, high-dose retinoic acid has been used for treatment of acute promyelocytic leukemia and can induce fever and hepatotoxicity (vitamin A "intoxication"), which mimic the symptoms of infection during neutropenia, requiring a careful differential diagnosis.

Intravesical administration of Calmette-Guérin bacillus (BCG), a live strain of Mycobacterium bovis, is adopted as a treatment of non–muscle-invasive bladder cancer, and disseminated genitourinary or osteomuscular infections have been reported.67

The main infectious complications associated with the use of biologic and targeted therapies are reported in Table 311.2.65,66,68–73

Chimeric Antigen Receptor T-Lymphocyte Therapy

CAR T-cell therapy is being increasingly used to treat lymphocyte-derived malignancies such as acute lymphoblastic leukemia (ALL) and B-cell malignancies such as lymphoma and multiple myeloma and is being studied in other tumors. There are two major early complications: CRS and CAR T-cell–related encephalopathy (CRES), which have been associated with an increased risk of infectious complications.

TABLE 311.2

Selected Biologic Drugs for Treatment of Hematologic Malignancies and Solid Tumors and Possible Infectious Complications

CLASS/SITE AND MECHANISM OF ACTION	DRUG	POSSIBLE INFECTIOUS COMPLICATIONS
Monoclonal Antibodies (mAbs) Targeting Surface Antigens on Lymphoid Cells		
Bispecific T-cell engager (BiTE): anti-CD19 mAb conjugated to anti-CD3 mAb leading to T-mediated lysis of CD19+ cells	Blinatumomab	CVC-associated infections due to continuous prolonged intravenous infusion. Severe and prolonged hypogammaglobulinemia and possible neutropenia, with consequent infectious risk. VZV and HSV infections, pneumocystosis, and HBV reactivation.
Anti-CD20 mAbs	Rituximab	Possible hypogammaglobulinemia of variable duration, not associated with an increased risk of bacterial infections. HBV reactivation (both of chronic inactive and resolved infection); exacerbation of HCV infection. Possible neutropenia; severe viral respiratory tract infection. Cases of severe enteroviral infections, PML, VZV, CMV parvovirus infections. Humoral vaccine response is almost absent in the first 6 months after rituximab administration. Higher risk of infections when used in association with other drugs (e.g., bendamustine).
	Obinutuzumab	Similar to rituximab and frequently used in combination therapies. Neutropenia and severe enteroviral infections have been reported.
	Ofatumumab	Probably similar to rituximab. Reported neutropenia, HSV, viral respiratory infection, fatal case of HBV reactivation.
Anti-CD20 mAbs conjugated with a radioactive isotope	^{90}Y-Ibritumomab tiuxetan ^{131}I-tositumomab	Higher rate of cytopenias, such as lymphopenia or neutropenia, due to combination with radioactive isotope. Cytopenia risk is higher if bone marrow is infiltrated by CD20+ hematologic malignancy. General safety profile probably similar to rituximab.
Anti-CD22 mAb	Epratuzumab	No significant increase in infections in placebo-controlled trials. Profile of infections similar to anti-CD20 mAbs: respiratory tract infections (viral and bacterial), pneumonia.
Anti-CD22 mAb conjugated with the DNA-damaging agent calicheamicin	Inotuzumab ozogamicin	
Anti-CD22 mAb conjugated with *Pseudomonas* exotoxin A	Moxetumomab pasudotox	
Anti-CD30 mAb conjugated with the microtubule-disrupting agent monomethyl auristatin E	Brentuximab vedotin	Herpetic infections, including CMV; pneumocystosis; cases of PML.
Anti-CD33 mAb conjugated with a calicheamicin agent	Gemtuzumab ozogamicin	No significant increase in infections.
Anti-CD38 mAb	Daratumumab	Slightly higher risk of pneumonia and upper respiratory tract infections, including severe pneumonia. Higher rate of neutropenia. Increased risk of HBV and VZV reactivation (HZ prophylaxis warranted during treatment and for 3 months after the end of treatment).
Anti-CD40 mAb	Dacetuzumab	Higher rate of neutropenia vs. placebo. Opportunistic infections similar to those observed in hyper-IgM syndrome (e.g., pneumocystosis, CMV reactivation) might be theoretically expected.
Anti-CD52 mAb	Alemtuzumab	Severe viral infections (particularly HSV, VZV, and CMV, but also HHV6, BKV, parvovirus, and adenovirus). Pneumocystosis and other fungal infections, toxoplasmosis, tuberculosis and other mycobacterial infections, reactivation or exacerbation of HBV infection, listeriosis, infections due to acanthamoebae or *Balamuthia mandrillaris* have also been described.
Anti-CD139 (signaling lymphocytic activation molecule F7 [SLAMF7]) mAb	Elotuzumab	Lymphopenia and increased risk of opportunistic infections, particularly due to VZV.
Anti–C-C motif of chemokine receptor 4 (CCR4) mAb	Mogamulizumab	Infectious risk difficult to distinguish from the intrinsic effect of underlying diseases and concomitant lymphotoxic therapies. Pneumocystosis, HSV, VZV, HBV reactivation (risk of severe hepatitis in patients with high-level HBV DNA due to T-regulatory-cell downregulation).
Tyrosine Kinase Inhibitors		
Breakpoint Cluster Region–Abelson Murine Leukemia (BCR-ABL) Signaling Pathway		
Inhibitor of tyrosine kinase BCR-ABL, c-KIT, and platelet-derived growth factor (PDGF) receptor	Imatinib	Febrile neutropenia has been reported. Cases of pneumocystosis and viral diseases, including reactivation of hepatitis.
Inhibitor of tyrosine kinase BCR-ABL	Nilotinib	Cases of pneumocystosis and viral diseases, including reactivation of hepatitis.
Inhibitor of tyrosine kinase BCR-ABL, SRC family, c-KIT, and others	Dasatinib	Cases of pneumocystosis and viral diseases, including reactivation of HBV. Noninfectious pleural effusions, less often pneumonitis (in differential diagnosis with infections).
Bruton Tyrosine Kinase Signaling Pathway		
Bruton tyrosine kinase inhibitor	Ibrutinib (similar infectious risk of novel agents such as acalabrutinib or zanubrutinib)	Bacterial pneumonia, urinary tract infection and cellulitis, noninfectious pneumonitis, invasive fungal infections, also involving central nervous system.

Table Continued

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CLASS/SITE AND MECHANISM OF ACTION	DRUG	POSSIBLE INFECTIOUS COMPLICATIONS
Janus Kinase–Signal Transducers and Activators of Transcription (JAK-STAT) Signaling Pathway		
Inhibitor of the JAK pathway (kinases 1 and 2)	Ruxolitinib, possibly also fedratinib	VZV reactivation, cases of tuberculosis, PML, pneumocystosis, cryptococcosis, and other opportunistic infections associated with T-cell dysfunction.
FMS-Like Receptor Tyrosine Kinase 3 (FLT3) Signaling Pathway		
Multi-tyrosine kinase inhibitor (including FLT3)	Midostaurin	Usually used with standard induction chemotherapy regimens for AML, possible prolonging the neutropenia and similar rate of infections in RCT: bacterial (BSI and pneumonia) and IFDs. Midostaurin is metabolized by CYP3A4, so posaconazole prophylaxis in this setting should be weighed against the risk of increased exposure (see text) and potential toxicity of midostaurin because posaconazole is a strong CYP3A4 inhibitor; interactions with other drugs should be also evaluated.
FLT3 inhibitor (active also against receptors of other tyrosine kinases)	Gilteritinib	In RCT compared with azacitidine, higher rate of fever, diarrhea, and febrile neutropenia but not pneumonia or sepsis Gilteritinib is metabolized by CYP3A4, with expected increase in gilteritinib levels, and potentially toxicity, in case of coadministration with strong CYP3A4 inhibitors (such as most triazoles).
Selective FLT3 inhibitor	Quizartinib	When used in monotherapy vs. standard regimen, similar rate of sepsis or septic shock; pneumonia grade ≥3: 12% vs. 8% Quizartinib is metabolized by CYP3A4, in case of coadministration with strong CYP3A4 inhibitors (such as most triazoles) expected increase in quizartinib levels and possible toxicity (QT prolongation).
Phosphatidylinositol 3-Kinase (PI3K) Delta Isoform Signaling Pathways		
Inhibitor of PI3Kδ	Idelalisib	Pneumonia, sepsis (20%), opportunistic infections, along with noninfectious pulmonary and gastrointestinal inflammation. Prophylaxis of HBV reactivation and pneumocystosis and preemptive management strategy for CMV are warranted.
Others		
Multikinase inhibitors	Lapatinib	Possible skin infections because of skin toxicity.
	Pazopanib **Regorafenib**	No significant data available for evaluation of the risk of severe infectious complications.
B-Cell Lymphoma (BCL)-2 Protein Signaling Pathway		
B-cell lymphoma (BCL)-2 protein inhibitor	Venetoclax	In RCTs when used in combination in AML, CLL, and relapsed/refractory multiple myeloma, higher rate of neutropenia in some, but same rate of grade 3 infections as placebo arms. Venetoclax is metabolized by CYP3A4, with expected increase in drug levels in case of coadministration with strong CYP3A4 inhibitors (most triazoles) and inducers; reduce the dose by 75% when coadministered with posaconazole for prophylaxis, which is recommended in case of combination therapy with hypomethylating agents.
Vascular Endothelial Growth Factor (VEGF) Signaling Pathway		
Anti–human VEGF mAb	Bevacizumab	Infections in cases of chemotherapy-induced neutropenia: sepsis and pneumonia. Cases of intestinal perforation (with or without abscess formation), probably from the inhibition of endothelial cell proliferation and new blood vessel formation. Possible endophthalmitis due to intravitreal injections, but an increased risk of this type of infection has not been clearly demonstrated.
Recombinant protein circulating antagonist that prevents VEGF receptor binding	Aflibercept	Severe infections: incidence 7.3% (95% CI, 4.3–12.0%) with increased risk (RR, 1.87; 95% CI, 1.52–2.30). Mortality: 2.2% (95% CI, 1.5–3.1%), with increased risk (OR, 2.16; 95% CI, 1.14–4.11). The risk of infections with aflibercept substantially higher than with bevacizumab.
Inhibitor of tyrosine kinases on receptors for vascular endothelial growth factor (VEGFR), platelet-derived growth factor (PDGFR), and isoform B of tyrosine kinase (known as rapidly accelerated fibrosarcoma [B-RAF])	Sorafenib	Infections and gastrointestinal perforations are reported in less than 1% of treated patients.
Inhibitors of tyrosine kinases on PDGFRs and VEGFRs	Sunitinib	Cases of necrotizing fasciitis, respiratory infections, and sepsis.
Inhibitor of vascular endothelial growth factor receptor-2 (VEGFR-2), epidermal growth factor receptor (EGFR), and rearranged during transfection (RET) tyrosine kinase	Vandetanib	No significant increase in infections.
Tyrosine kinase inhibitor that inhibits angiogenesis blocking VEGFR-1–VEGFR-3, c-KIT, and PDGFR	Axitinib	No significant data available for evaluation of the risk of severe infectious complications.
Inhibitor of tyrosine kinases c-Met and VEGFR-2 and also AXL and RET	Cabozantinib	No significant data available for evaluation of the risk of severe infectious complications.

Table Continued

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CLASS/SITE AND MECHANISM OF ACTION	DRUG	POSSIBLE INFECTIOUS COMPLICATIONS
Inhibitor of tyrosine kinases VEGFR-1–VEGFR-3	Lenvatinib	No significant data available for evaluation of the risk of severe infectious complications.
Epidermal Growth Factor (EGF) Signaling Pathway		
Anti–EGF receptor (ErbB1 or EGFR) mAb	Panitumumab (IgG2) Cetuximab (IgG1, chimeric)	Cases of necrotizing fasciitis, abscesses sepsis caused by *Staphylococcus aureus*. Important dermatologic toxicity, such as rash, skin drying and fissuring, or paronychial inflammation, with infectious complications (abscess, bacteremia) in up to 30% of patients, including sepsis caused by *S. aureus*. More severe infections have been described when combined with neutropenia-inducing chemotherapy.
	Nimotuzumab (humanized)	No significant data available for evaluation of the risk of severe infectious complications.
mAb against human EGF receptor 2 (human ErbB-2 or HER2)	Trastuzumab	Infections are generally mild and reported as URTIs or UTIs. Increased risk of infections has been described in combination with chemotherapy. Exacerbation of chemotherapy-induced neutropenia. Infection ≥grade 3: 8.5% of patients (95% CI, 4.5–15.4), RR vs. control, 1.21 (95% CI, 1.07–1.37). Febrile neutropenia in the neoadjuvant/adjuvant and combination therapies: 12.0% (95% CI, 8.1–17.4), RR vs. control, 1.28 (95% CI, 1.08–1.52).
	Pertuzumab	>10% of febrile neutropenia, URTIs (associated with chemotherapy).
Inhibitor of epidermal growth factor receptor–tyrosine kinase inhibitor (EGFR-TKI), possibly also of mutated JAK2	Erlotinib	Approximately 9% higher rate of infections compared with control arms for erlotinib. In a meta-analysis of over 6000 patients with NSCLC: any infection, 7% (95% CI, 4.7–10.3), severe infection, 2.1% (95% CI, 1.7–2.8), and fatal, 0.7% (95% CI, 0.4–1.0). There is a trend toward a higher risk with longer treatment.
Inhibitor of EGFR-TKI	Gefitinib	
Immune Checkpoint Inhibitors		
mAbs inhibiting programmed cell death protein 1 (PD-1)	Nivolumab	No significant increase in infectious complications, but immune-mediated complications are frequent and should be considered in differential diagnosis (pneumonitis, colitis, hepatitis).
	Pembrolizumab	No significant data available for evaluation of the risk of severe infectious complications.
mAbs inhibiting programmed cell death ligand 1 (PD-L1)	Atezolizumab	Cases of fever and UTIs.
	Avelumab **Durvalumab**	No significant data available for evaluation of the risk of severe infectious complications.
mAb inhibiting cytotoxic T-lymphocyte–associated antigen 4 (CTLA-4)	Ipilimumab	No significant data available for evaluation of the risk of severe infectious complications.
	Tremelimumab	
Immunomodulating Drugs		
Angiogenesis inhibitor	Thalidomide	Neutropenia and lymphopenia described, reported increased risk of infections (RR, 1.64; 95% CI, 1.40–1.92).
Apoptosis inducer in vivo, with antiangiogenic and osteoclastogenic effects	Lenalidomide	A meta-analysis reported 14% (95% CI, 12.08–16.90) incidence of severe infections, with RR 2.23 (95% CI, 1-71–2.91). May increase risk and duration of cytopenia if combined with chemotherapy.
Immunomodulatory, like thalidomide and lenalidomide, based on its use in multiple myeloma	Pomalidomide	Severe infectious complications in 23%, frequently in the absence of neutropenia. Pneumonia is the most frequently reported localization. Antibacterial prophylaxis has been recommended for patients receiving this drug, but fluoroquinolones could be contraindicated due to the risk of drug interactions.
Activator of monocytes and macrophages	Mifamurtide	Fever frequently described, but no specific data on severe infectious complications.
Proteasome inhibitors	Bortezomib	Increased risk of herpes zoster.
	Carfilzomib	Similar rate compared with bortezomib arms. Low risk of febrile neutropenia. Pneumonia (13%) and URTIs (28%) observed in patients receiving single-drug therapy, but approximately 2%–6% had severe infections. Increased risk of herpes zoster.
	Ixazomib	Pneumonia (10.2% vs. 7.4% in placebo arm) in newly diagnosed MM patients; Increased risk of herpes zoster in those not receiving prophylaxis and overall 10% higher risk of any infection.
Others		
mAb against disialoganglioside (expressed on a variety of embryonal cancers, e.g., neuroblastoma, retinoblastoma, osteosarcoma, Ewing sarcoma, rhabdomyosarcoma)	Dinutuximab (anti-GD2)	Fever is frequently observed, but there are data available for significant infectious complications.

Table Continued

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CLASS/SITE AND MECHANISM OF ACTION	DRUG	POSSIBLE INFECTIOUS COMPLICATIONS
Cyclopamine-competitive antagonist of the smoothened (a transmembrane protein) receptor that is a part of the hedgehog signaling pathway that is involved in proper cell differentiation	Vismodegib	No significant data available for evaluation of the risk of severe infectious complications.

CRS most often begins 2 to 7 days after cell infusion and peaks within 2 weeks. CRS usually presents with fever, myalgia, anorexia, and nausea; if it progresses, capillary leakage leads to dyspnea, hypoxia, diffuse pulmonary infiltrates, hypoxemia, and hypotension. Further progression can result in damage to the heart, liver, and kidneys. Coagulopathy and disseminated intravascular coagulation may occur. Page 3649Manifestations of CRS have many features in common with sepsis or septic shock and hemophagocytic lymphohistiocytosis/macrophage activation syndrome (HLH/MAS), with elevated cytokine levels, cytopenias, hypofibrinogenemia with elevated D-dimers, hyperferritemia, and elevated C-reactive protein (CRP).

Surgery

Surgical site infections and deep organ abscesses are not uncommon in cancer patients undergoing urologic, gynecologic, and abdominal surgeries, but it is difficult to know if this happens significantly more often in oncologic versus nononcologic patients.79,80 In patients with solid tumors, surgery, together with prolonged hospital stays and use of third-generation cephalosporins and glycopeptides, have been associated with an increased risk of infections caused by MDR pathogens.81

Several studies reported the rates of postsurgical infections in different cancer populations. Among patients with peritoneal carcinomatosis undergoing peritonectomy and intraperitoneal hyperthermic chemotherapy, the proportion of infectious complications ranged from 24% to 36%, with more than two infectious episodes per patient.79,80,82 In breast cancer, surgical site infection occurs in 4% to 8% of cases, depending on whether breast reconstruction is performed in one or two steps and whether surgery follows prior chemotherapy.83,84 In case of malignant biliary obstruction, 6.5% of patients who undergo percutaneous biliary stent insertion develop early infections.85 A similar incidence was reported in patients undergoing surgery for hepatocellular or metastatic carcinoma (3%–11%), although apparently lower than with surgery for nonmalignant conditions such as hepatolithiasis (24%).86 The rate of infectious complications after hepatectomy for hepatocarcinoma was associated with surgical risk factors such as bile leakage and blood loss.87 A similar incidence of surgical site infections was reported after elective colon and rectal surgery (9% and 18%, respectively) and after orthopedic surgery (9.5%).88,89 Of interest, in the latter study the use of an implant or allograft did not represent a risk factor for infectious complications.89 Finally, postoperative respiratory infections have been reported in nearly 4% of patients undergoing surgery for lung cancer, and they occur more frequently with advanced age, impaired respiratory function, advanced pathologic stage, and induction chemotherapy.90

In conclusion, surgical and ICU-related risks are important mediators of infectious complications in solid tumor patients.

Etiology

Surveillance studies on pathogens causing infections in cancer patients are of the utmost importance for the implementation of management strategies. Large-scale studies are obviously crucial because they can provide information about worldwide trends, but also single-center surveillance reports may be useful, because every geographic region, country, or even single center may have peculiarities related to the type of patient, type of care, and previous local antibiotic policies. Most of the available information concerns bacterial and fungal pathogens isolated in the bloodstream, whereas the role of deep-seated infections, in addition to the impact of viral infections, is less well described.

Bacterial Infections

Over the last 30 years, gram-positive bacteria were the most frequent pathogens causing BSIs in cancer patients. However, an increase in the frequency of bacteremias caused by gram-negative rods has been reported, with gram-negative pathogens becoming either predominant or at least as frequently isolated as the gram-positives.12,91,92 This trend has also been reported in a retrospective literature review and Page 3653contemporary surveillance study performed in 2011 in 39 European hematology centers from 18 countries belonging to the network of the European Conference of Infections in Leukemia (ECIL).93 As shown in Fig. 311.1, this study found that gram-negative pathogens were almost as frequently isolated as the gram-positives, with gram-positive/gram-negative ratios in BSIs of 60%/40% and 55%/45% in the literature review and the

ECIL-4 surveillance, respectively. The detailed etiology was similar (see Fig. 311.1) in the literature review and the surveillance study, with a slightly higher rate of enterococci and Enterobacterales and a decreased rate of Pseudomonas aeruginosa in the surveillance study.93 These changes in etiology appear to be associated with an important increase in the proportion of resistant pathogens, such as ESBL-producing Enterobacteriaceae, VRE, MDR P. aeruginosa, and the most worrisome, carbapenem-resistant Enterobacterales (CRE). In addition, in leukemic patients, most staphylococci are resistant to methicillin, whereas most gram-negative pathogens are resistant to FQs.93

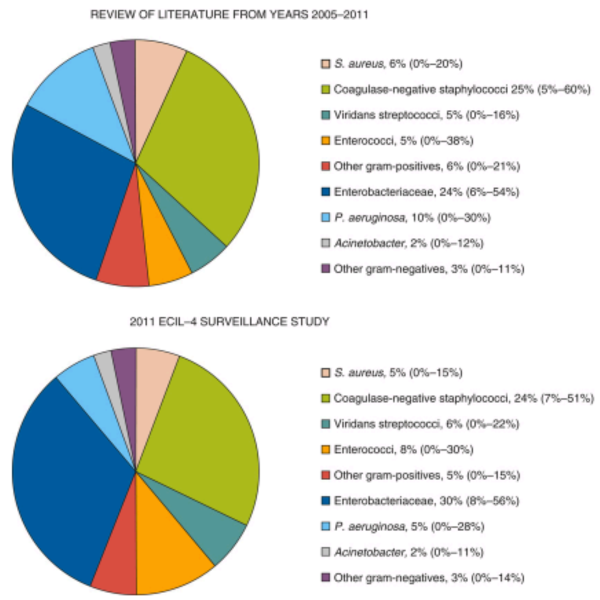

FIG. 311.1 Etiology of bloodstream infections in cancer patients.*Top*, Data from recent literature review. *Bottom*, Data from surveillance study for the Fourth European Conference of Infections in Leukemia (ECIL-4) in 2011. *(Modified from Mikulska M, Viscoli C, Orasch C, et al. Fourth European Conference on Infections in Leukemia Group (ECIL-4), a joint venture of EBMT, EORTC, ICHS, ELN and ESGICH/ESCMID. Aetiology and resistance in bacteraemias among adult and paediatric haematology and cancer patients.* J Infect. *2014;68:321–331.)*

Follow for extended description
Of note, the rates of resistance were generally higher in southern and eastern than in northern and western Europe, and this trend is also evident in the non–hematologic cancer population.93,94,95 However, the problem of increasing infections due to resistant pathogens is worldwide.

The increase in infections caused by resistant strains usually stems from an increase in colonization with these pathogens. Indeed, colonization with resistant bacteria is one of the most important risk factors for infection with resistant bacteria (Table 311.3). The association between colonization and subsequent infection has been reported for VRE, ESBL-producing Enterobacterales, P. aeruginosa, Stenotrophomonas maltophilia, and carbapenem- or colistin-resistant Klebsiella pneumoniae.96–99 These data may differ in pediatrics. A recent multinational survey on antibiotic-resistant bacteremias in children receiving Page 3654antineoplastic chemotherapy or HSCT showed that colonization was a significant risk factor for MRSA bacteremia, whereas it was not associated with an increase in the risk of antibiotic-resistant gram-negatives, with the exception of resistance to piperacillin-tazobactam.5 This observation can have important management implications because patients carrying MRSA can be successfully

TABLE 311.3

Risk Factors for Infections Caused by Resistant Bacteria in Cancer Patients

1. Previous infection or previous or current colonization by resistant bacteria, in particular:
    - Enterobacteriaceae resistant to third-generation cephalosporins or carbapenems
    - Multidrug-resistant, nonfermenting, gram-negative rods: *Pseudomonas aeruginosa, Acinetobacter baumannii, Stenotrophomonas maltophilia*
    - Vancomycin-resistant enterococci
    - Methicillin-resistant *Staphylococcus aureus*
2. Previous exposure to broad-spectrum antibiotics, in particular to third-generation cephalosporins
3. Health care–related infection
4. Prolonged hospital stay and/or repeated hospitalizations
5. Presence of urinary catheter
6. Older age
7. Intensive care unit stay

Obviously, not all colonized patients will develop infection due to the colonizing strain, because the risk depends on multiple host- and chemotherapy-related variables. However, colonization-informed choice of empiric treatment of febrile neutropenia targeting the MDR colonial decolonized100 and might benefit from the use of a glycopeptide in the initial empirical therapy of febrile neutropenia (see later).

Obviously, not all colonized patients will develop infection due to the colonizing strain, because the risk depends on multiple host- and chemotherapy-related variables. However, colonization-informed choice of empiric treatment of febrile neutropenia targeting the MDR colonial might prevent breakthrough infections. In a study on carbapenem-resistant KPC-producing K. pneumoniae, the introduction of prompt screening and a targeted empirical treatment lowered the rate of BSI due to KPC-producing K. pneumoniae among colonized patients from 67% to 11%, and the mortality of BSI due to the resistant strain was also significantly reduced.101 Although early detection of colonization with MDR gram-negative bacteria is useful, the impact of screening for VRE is less clear. VRE BSI has been associated with lower survival at 3 and 12 months after allogeneic HSCT102; however, there was no benefit of VRE-targeting empirical therapy in colonized patients with VRE BSI, suggesting that VRE BSI might be a marker of mortality risk as opposed to causative.103

In a recent cross-sectional study in adults from the United States, the rate of critical illness (defined as ventilation, vasopressor therapy, or death) within 30 days from BSI was not influenced by FQ prophylaxis, organism type, initial antibiotics, or adequacy of coverage, although in these patients approximately 10% of gram-negatives were susceptible to the standard empirical therapy with cefepime or piperacillin-tazobactam.104 In a multicenter pediatric study in which over 20% of gram-negatives were resistant to cefepime or piperacillin-tazobactam, the risk of ICU admission or death was related to antibiotic resistance and prior antibiotic exposure.5 In both these studies the treatment center played an important role in the distribution of gram-positive versus gram-negative species and development of infections due to resistant pathogens.

Anaerobic bacteria are isolated in less than 1% of positive blood cultures in cancer patients, but the proportion increases to 3% among those undergoing abdominal surgery.79 Anaerobes are usually isolated in polymicrobial bacteremias, especially together with gram-negative rods, with a rate that seems to be higher than that observed in non-oncology patients undergoing similar surgery (0.597 vs. 0.033 per 1000 hospital days, respectively).105 Differences in the etiology of bacterial infection between neutropenic and nonneutropenic cancer patients have been reported.12

CVC-related bacteremias are generally caused by gram-positive cocci (especially coagulase-negative staphylococci), which are isolated in more than 50% of the episodes compared with 25% to 40% for gram-negative rods.7,106–113 The source of infection is likely to be partially different in cases of gram-positive and gram-negative CVC-related bacteremias. Infusate contamination is rare but possible, particularly with Klebsiella, Enterobacter, Citrobacter, Achromobacter, Serratia, Ralstonia, and Pseudomonas (other than P. aeruginosa). Polymicrobial infections are not uncommon, especially with gram-negative bacteria.

Bacterial gastroenteritis caused by classic enteric pathogens (Salmonella and Shigella) is a rare event in patients with acute leukemia, involving less than 1% of acute enteritis after chemotherapy.114 In contrast, C. difficile is common in cancer patients, with an incidence that is twofold higher than in the noncancer population115 and up to sixfold higher in hematology patients.116 Helicobacter pylori has also been described as a possible cause of gastrointestinal disease in this patient population.

Legionellosis and nocardiosis are rare but potentially life-threatening infections, with Legionella pneumonia sometimes presenting with unusual radiologic patterns. Mycoplasma pneumoniae and Chlamydia pneumoniae have been rarely described as a cause of pneumonia in cancer patients, but it is possible that their incidence is underreported.

Finally, TB might be underestimated and underdiagnosed in cancer populations. Older data showed a rate of approximately 90 cases per 100,000 persons (i.e., ninefold higher than in the general population in developed countries).117 However, a recent review found very low rates of TB in hematology settings, with no evidence that universal screening with immunoassays and treatment of latent TB provides clinical benefits in people with low risks in nonendemic areas.118 Patients from high-incidence countries or communities account for most of the cases of TB in the cancer population. Findings of pleuroparenchymal imaging abnormalities (mainly on the upper lobes) suggestive of previous TB and people with reported or predicted exposures should be screened for prior infection using immunoassays, with consideration for anti-TB therapy.118 Routine screening may be warranted in people scheduled to undergo therapy that causes prolonged T-cell depletion. Infections caused by nontuberculous mycobacteria are rare outside of cases of CVC-related BSIs. Local or even disseminated M. bovis infections can occur in patients receiving BCG immunotherapy for bladder cancer.

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Obviously, not all colonized patients will develop infection due to the colonizing strain, because the risk depends on multiple host- and chemotherapy-related variables. However, colonization-informed choice of empiric treatment of febrile neutropenia targeting the MDR colonial might prevent breakthrough infections. In a study on carbapenem-resistant KPC-producing K. pneumoniae, the introduction of prompt screening and a targeted empirical treatment lowered the rate of BSI due to KPC-producing K. pneumoniae among colonized patients from 67% to 11%, and the mortality of BSI due to the resistant strain was also significantly reduced.101 Although early detection of colonization with MDR gram-negative bacteria is useful, the impact of screening for VRE is less clear. VRE BSI has been associated with lower survival at 3 and 12 months after allogeneic HSCT102; however, there was no benefit of VRE-targeting empirical therapy in colonized patients with VRE BSI, suggesting that VRE BSI might be a marker of mortality risk as opposed to causative.103

In a recent cross-sectional study in adults from the United States, the rate of critical illness (defined as ventilation, vasopressor therapy, or death) within 30 days from BSI was not influenced by FQ prophylaxis, organism type, initial antibiotics, or adequacy of coverage, although in these patients approximately 10% of gram-negatives were susceptible to the standard empirical therapy with cefepime or piperacillin-tazobactam.104 In a multicenter pediatric study in which over 20% of gram-negatives were resistant to cefepime or piperacillin-tazobactam, the risk of ICU admission or death was related to antibiotic resistance and prior antibiotic exposure.5 In both these studies the treatment center played an important role in the distribution of gram-positive versus gram-negative species and development of infections due to resistant pathogens.

Anaerobic bacteria are isolated in less than 1% of positive blood cultures in cancer patients, but the proportion increases to 3% among those undergoing abdominal surgery.79 Anaerobes are usually isolated in polymicrobial bacteremias, especially together with gram-negative rods, with a rate that seems to be higher than that observed in non-oncology patients undergoing similar surgery (0.597 vs. 0.033 per 1000 hospital days, respectively).105 Differences in the etiology of bacterial infection between neutropenic and nonneutropenic cancer patients have been reported.12

CVC-related bacteremias are generally caused by gram-positive cocci (especially coagulase-negative staphylococci), which are isolated in more than 50% of the episodes compared with 25% to 40% for gram-negative rods.7,106–113 The source of infection is likely to be partially different in cases of gram-positive and gram-negative CVC-related bacteremias. Infusate contamination is rare but possible, particularly with Klebsiella, Enterobacter, Citrobacter, Achromobacter, Serratia, Ralstonia, and Pseudomonas (other than P. aeruginosa). Polymicrobial infections are not uncommon, especially with gram-negative bacteria.

Bacterial gastroenteritis caused by classic enteric pathogens (Salmonella and Shigella) is a rare event in patients with acute leukemia, involving less than 1% of acute enteritis after chemotherapy.114 In contrast, C. difficile is common in cancer patients, with an incidence that is twofold higher than in the noncancer population115 and up to sixfold higher in hematology patients.116 Helicobacter pylori has also been described as a possible cause of gastrointestinal disease in this patient population.

Legionellosis and nocardiosis are rare but potentially life-threatening infections, with Legionella pneumonia sometimes presenting with unusual radiologic patterns. Mycoplasma pneumoniae and Chlamydia pneumoniae have been rarely described as a cause of pneumonia in cancer patients, but it is possible that their incidence is underreported.

Finally, TB might be underestimated and underdiagnosed in cancer populations. Older data showed a rate of approximately 90 cases per 100,000 persons (i.e., ninefold higher than in the general population in developed countries).117 However, a recent review found very low rates of TB in hematology settings, with no evidence that universal screening with immunoassays and treatment of latent TB provides clinical benefits in people with low risks in nonendemic areas.118 Patients from high-incidence countries or communities account for most of the cases of TB in the cancer population. Findings of pleuroparenchymal imaging abnormalities (mainly on the upper lobes) suggestive of previous TB and people with reported or predicted exposures should be screened for prior infection using immunoassays, with consideration for anti-TB therapy.118 Routine screening may be warranted in people scheduled to undergo therapy that causes prolonged T-cell depletion. Infections caused by nontuberculous mycobacteria are rare outside of cases of CVC-related BSIs. Local or even disseminated M. bovis infections can occur in patients receiving BCG immunotherapy for bladder cancer.

Fungal Infections

Aspergillus spp. and Candida spp. are the most common fungal pathogens in cancer patients, with the former now seen more frequently than the latter. Other fungal pathogens include P. jirovecii, cryptococci, molds such as Mucorales or Fusarium, and rare yeasts.

Among yeasts, Candida is the most frequently isolated, usually in BSIs, with an increasing proportion of non-albicans strains. This phenomenon, known since 1999, was confirmed by a study from the European Organization for Research and Treatment of Cancer (EORTC), which reported that in cancer patients the overall incidence of fungemia was 2.3%, with C. albicans responsible for 72% of candidemia cases in the whole study population, but only 27% in patients with hematologic malignancies. This is due to fluconazole prophylaxis, which reduced the overall rate of candidiasis but favored breakthrough infections with fluconazole-resistant species.119,120 Similar proportions were observed recently in pediatric patients.121 The majority of candidemias were observed in nonneutropenic patients. In general, yeasts belonging to the Candida parapsilosis complex are usually associated with CVC contamination, whereas other Candida species typically originate from the gastrointestinal tract after selection and translocation. C. glabrata is increasing in frequency and resistance, not only to fluconazole but also to echinocandins.122 Less common is infection with Candida auris, a frequently drug-resistant species causing outbreaks worldwide, mainly due to its ability to form biofilms and to persist on surfaces.

Among molds, Aspergillus represents the most frequently isolated or suspected. The majority of episodes are caused by Aspergillus fumigatus, although some centers report a predominance of infections caused by Aspergillus flavus and Aspergillus terreus.126,127 Invasive aspergillosis in patients with malignancy or HSCT is relatively common and usually community acquired, but outbreaks can occur with environmental exposures, both in and outside the hospital. The incidence of invasive aspergillosis depends mainly on the underlying malignancy and its treatment. It is highest in patients with prolonged neutropenia, followed by those receiving high doses of steroid therapy.128 In a multicenter Italian study, the incidence of aspergillosis among hematologic cancer patients varied from 7.9% in acute nonlymphoblastic leukemia to 4.3% in ALL, 2.3% in chronic myelogenous leukemia, and less than 1% in CLL, Hodgkin and non-Hodgkin lymphoma, and multiple myeloma.126 Additionally, the risk of aspergillosis in adult patients with ALL was reported to be as high as 11.7%.129 Recent pediatric data showed a higher incidence of invasive mycoses in high-risk ALL (up to 15% in the absence of pharmacologic prophylaxis) occurring since the beginning of treatment (which includes the use of high-dose steroids) and in all the most aggressive phases.130,131 Patients with chronic lymphoproliferative disorders represent another at-risk group in which aspergillosis can be seen, probably as a result of more intensive treatment protocols. Despite the high number of patients with these diseases, the rate of IFD is <5%.127,128 The incidence of invasive aspergillosis after autologous HSCT is low (0.3%–2%), and it typically occurs during pre-engraftment neutropenia.132

Increased attention to azole resistance among Aspergillus species, especially members of the A. fumigatus species complex, is necessitated by recognition of organisms harboring mutations in erg11 along with Page 3655recognition of MDR Aspergillus species (e.g., A. calidoustus) and the recently described “cryptic” species (e.g., A. lentulus).133 Rates of infections with these organisms can vary depending on geographic region and agricultural use of azole antifungals. Despite the fact that the phenomenon is so far relatively limited in most settings, knowledge of the frequency of azole resistance at the country and hospital level is fundamental. Secondary azole resistance, with different genetic resistance mutations, has been reported in patients with chronic aspergillosis after prolonged treatment, particularly in the case of suboptimal blood levels or high fungal burden (see Chapter 263).134

Mucormycosis is increasingly reported in some centers, especially in the United States. It is unclear whether this represents a general trend influenced by geography or prior antifungal therapies or if these infections are simply diagnosed more often because of increased awareness or improved survival. Other fungi, such as Fusarium and Scedosporium, are reported sporadically, with some organisms having higher prevalence in certain regions.135

P. jirovecii is a well-known cause of pneumonia in cancer patients not receiving trimethoprim-sulfamethoxazole (TMP-SMX) prophylaxis, especially if treated with high-dose and prolonged steroid therapy or certain antineoplastic drugs such as fludarabine, alemtuzumab, idelalisib, or temozolomide. Attack rates vary from 6.5% to 43% in ALL, 4% to 25% in rhabdomyosarcomas, and nearly 1% in Hodgkin lymphoma and primary or metastatic central nervous system tumors.136,137

Finally, infections with Cryptococcus spp. and geographically influenced “endemic” fungi, such as Histoplasma or Coccidioides, are possible in patients who live or used to live in endemic areas.

Viral Infections

Apart from herpes simplex virus (HSV) reactivation, which occurs in up to 60% of HSV-seropositive patients with acute leukemia, there are few data describing viral infections in febrile neutropenia.3 Cytomegalovirus (CMV) reactivation, as measured by pp65 antigenemia or PCR, occurs in people with severe illness, without necessarily being followed by CMV disease.138 For this reason, routine monitoring of CMV reactivation and preemptive therapy is not considered necessary in cancer patients other than HSCT recipients. A recent study evaluating febrile neutropenia in children and adults receiving intensive chemotherapy showed viremia in 15% of episodes, with the highest frequency in children and the highest viral loads in patients otherwise classifiable as fever of unknown origin (FUO), suggesting that at least some cases of febrile neutropenia in children could be attributable to viral infection/reactivation.139 The risk of viral reactivation (herpes simplex virus [HSV], VZV, CMV) and disease increases significantly with use of T-cell–suppressing agents or drug combinations, such as alemtuzumab or idelalisib or a combination of bendamustine and rituximab.59,140–142

Community-acquired respiratory viruses, such as influenza, parainfluenza, respiratory syncytial virus (RSV), human metapneumovirus, adenoviruses, rhinoviruses, and coronaviruses, are a frequent cause of respiratory disease in cancer patients and are probably underestimated as a cause of fever. Whereas most cancer patients experience self-limited upper respiratory illness, those with severe immune deficits, which can be estimated using dedicated scores that take into consideration lymphopenia, steroid use, hypogammaglobulinemia and recent transplant, and B-cell or T-cell depletion,143 are at increased risk for progression from upper respiratory tract infection to pneumonia, with possible respiratory failure.144 The rate of viral respiratory infections in cancer patients depends on the ongoing rate of infections in the community and on the testing policy. In cancer patients with viral respiratory disease, deferral of chemotherapy may be considered. Specific treatment is warranted for influenza and in some cases for RSV infection (e.g., in leukemic patients with risk factors for RSV-related mortality; see Chapter 165).144

SARS-CoV-2 infection has caused significant morbidity and mortality in cancer patients—much higher than in the general population.145,146 In the cancer setting, the additional negative effect of even mild SARS-CoV-2 infection is caused by interruption of chemotherapy, which can be extended in case of prolonged viral shedding. Patients with non-Hodgkin lymphoma and CLL, particularly if treated with B-cell–depleting therapy, are at the highest risk of prolonged or relapsing infection, and optimal management remains still to be defined. Early treatment is warranted to prevent progression to severe disease and potentially shorten the duration of infection.146

Viral gastroenteritis, mainly caused by rotavirus but also norovirus or sapovirus, may be a frequent complication in pediatric oncology, with a potential to cause outbreaks in cancer centers because of persistent gastrointestinal shedding in immunocompromised hosts. Both adenoviruses and parvovirus B19 have been reported as rare causes of severe gastrointestinal disease in cancer patients.

Finally, the reactivation/exacerbation of infections due to hepatotropic viruses (HBV and hepatitis C virus [HCV]) represents an important problem in areas of high endemicity. HBV reactivation is frequent in cancer patients with chronic inactive HBV infection (HBsAg positive, with negative or low-level serum HBV DNA), but it can also occur in patients with resolved HBV infection (HBsAg negative, HBcAb positive, or with low-level serum HBV DNA), particularly after treatment with rituximab (up to 40% of patients).60 Antiviral prophylaxis, preferably with a high barrier to resistance such as entecavir or tenofovir, is usually given to these patients for at least 6 to 12 months after the end of chemotherapy.

HCV flares might occur in cancer patients undergoing chemotherapy, and HCV eradication could lower the risk of hepatic toxicity and even improve the outcome in certain lymphomas.147 The possibility of chronic hepatitis E virus infection due to genotype 3, which is endemic in certain industrialized regions, has been reported in patients with hematologic malignancies and transplant recipients.148,149 Repeated transfusions might be a risk factor for hepatitis E virus infection.150,151

Other Pathogens

The risk of rare infections or reactivations caused by protozoa (leishmaniasis, South American trypanosomiasis, and malaria), helminths (strongyloidiasis), endemic fungi, and other tropical diseases should be considered in patients who live or lived in endemic areas. Obtaining a detailed history to identify potential exposures is the most important screening tool, with additional serology to document past exposure. However, in the case of strongyloidiasis, for example, the suboptimal performance of stool examination or serologic screening may warrant empirical treatment with ivermectin in patients who present with unexplained eosinophilia and who lived in endemic areas, such as the tropics, the subtropics, or the southeastern United States and Europe.

Prevention of Infections in Cancer Patients

Prevention is desirable, given the remarkable mortality and morbidity associated with infections in cancer patients. In general, to evaluate the cost-effectiveness of a prophylactic protocol, one should know the rate of the target infection in specific patient populations and the number of patients needed to treat to prevent a single infectious episode, severe infection, and attributable death for each center. The rate of severe side effects, reported as the number needed to harm, must also be taken into consideration.

Table 311.4 summarizes different regimens for primary chemoprophylaxis and other approaches appropriate in cancer patients based on clinical trials and guidelines. Advantages and disadvantages of these procedures are discussed here.

Prevention of Bacterial Infections

The use of antibiotics to prevent bacterial infections should be weighed against efficacy, toxicity, and impact on the development of resistance. Chemoprophylaxis for the prevention of bacterial infections was first proposed in clinical practice based on the recognition that 80% of bacterial pathogens causing infection in neutropenic cancer patients originated from the endogenous flora, with approximately half of infections acquired during the hospital stay. The first approach relied on the oral administration of nonabsorbable antibiotics (gentamicin, vancomycin, and nystatin) aimed at totally (including anaerobes) or partially (excluding anaerobes) suppressing the intestinal bacterial flora and preventing the acquisition of exogenous organisms (known also as gut decontamination). Subsequently, absorbable drugs such as TMP-SMX, usually given in combination with oral nystatin and then FQs, were introduced. Large meta-analyses of quinolone prophylaxis in cancer patients have confirmed a statistically significant reduction in febrile Page 3656neutropenia, BSIs, and mortality, with the number needed to treat (NNT) to prevent one death ranging from 24 to 50, with no untoward effect in induction of resistance.152,153 Most guidelines recommended FQ prophylaxis for adult patients with expected neutropenia longer than 7 days. These results were based on studies published between 1973 and 2011, largely before the spread of bacterial resistance.154

Given the alarming increase in bacterial resistance worldwide, two issues need to be addressed: (1) whether the benefit of FQ prophylaxis can still be expected and (2) what is the contemporary impact of routine prolonged administration of broad-spectrum agents, such as FQ, on the selection of resistant strains and local epidemiology. Meta-analyses of recent studies did not confirm the benefit on mortality, although prophylaxis was still associated with lower rates of BSI and febrile episodes.154,155 The reason might be that the sample size was insufficient to show a difference in mortality, because the outcome of febrile neutropenia has significantly improved. Alternatively, infections prevented by FQ may be associated with low mortality, because they can be effectively treated with standard empirical antibiotics.95,156,157 As demonstrated by studies performed in both children and adults, BSI-associated mortality is mainly driven by MDR pathogens, which are unlikely to be prevented by FQs.5,95,158,159 This might explain the fact that FQ prophylaxis may no longer have any effect on overall mortality, despite efficacy in preventing some infections (i.e., those due to organisms that are effectively treated with classic empirical therapy).154,156

Some studies have suggested that FQ prophylaxis may increase rates of MDR infections.154 Finally, FQ can be associated with toxicities, including QT prolongation, tendon rupture, and interaction with drugs such as cyclophosphamide.160 Therefore, some recent guidelines do not support the use of FQ prophylaxis.161–163 In patients with solid tumors, who usually experience low rates of BSI and shorter (<7 days) neutropenia, FQ prophylaxis is not recommended.20

Additional strategies for prevention of MDR bacterial infection include decolonization with orally nonabsorbable antibiotics and fecal microbiota transplantation (FMT), which reduce intestinal burden of MDR strains. Although decolonization with gentamicin or colistin can decrease infections with MDR pathogens, this strategy may also increase resistance among colonizing bacteria.164 FMT has been evaluated in a small prospective trial of MDR-colonized patients with hematologic malignancies; partial-to-complete decolonization was reported.165 The FMT procedure was more effective in the absence of concomitant antibiotic administration (79% vs. 36%), and no severe adverse events were observed.165 Further studies are warranted to determine all the potential benefits and real-life applications of manipulation of gut microbiota.

Prevention of Tuberculosis

Risks for reactivation of latent TB are highest among people with sustained defects in T-cell immunity and are not clearly increased in people with short or intermittent episodes of neutropenia. Few data exist, and they depend on geographic prevalence.118 Standard regimens for latent TB are recommended (rifampin 4 months, rifampin and isoniazid 3 months, isoniazid 9 months), but drug-drug interactions between rifampin and hematologic treatments, combined toxicities (hepatic and neurologic), and the negative impact of delaying hematologic treatment118 must be carefully weighed.

Prevention of Central Venous Catheter–Related Infections (See Chapter 306)

Prevention of CVC-related infections starts with the correct insertion and manipulation of the devices, not only by health care workers but also by caregivers who perform CVC maintenance procedures (and sometimes drug infusions) at home. Aseptic bundles during insertion and postinsertion care, which include assessment of need for CVC, daily inspection of the site of insertion, change of dressing at least weekly or whenever wet or soiled, use of chlorhexidine gluconate–impregnated sponges, application of alcohol scrub for 15 seconds before each access of the line, and hand hygiene, are effective measures to prevent infectious complications.41 Continuous education and regular audits of bundle implementation represent other helpful activities.166 Antibiotic catheter flushing has also been suggested to reduce the risk of infections, but this could represent a risk for selection of resistant bacteria. On the contrary, taurolidine lock has been associated with a significant reduction of CVC-related BSI, even if the susceptibilities of gram-positives and gram-negatives to taurolidine are indeterminate due to limited data.167

Prevention of Fungal Infections
Primary Antifungal Chemoprophylaxis

Although fungal infections usually represent no more than 10% of all infections (see Table 311.2), their associated morbidity and mortality remain high. Two main drugs used for antifungal prophylaxis in patients with acute leukemia include fluconazole (active against yeasts such as Candida but not molds) and posaconazole (mold-active prophylaxis). Fluconazole is effective in preventing Candida infections in people with acute myeloid leukemia and may benefit other populations that have an incidence of invasive candidemia higher than 7% to 10%.128 Mold-active

Suggested Prophylaxis for Infections in Cancer Patients

	DRUG	SCHEDULE	COMMENTS
Antibacterial	Ciprofloxacin	500 mg twice daily	Use of FQs should be based on local epidemiology and careful evaluation of potential drawbacks of FQ prophylaxis. Probably not active anymore in many centers. Some studies showed possible effects on increasing resistance. Traditionally administered to adults receiving chemotherapy for acute leukemia with expected neutropenia >7–10 days; starting with chemotherapy and continuing until resolution of neutropenia or initiation of empirical antibacterial therapy for febrile neutropenia.
	Levofloxacin	500 mg once daily; children 10 mg/kg twice daily	
Antifungal	Posaconazole	100-mg tablets: 3 tablets twice daily on the first day, then 3 tablets daily As alternative, oral solution 200 mg tid orally with a (fatty) meal or acidic drink IV: 300 mg twice daily on the first day, then 300 mg daily	Patients receiving high-intensity chemotherapy or certain combination therapies with targeted agents for acute myelogenous leukemia or myelodysplastic syndrome. Therapeutic drug monitoring is warranted in case of oral solution.
	Fluconazole	400 mg once daily	Patients at risk for candidiasis but not aspergillosis.
	Other		Secondary prophylaxis according to isolated pathogen, clinical presentation, or both.
Anti–*Pneumocystis jirovecii*	Trimethoprim-sulfamethoxazole (TMP-SMX)	One double-strength tablet (160/800 mg) three times weekly, or 25 mg/kg TMP-SMX (5 mg/kg of TMP), in 2 divided doses for 3 consecutive days/wk	All patients receiving chemotherapy with steroids, including those with solid tumors (e.g., brain cancer); treatment with agents like fludarabine, alemtuzumab, or idelalisib; or some combination therapies (e.g., bendamustine plus anti-CD20).
	Dapsone	100 mg daily in adults, or 2 mg/kg/day (max. 100 mg) on alternate days three times/wk	In patients who cannot tolerate TMP-SMX.
	Aerosolized pentamidine	300 mg once a month with nebulizer	In patients who cannot tolerate TMP-SMX; effective, but it is more difficult to administer.
	Atovaquone	750 mg twice daily or 1500 mg once daily	In patients who cannot tolerate TMP-SMX.
Antiviral	Acyclovir or	40 mg/kg in children in 2 divided doses In adults >40 kg: 800 mg twice daily	Patients with positive anti-HSV antibodies and severe mucositis or receiving treatment for acute leukemia or other drugs which increased the risk for herpes zoster. VZV-susceptible patients exposed to chickenpox who did not receive prompt administration of specific immunoglobulins.
	Valacyclovir	For HSV or VZV prophylaxis: 500 mg twice daily or 1000 mg daily For VZV exposure: 1 g three times daily (see text)	
	Lamivudine	100 mg once daily	Patients with resolved HBV infection (HBsAg negative and HBcAb positive).
	Entecavir	0.5 mg daily	HBsAg-positive/HBV-DNA <2000 IU/mL or high-risk (e.g. receiving anti-CD20 antibodies) patients with resolved HBV infection.
	Tenofovir disoproxil fumarate (TDF)	300 mg daily	HBsAg-positive/HBV-DNA <2000 IU/mL.
Antituberculosis	Isoniazid	5 mg/kg (max. 300 mg) daily in adults, 10 mg/kg (max. 300 mg) daily in children for 9 months	Patients with latent tuberculosis. Efficacy not specifically evaluated in cancer patients. Add pyridoxine (50 mg/daily for adults).
	Rifampicin	600 mg/daily for 4 months20 mg/kg daily (max. 600 mg) in children	Patients with latent TB who do not receive concomitant medications subject to drug interactions. This regimen has been associated with the lowest toxicity and highest compliance rate.
	Isoniazid and rifampicin	Dose as noted earlier For 3 months	Patients with latent TB who do not receive concomitant medications subject to drug interactions of rifampicin.
Anti–CVC-associated infection	None	Good skin preparation and the use of sterile technique at time of device insertion. Good maintenance procedures.	All patients with indwelling central venous catheter.
			Table Continued

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	DRUG	SCHEDULE	COMMENTS
Others	Growth factors	Filgrastim either subcutaneously or as an intravenous infusion over at least 1 h, or pegylated filgrastim	For the prevention of febrile neutropenia in patients who have a high risk of this complication based on age, medical history, disease characteristics, and myelotoxicity of the chemotherapy regimen. Secondary prophylaxis with G-CSFs recommended for patients who experienced a neutropenic complication from a prior cycle of chemotherapy (for which primary prophylaxis was not received), in whom a reduced dose of chemotherapy may compromise disease-free or overall survival or treatment outcome. Efficacy not fully demonstrated for pegylated filgrastim.
	Immunoglobulins	Polyclonal immunoglobulins: 400 mg/kg every 21–28 days	Patients with chronic lymphocytic leukemia after the second episode of severe bacterial infection. Patients with leukemia or lymphoma with hypogammaglobulinemia (<400 g/dL) and severe bacterial infections (reasonable, but not proved).
		Specific anti-VZV (VariZIG): 125 IU for every 10 kg of body weight (max., 625 IU); adults 650 IU	In high-risk contact with a negative history of varicella preferably within 96 h after exposure to chickenpox.
	Vaccines	Influenza	Influenza vaccination of patients, especially during less aggressive treatment phases. Vaccination of household contact and health care workers.
		Pneumococcus	Conjugated (PCV) vaccine recommended; with new 15-valent or 20-valent PCV vaccines available, the role of PPSV23 to be determined.
		Varicella	VZV-seronegative household contacts and health care workers.
		Herpes zoster	Recombinant inactivated 2-dose vaccine in VZV seropositive patients.
		COVID-19	Vaccinate with 3-dose primary cycle, and provide boosters following updated recommendations.
	Isolation procedures	Perform hand hygiene with an alcohol-based hand rub or by washing hands with soap and water if soiled, before and after all patient contacts or contact with the patients' potentially contaminated equipment or environment. Use contact precautions (gowns and gloves). Ensure adherence to standard environmental cleaning with an effective disinfectant.	Patients colonized or infected with multidrug-resistant pathogens (e.g., VRE, CRE) or infected with other pathogens for which contact isolation precautions are advisable (e.g., *C. difficile*, norovirus); of note, alcohol-based hand rubs are not cidal against *C. difficile* or norovirus.

CRE, Carbapenem-resistant Enterobacteriaceae; *CVC*, central venous catheter; *FQ*, fluoroquinolone; *G-CSFs*, granulocyte colony-stimulating factors; *HBcAb*, hepatitis B core antibody; *HBsAg*, hepatitis B surface antigen; *HBV*, hepatitis B virus; *HSCT*, hematopoietic stem cell transplantation; *HSV*, herpes simplex virus; *IU*, international unit; *max.*, maximum; *VRE*, vancomycin-resistant enterococci; *VZV*, varicella-zoster virus.

prophylaxis with posaconazole in adults receiving multiple cycles of chemotherapy for acute myeloid leukemia or myelodysplastic syndrome reduces the incidence of invasive mycosis from 8% to 2% (primary end point) compared with standard prophylaxis with fluconazole or itraconazole, with improved survival.168 The NNT to prevent one case of proven/probable invasive mycosis was 16, and the NNT to prevent one fungal infection–related death was 27.169 However, if the incidence of invasive mycoses is lower than 8%, the NNT to prevent one infection would be higher, and this underlines the requirement to tailor needs in different hosts. Specific recommendations on drug dosing and formulations are summarized in Table 311.4 and Table 311.5. Recently introduced targeted agents in the treatment of AML, such as venetoclax,

TABLE 311.5

Antibacterial and Antifungal Agents Usually Used in Cancer Patients

PATIENTS' TYPE OF THERAPY	DRUG	ROUTE OF ADMINISTRATION	DAILY PEDIATRIC DOSAGE	USUAL DAILY ADULT DOSAGE	NO. OF DAILY DIVIDED DOSES
Antibacterial, intravenous	Piperacillin-tazobactam	IV	400 mg/kg (as piperacillin)	13.5–18 g (as both drugs)	3–4 in adults, 4 in children; preferably in prolonged or continuous infusion with a loading dose
	Ceftazidime	IV	100 mg/kg	6000 mg	3
	Cefepime	IV	100 mg/kg	6000 mg	3
	Meropenem	IV	60 mg/kg	3000–4000 mg	3, infuse over 3–6 hours
	Ceftolozane-tazobactam	IV	90 mg/kg, up to a maximum dose of 4.5 g [a] (as both drugs)	4500 mg (as both drugs) 9000 mg for nosocomial pneumonia	3, infuse over 1 hour
	Ceftazidime-avibactam	IV	6 months to 18 years 187.5 mg/kg, up to a maximum of 7500 mg/daily (as both drugs) <3 months 150 mg/kg [a]	7500 mg (as both drugs)	3, infuse over 2 hours
	Ceftaroline	IV	2 months to <2 years: 8 mg/kg every 8 h ≥2 years to <18 years (≤33 kg): 12 mg/kg every 8 hours ≥2 years to <18 years (>33 kg): 400 mg every 8 hours or 600 mg every 12 hours	1200 mg	2, infuse over 5–60 minutes
	Ceftobiprole	IV	3 months to 12 years, <33/kg: 60 mg/kg, ≥33 kg 1500 mg; 12–18 years, <50 kg: 30 mg/kg, ≥50 kg 1500 mg	1500 mg (as ceftobiprole) (2 g for the first 7 days in case of *S. aureus* BSI)	4
	Imipenem-cilastatin	IV	60–100 mg/kg (as imipenem)	2000–4000 mg (as both drugs)	3–4
	Meropenem-vaborbactam	IV	ND	12 g (as both drugs)	3, infuse over 3 hours
	Ciprofloxacin	IV	15–30 mg/kg	800–1200 mg	2–3
	Ceftriaxone	IV	80 mg/kg	2000 mg	1
	Amikacin	IV	20 mg/kg	1000–1500 mg	1
	Vancomycin	IV	40–60 mg/kg	2000 mg	2 or as continuous infusion after a loading dose
	Teicoplanin	IV; not available in United States	10 mg/kg (loading dose of 10 mg/kg bid on first day of treatment)	600–1200 mg (loading dose of 600 mg bid on first day of treatment)	1 (2 on first day of treatment)
	Daptomycin	IV	10 mg/kg	6–12 mg/kg [b]	1
	Linezolid	IV (also available oral)	30 mg/kg in three doses if <12 years, then 20 mg/kg	1200 mg	2
	Tedizolid	Oral/IV	ND	200 mg	1
	Dalbavancin	IV	Birth to <6 years 22.5 mg/kg (max. 1500 mg) 6–18 years 18 mg/kg (max. 1500 mg)	1500 mg	Single-dose regimen
	Tigecycline	IV	2.4 mg/kg loading dose, then 2.4 mg/kg (proposed)	100 mg loading dose, then 100 mg	2
	Colistimethate sodium (colistin)	IV	150,000–200,000 IU/kg loading dose, then 150,000–200,000 IU/kg [c]	9,000,000 IU loading dose, then 9,000,000 IU [c]	2
	Polymyxin B	IV	Infants: up to 40,000 U/kg Children: 15,000–25,000 U/kg	15,000–25,000 U/kg (= 1.25–2.5 mg/kg, 1 mg = 10,000 U; a loading dose might be indicated)	
	Fosfomycin	IV; not available in United States	300 mg/kg	15–24 g	3–4
	Cefiderocol	IV	60 mg/kg	6 g (8 g if augmented renal clearance)	3–4
	Aztreonam-avibactam	IV	ND	Loading dose 2670 mg (as both drugs) then 8000 mg daily	4, infuse over 3 hours
	Imipenem-cilastin-relebactam	IV	ND	5000 mg (as all three components, vial 500/500/250)	4, infuse over 30 minutes
	Sulbactam-durlobactam	IV	ND	4000 mg (as of both)	4, infuse over 3 hours

Table Continued

PATIENTS' TYPE OF THERAPY	DRUG	ROUTE OF ADMINISTRATION	DAILY PEDIATRIC DOSAGE	USUAL DAILY ADULT DOSAGE	NO. OF DAILY DIVIDED DOSES
Antibacterial, oral	Amoxicillin-clavulanate	Oral	60–80 mg/kg (as amoxicillin)	2–3 g (as both)	2–3
	Ciprofloxacin	Oral	30 mg/kg	1000–1500 mg	2
	Cefixime	Oral	6–8 mg/kg	400 mg	2–3
	Moxifloxacin	Oral, IV	Based on age, 4–6 mg/kg twice daily [285]	400 mg	1
Antifungal	Amphotericin B deoxycholate [d]	IV	0.5–1 mg/kg	0.5–1 mg/kg	1
	Liposomal amphotericin B	IV	3–5 mg/kg	3–5 mg/kg	1
	Amphotericin B lipid complex	IV	5 mg/kg	5 mg/kg	1
	Caspofungin	IV	50 mg/m^2 for age <17 years	70 mg the first day, then 50 mg the following days	1
	Micafungin	IV	2–4 mg/kg	100 mg	1
	Anidulafungin	IV	3 mg/kg q24h the first day, then 1.5 mg/kg q24h	200 mg the first day, then 100 mg the following days	1
	Rezafungin	IV	ND	400 mg loading dose, then from day 8 200 mg once/wk	
	Voriconazole [f]	IV, oral	9 mg/kg bid the first day, then 8 mg/kg bid [e]	6 mg/kg bid the first day, then 4 mg/kg bid	2
	Fluconazole	IV, oral	10 mg/kg	400 mg	1
	Posaconazole [f]	Oral solution (doses are different for tablets and IV)	If >12 years, dose as adults	800 mg (with a fatty meal or acidic carbonated drink)	4
		Oral delayed-release tablets	Not approved for <18 years but PK/PD data available for per-kg dosing [286, 287]	600 mg/day divided in 2 doses, then 300 mg daily	1
	Isavuconazole [f]	IV/oral	5.2 mg/kg/dose as isavuconazole	200 mg q8h for 6 doses, then 200 mg daily (372 mg if ordered as isavuconazonium sulfate)	3 during the first 48 hour, 1 thereafter

ND, No established dosing for children; *PK/PD*, pharmacokinetic/pharmacodynamic.

[a] Approved for abdominal and urinary tract infection only, not pneumonia.

[b] High doses necessary for enterococcal infections due to high epidemiological cutoff for *E. faecium* (4 mg/L). [292]

[c] An important and potentially confusing characteristic is colistin's dosage expression. Colistin is available in two salt forms: colistin sulfate and colistimethate sodium. In Europe, colistimethate sodium (salt) is available, and dosing is expressed usually in international units (IU) and sometimes in milligrams of colistimethate sodium, whereas in the United States, the dosage of US Food and Drug Administration–approved colistimethate sodium is defined in milligrams of colistin base activity. Thus particular attention should be paid to avoid dosage errors. Approximate dose conversion: 1,000,000 IU = 80 mg of colistimethate sodium = 30 mg colistin base. [288] The optimal dosing of colistin remains to be established. Recent studies in adults with normal renal function reported the use of 6,000,000–9,000,000 IU daily, with pharmacologic analyses supporting the use of a loading dose of 9,000,000 IU, followed by 4,500,000 IU every 12 hours. [289–291] See **Chapter 31** for more details.

[d] Contraindicated in the presence of risk factors for renal toxicity (e.g., impaired renal function at baseline; nephrotoxic comedication, including aminoglycoside antibiotics; and history of previous toxicity).

[e] In patients 2–12 years of age and 12 to 14 years if weighing less than 50 kg; otherwise, dose as for adults if 12 years of age and older. Therapeutic drug monitoring is highly recommended. Children metabolize voriconazole more rapidly than adults.

[f] Therapeutic drug monitoring might be useful to assess if trough levels are in the range for efficacy in case of oral solution.

midostaurin, quizartinib, or gilteritinib, have led to challenges in the management of antifungal prophylaxis Page 3658because coadministration of these agents with posaconazole leads to significant increase in their blood levels, which carries the risk of toxicity (see Table 311.2). Posaconazole prophylaxis is still recommended when these agents are used in combination with other chemotherapy drugs, and dose reduction has been proposed for some (venetoclax), whereas careful monitoring was recommended with a low level of evidence of others (midostaurin).66

Voriconazole prophylaxis in patients with AML has been studied largely in small, single-center and retrospective studies with similar efficacy and toxicity to comparators. Therefore voriconazole has been only moderately recommended, with a lower level of evidence than posaconazole, fluconazole, or itraconazole, in patients with AML or MDS undergoing intensive remission-induction chemotherapy.170 In addition to CYP3A4-mediated drug-drug interactions typical for most of the azoles, voriconazole metabolism is also influenced by the activity of CYP2C19, resulting in significant interpatient variability.171 As certain genetic variations in CYP2C19 may result in increased metabolic activity of voriconazole leading to subtherapeutic blood levels, a strategy based on genetic identification of rapid and ultrarapid metabolizers leading either to increased dosing of voriconazole or alternative prophylaxis reduced the risk of subtherapeutic trough concentrations.172 Isavuconazole prophylaxis has been studied in single-center, mainly retrospective, trials. The three largest studies that included exclusively or mainly AML patients reported the rate of breakthrough proven/probable IFDs ranging from 3%173 to 6% to 7%.174,175

Prophylaxis with nebulized liposomal amphotericin B plus systemic fluconazole, compared with fluconazole only, was effective in reducing proven/probable IFD during repeated periods at risk after chemotherapy for acute leukemia in adults (a 10% reduction in IFD events).176 Systemic antifungal prophylaxis with liposomal amphotericin B at 2.5 mg/kg twice weekly was feasible and safe in high-risk pediatric cancer patients compared with a historical control group.177 On the other hand, a randomized controlled trial that included 355 adult patients.

undergoing remission-induction chemotherapy for newly diagnosed ALL demonstrated a high incidence of IFD in this population but failed to demonstrate the efficacy of liposomal amphotericin B at 5 mg/kg twice weekly (IFD 7.9% in prophylaxis arm vs. 11.7% in placebo arm).4 In a randomized study in children and young adults with AML, caspofungin prophylaxis during neutropenia resulted in a lower rate Page 3660of proven/probable IFDs and aspergillosis compared with fluconazole (respectively, 3.1% vs. 7.2% and 0.5% vs. 3.1%).178

Prophylaxis Against Pneumocystis jirovecii
In recent years, P. jirovecii pneumonia has been described with increasing frequency in patients without HIV.179 Certain underlying diseases, particularly lymphoproliferative ones, and their treatment confer a high risk of pneumocystosis. These include ALL, non-Hodgkin lymphoma, multiple myeloma, and CLL treated with standard chemotherapy or undergoing autologous transplantation.137 Other patients at risk for this complication are those with central nervous system solid tumors, associated with the prolonged use of high-dose steroids and the alkylating agent temozolomide, or those receiving bendamustine for breast cancer.179 Several other drugs affecting cell-mediated immunity have also been associated with the risk of P. jirovecii pneumonia, including fludarabine, ara-C (cytarabine), methotrexate, D-actinomycin, bleomycin, and L-asparaginase. In addition, pneumocystosis has been associated with alemtuzumab, which causes profound depletion of T lymphocytes; rituximab if used in combination with bendamustine or CHOP (cyclophosphamide, doxorubicin [hydroxydaunomycin], vincristine [Oncovin], and prednisolone) chemotherapy every 14 days; with idelalisib and ibrutinib (rare); and, in two cases, with the tyrosine kinase inhibitor dasatinib.59,137 It remains to be determined whether novel biologic drugs are responsible for an increased infection risk or whether other concomitant or previous treatments (purine analogues, alkylating agents, or steroids) contribute significantly to the risk of pneumocystosis. Finally, the duration of risk after treatment discontinuation is not known. Indications for prophylaxis of pneumocystosis include certain diseases (e.g., ALL) and treatments (e.g., fludarabine, alemtuzumab, or corticosteroid at the dose of >20 mg of prednisone daily for at least 4 weeks).180 An absolute CD4+ lymphocyte count of 200/mm3 or less, or a proportion of 15% of all T cells or less (or similar age-related values for children), has been suggested as an indication for prophylaxis, at least after HSCT, but qualitative defects might be present, leading to pneumocystosis even with higher CD4+ levels. Because the efficacy and tolerability of 160/800 mg of TMP-SMX given three or seven times a week are comparable, one 160/800-mg TMP-SMX tablet three times weekly remains the best prophylactic option, provided the patient is strictly compliant. Other drugs, such as daily oral dapsone or atovaquone or monthly aerosolized pentamidine, have demonstrated efficacy and are alternatives for patients who cannot tolerate TMP-SMX (see Chapter 275).180

Prevention of Viral Infections
For the prevention of HSV or VZV reactivation, antiviral prophylaxis with acyclovir or valacyclovir is recommended for seropositive patients undergoing therapy for acute leukemia or treatment with certain new agents such as bortezomib, alemtuzumab, idelalisib, or daratumumab. In patients not on active prophylaxis but who are exposed to varicella, intravenous administration of anti-VZV immunoglobulins (VariZIG) within 72 to 96 hours of the exposure is recommended, although protection is not guaranteed181 (see Chapter 143). If anti-VZV immunoglobulins are not available or in case of delayed notification of the risk contact, prophylaxis with valacyclovir (1 g three times daily for adults weighing more than 40 kg) or acyclovir (in children orally 20 mg/kg per dose, administered four times per day, with a maximum daily dose of 3200 mg; in adults 800 mg five times a day) is an option, although the efficacy has not been established. Prophylactic antiviral chemotherapy should be started on day 7 after exposure and continued for 7 days thereafter.

Routine CMV prophylaxis is not recommended outside of the transplantation setting. Blood CMV DNA testing should be performed in all cancer patients with signs or symptoms compatible with CMV reactivation (cytopenias, unexplained fever) or disease (gastroenteritis, pneumonia, retinitis, hepatitis, encephalitis) (see Chapter 144).141 In certain settings, such as patients receiving alemtuzumab for hematologic malignancies, idelalisib, rituximab and bendamustine, and possibly also brentuximab, the risk of CMV reactivation and subsequent disease may warrant CMV monitoring.141

All cancer patients should be screened for active or past HBV infection (with at least HBsAg and HBcAb) before starting chemotherapy because viral reactivation and acute hepatitis have consequences ranging from delaying chemotherapy to fatal fulminant infection. Patients with chronic hepatitis should receive antiviral treatment, whereas those with inactive infection (i.e., HBsAg positive, HBV DNA negative) should receive prophylaxis to prevent reactivation if they are scheduled to receive intensive chemotherapy, particularly rituximab. Lamivudine (100 mg/day) used to be the mainstay of antiviral prophylaxis, but antivirals with a higher barrier to resistance are preferred. Limited data on new antivirals are available in this setting, with only one recent randomized, open-label trial showing entecavir (0.5 mg/day) to be superior to lamivudine in preventing clinical hepatitis in patients with chronic inactive HBV infection undergoing chemotherapy, including rituximab with CHOP (R-CHOP) for lymphoma.182 Based on data from immunocompromised patients treated for chronic HBV infection, most guidelines recommend agents with a high genetic barrier (such as entecavir or tenofovir in any of the two formulations) for prophylaxis in HBsAg-positive patients.60 Prophylaxis should be started before chemotherapy if possible, but chemotherapy should not be delayed, and prophylaxis should be continued for at least 6 to 12 months after the end of chemotherapy (18 months for rituximab), depending on the predicted duration and intensity of immunosuppression. Monitoring of reactivation with HBV-DNA and HBsAg is warranted after discontinuing the prophylaxis. In a recent study of 77 noncirrhotic HBsAg-positive patients who received mainly entecavir treatment after rituximab-containing chemotherapy, clinical relapse of HBV occurred in 32.5%, with hepatic decompensation in 14.3% after drug discontinuation, with most cases occurring during the first year after discontinuation.183

HBV reactivation is also possible, although less likely, in patients with resolved HBV infection (HBsAg negative, HBcAb positive). The risk of reactivation can be as high as 42% in patients with lymphoma treated with rituximab and 20% in CLL, 13% in T-cell leukemia, and 5% in multiple myeloma.60 The options are either to provide lamivudine prophylaxis in those with high risk (i.e., >10%) of reactivation (e.g., patients receiving anti-CD20 agents) or to monitor them closely for HBV DNA and treat at the first sign of reactivation.

For influenza, vaccination is the most effective preventive measure globally. During community or nosocomial outbreaks of influenza infection, prophylaxis with oseltamivir 75 mg once daily for 10 days or postexposure prophylaxis with 75 mg twice daily for 5 days should be considered, especially in severely immunocompromised patients, because of the risk of severe infection and its complications.

Vaccination is also the most effective preventive measure for SARS-CoV-2 infection. All cancer patients (and their caregivers) should receive a full primary immunization schedule (three doses for the immunocompromised) and all recommended booster doses.146 In addition to vaccination, passive immunization with antispike monoclonal antibodies (mAbs) has been successfully studied, either as preexposure or postexposure prophylaxis. However, the neutralizing activity and consequent efficacy of approved antispike mAbs have been significantly reduced or abolished by mutations in spike proteins of novel viral variants. There are currently no antibodies for preexposure prophylaxis in high-risk patients or early treatment, but this may change as new viruses emerge.

Currently no antiviral has been approved for prophylaxis of SARS-CoV-2 infection. Oral nirmatrelvir/ritonavir, intravenous remdesivir, and oral molnupiravir are approved for treatment of mild-to-moderate SRS-CoV-2 infection in cancer patients, with the aim to prevent progression to severe COVID-19146 (see Chapter 136).

Other Prophylactic Strategies
Role of Colony-Stimulating Factors in Prophylaxis

Colony-stimulating factors (CSFs) are used as primary or secondary prophylaxis with the aim of facilitating dose-intense treatments and decreasing infections by preventing the development of neutropenia or reducing its duration. Granulocyte colony–stimulating factor (G-CSF) is the most widely used CSF, and many compounds with this activity are now available.184 A systematic review and meta-analysis on the efficacy and safety of 11 different CSFs showed a higher risk of febrile neutropenia for short-acting G-CSF filgrastim compared Page 3661with prolonged-release molecules of pegfilgrastim.185 Among the new molecules and biosimilars, mecapegfilgrastim, lipegfilgrastim, and balugrastim significantly reduced the incidence of febrile neutropenia (cumulative probabilities: 58%, 15%, and 11%, respectively), but empegfilgrastim (a short-acting G-CSF) and a long-acting G-CSF biosimilar were best in reducing severe neutropenia (cumulative probabilities: 21%, 20%, 15%, respectively) after cytotoxic chemotherapy. Unfortunately, results from different studies in different underlying conditions are not reported uniformly or with similar end points, and therefore comparison is difficult.186 Moreover, the use of G-CSF is frequently performed outside of guidelines,186,187 with reports of efficacy in reducing febrile neutropenia incidence and duration of hospitalization in people with solid tumors.186,187 A systematic review188 demonstrated improved survival in patients receiving intensified chemotherapy with primary G-CSF prophylaxis, but this was also associated with an increased risk of secondary neoplasia, including AML and myelodysplastic syndrome.

Finally, the effectiveness of G-CSF versus FQ for preventing febrile neutropenia was assessed in two studies that reported a significant difference in favor of antibiotics.189 In a recent study of patients with breast cancer randomized to receive ciprofloxacin or G-CSF, a nonsignificant difference was observed in favor of G-CSF in reducing febrile neutropenia and non-febrile neutropenia–related hospitalization.190

Role of Immunoglobulins in Prophylaxis

Based on the experience in patients with primary immunodeficiency syndromes, the administration of immunoglobulins has been recommended in those with secondary iatrogenic immunoglobulin deficiencies, such as in patients with acute and chronic lymphocytic leukemia or non-Hodgkin lymphoma, and in those receiving rituximab. Keeping the immunoglobulin G (IgG) level higher than 600 mg/dL has been associated with a reduction of infectious episodes due to bacteria, provided the treatment could be given for at least 6 months, but with no effect on viral and fungal diseases.191 Similarly, a meta-analysis of studies in patients with CLL or multiple myeloma receiving immunoglobulins reported a reduction in the occurrence of infections but without any impact on mortality.192

Among 4479 rituximab-treated patients, 4.5% of 3478 patients with cancer and 9.7% of 340 patients with a hematologic disorder received replacement therapy with immunoglobulins, and a higher cumulative dose of administered immunoglobulins was associated with a reduced risk of serious infectious complications (hazard ratio [HR], 0.98; 95% CI, 0.96–0.99; $p = 0.002$).193 Patients who benefit from regular IgG replacement therapy are those with CLL and recurrent infections (mainly pneumonia or sinusitis) and an IgG level less than 500 mg/dL.194 Subcutaneous administration of immunoglobulins has been demonstrated to be safe and effective.195

Infection Control: Isolation and Antimicrobial Stewardship

In consideration of growing antimicrobial resistance and a shortage of new antibiotics, infection control and antimicrobial stewardship programs should be implemented in all cancer centers.196 Infection control practices include surveillance, promoting and auditing the use of standard and transmission mode–based precautions, appropriate screening for colonization with resistant pathogens, and preventive measures, such as proper hand hygiene and contact precautions for patients colonized or infected with resistant bacteria (see Chapter 304). Active surveillance—for example, with rectal swabs for colonization with carbapenemase-producing Enterobacterales or VRE and nasal and skin colonization by MRSA—should be performed in institutions where these pathogens are regularly encountered. Additional transmission mode–based precautions include the use of high-level masks/respirators for health care personnel, such as N95 or higher level or free-flow pressure 3 (FFP3) masks. These airborne precautions should be applied, for example, in cases of active TB or measles. Contact precautions (hand hygiene, use of disposable gowns and gloves) are indispensable when caring for patients colonized or infected with MDR bacteria. In these cases, patients should be isolated in single rooms, possibly with staff cohorting. If single rooms are not available, then patient cohorting is recommended. In addition, in cases of patient transfer to another ward or hospital, it is of utmost importance to notify the receiving unit about carrier status and need for precautions. Droplet precautions, including disposable gown and gloves and a surgical mask, should be applied in cases of infection with respiratory viruses such as influenza. For all precaution measures, regular monitoring of adherence is warranted, and the facilities should ensure access to adequate hand hygiene stations (e.g., sinks, alcohol-based rubs) and personal protective equipment.

For neutropenic patients, it is not clear whether staying at home or in a hospital may have an impact on the development of chemotherapy-related complications. In the hospital, keeping the patient in a reverse isolation/protective environment is believed to reduce colonization or infection with various pathogens. Unlike for transmission-based precautions, there is no universal consensus on the appropriate protective environment for neutropenic patients, and standards may vary between institutions. However, they usually include single rooms and, for people entering the room, surgical masks (mainly to prevent respiratory viruses) and in some institutions also disposable gloves and gowns (to prevent contamination with contact-transmitted pathogens). General guidelines for preventing the diffusion of infectious diseases in health care facilities should be followed in any ward where cancer patients are admitted.197 Additionally, because aspergillosis and other mold infections are acquired via the respiratory route, the use of high-efficiency particulate air (HEPA) filters in rooms or wards where leukemic or transplant patients are hospitalized is recommended.198 The use of masks with adequate filtration power (FFP3 or free-flow pressure 2 [FFP2] or N95 respirators) could reduce Aspergillus colonization and infection when the patient is moved from the protective environment. This approach has been demonstrated effective, together with other physical barriers, during building renovations.199 These precautions should also be recommended at home when even small masonry works are performed. The use of laminar airflow rooms is not deemed necessary and does not have a substantial impact on the rate of infection. On the contrary, it negatively affects patient quality of life and the possibility for the health care providers to care for them properly.

Antimicrobial stewardship (see Chapter 54), including antifungal stewardship, should include four main aspects: (1) local surveillance of antibiotic and antifungal resistance, antibiotic and antifungal consumption, and patient outcomes in selected infections; (2) development and regular update of protocols and algorithms for the diagnosis, prevention, and treatment of infections; (3) updated diagnostic methods and prompt reporting of microbiologic results by the laboratory, allowing timely de-escalation of broad-spectrum empirical regimens and shortening of antibiotic therapy; and (4) optimization of dosing regimens, including therapeutic drug monitoring for voriconazole and posaconazole. These call for a multidisciplinary approach and close collaboration between the treating physicians, the microbiology laboratory, the infectious diseases consultation service, the infection control unit, the hospital pharmacy, and, if possible, a medical pharmacologist.200

Food and Lifestyle

A low-bacterial-count diet seems not to offer any benefit compared with a normal diet.201,202 A recent systematic review and meta-analysis showed that a neutropenic diet and a control diet had no effect on infection and mortality rates in neutropenic oncology patients.203 However, some precautions should be recommended, such as avoiding unpasteurized milk; unpasteurized or moldy cheese products; raw or undercooked (including insufficiently reheated) meat, fish, shellfish, tofu, or eggs; and unpeeled fruits and salad ingredients, unless properly washed at home. Listeria colonization and infection have been described in association with unpasteurized dairy products and ready-to-eat deli meats, whereas raw vegetable sprouts have been associated with outbreaks of E. coli, Salmonella, and Listeria. Transmission of toxoplasmosis can also occur with raw meat and vegetables contaminated with cat feces. Although probiotics (foods with live yeast or bacteria cultures) are advertised as useful in reducing the risk of antibiotic-associated diarrhea, BSIs from probiotic administration have been reported. Safety practices for food handling should be followed, and specific information for cancer patients is available online.204 Diet recommendations that are too restrictive may have a negative impact on the patient's nutritional status or quality of life, or both.

Cancer patients, especially during less intensive treatment, may seek information about the safety of traveling or other recreational Page 3662activities. Although few data exist that quantify the risk of travel or recreational activities, some considerations can be made. First, the assessment of the underlying conditions should be made, with particular attention to the stability of the patient's clinical condition and their potential need for rapid access to health care facilities, in which case, remote destinations or cruises are not recommended. Evaluation of any ongoing treatment that might constitute a contraindication to the disease prevention measures recommended for the proposed destination, such as vaccines or antimalaria prophylaxis, is necessary. In immunocompromised hosts, live-attenuated vaccines (such as those against yellow fever or Salmonella typhi) are contraindicated, whereas effectiveness of other vaccines, such as that against hepatitis A virus (HAV), might be reduced.

Cancer patients should not be advised to part with their pets, although some precautions are necessary. For example, a different household member should be assigned to scoop cat litter because of potential Toxoplasma cyst exposure. Aquaria should not be touched or maintained by patients because water in fish tanks may be contaminated with nontuberculous mycobacteria, and Salmonella can be acquired directly from reptiles or from fomites. Finally, large pet birds should be avoided because they may transmit Chlamydia psittaci. In general, the potential benefit of recommendations on the safety of food, pet care, travel, and other lifestyle measures should be weighed against the unclear value of such recommendations and their potential to have a negative impact on patient nutritional intake and quality of life.

Vaccination

Inactivated vaccines have been demonstrated to be safe and effective in cancer patients, especially during nonaggressive/maintenance treatment phases. The only pitfall is that the immune response can be poor, especially with certain drugs. Vaccinations against easily transmitted diseases (e.g., viral respiratory diseases such as influenza, SARS-CoV-2, or measles and varicella) should also be recommended for caregivers. International recommendations are available for vaccination of patients with hematologic malignancies and HSCT recipients.194,205–207

Influenza vaccination with inactivated vaccines is strongly recommended for all immunocompromised patients, with the possible exception/postponing of those treated with intensive chemotherapy (e.g., a remission induction regimen for acute leukemia) and those who received rituximab within the previous 6 months. This is not because of safety concerns, but because the probability of responding is very low. Both unvaccinated and vaccinated but severely immunocompromised patients should be offered empirical antiviral treatment in cases of clinical signs or symptoms of influenza during influenza season. Influenza vaccination is also recommended for household contacts and health care personnel caring for cancer patients.

Pneumococcal vaccination is strongly recommended for immunocompromised patients, even though response to vaccination may be suboptimal. The pneumococcal conjugate vaccine (PCV) is thought to elicit better responses than the pneumococcal polysaccharide vaccine (PPSV23). Vaccination at the onset of disease and before chemotherapy is recommended for those with chronic proliferative diseases. The new PCV20, containing antigens from 20 pneumococcal strains, may not require a subsequent PSV23 dose.208 See Chapter 136 for SARS-CoV-2 immunization.

Treatment of Complications in Neutropenic Cancer Patients

Empirical Antibacterial Therapy for Fever During Neutropenia

Fever during neutropenia has always been considered a medical emergency and should always be considered as due to an infection, unless proven otherwise. Febrile episodes during the course of neutropenia are classified according to infection status as (1) microbiologically documented infections with bacteremia (isolation of a significant pathogen from one or more blood cultures); (2) microbiologically documented infections without bacteremia (isolation of a significant pathogen from a well-defined site of infection, usually urine or respiratory secretions obtained with sterile procedures or fluid aspiration); (3) clinically documented infections in the presence of signs and symptoms clearly and objectively due to infection but without microbiologic proof; and (4) unexplained fever when both clinical and microbiologic proof are lacking but the clinical course is compatible with infection. For the purpose of starting empirical antibiotic therapy, fever is now defined as one oral or axillary temperature measurement ≥38.5°C or two oral or axillary temperature measurements ≥38.0°C separated by at least 1 hour. An ear temperature of 39°C is equivalent to 38.4°C and can be used to decide to start empirical antibiotics in a neutropenic pediatric patient, except for patients with acute myeloid leukemia or HSCT.209 Additionally, in neutropenic cancer patients, the development of other signs of infection (e.g., altered mental status, hypotension, skin or mucosal lesions, respiratory failure) without fever or even with hypothermia (temperature <35.5°C) should always raise the suspicion of an infection and must receive prompt attention, including blood cultures and empirical antibacterial treatment, in addition to appropriate diagnostic procedures.

The effectiveness of the empirical therapy approach has been clearly demonstrated, and empirical antibiotic therapy has contributed substantially to the impressive reduction in mortality from infectious complications over the last decades.210 Unfortunately, some recent studies on infection in patients with acute leukemia and HSCT show that the mortality rate associated with infections due to gram-negative rods, and possibly also VRE, may be increasing. This is driven by MDR isolates with peaks as high as 70% of BSI due to carbapenem-resistant K. pneumoniae in allogeneic HSCT recipients due to the absence of effective empirical threatment.5,157,164,211,212 Indeed, in the era of bacterial resistance and the shortage of new antibiotics, especially those active against gram-negative rods, infectious complications may once again significantly compromise the success of cancer treatment. Bacterial epidemiology and patterns of resistance may vary from patient to patient (in cases of individual colonization with resistant pathogens), from center to center (in cases of environmental colonization), and from country to country (different endemicity of resistant strains).5,95 Therefore the risk of a severe and complicated clinical course and the risk of infection caused by resistant pathogens should be evaluated individually and the treatment chosen accordingly.213 Fig. 311.2 summarizes a possible initial approach to a patient with febrile neutropenia. Table 311.5 summarizes the major antibiotics used for empirical or targeted therapy in febrile neutropenia.

Etiology

Surveillance studies on pathogens causing infections in cancer patients are of the utmost importance for the implementation of management strategies. Large-scale studies are obviously crucial because they can provide information about worldwide trends, but also single-center surveillance reports may be useful, because every geographic region, country, or even single center may have peculiarities related to the type of patient, type of care, and previous local antibiotic policies. Most of the available information concerns bacterial and fungal pathogens isolated in the bloodstream, whereas the role of deep-seated infections, in addition to the impact of viral infections, is less well described.

Bacterial Infections

Over the last 30 years, gram-positive bacteria were the most frequent pathogens causing BSIs in cancer patients. However, an increase in the frequency of bacteremias caused by gram-negative rods has been reported, with gram-negative pathogens becoming either predominant or at least as frequently isolated as the gram-positives.12,91,92 This trend has also been reported in a retrospective literature review and Page 3653contemporary surveillance study performed in 2011 in 39 European hematology centers from 18 countries belonging to the network of the European Conference of Infections in Leukemia (ECIL).93 As shown in Fig. 311.1, this study found that gram-negative pathogens were almost as frequently isolated as the gram-positives, with gram-positive/gram-negative ratios in BSIs of 60%/40% and 55%/45% in the literature review and the ECIL-4 surveillance, respectively. The detailed etiology was similar (see Fig. 311.1) in the literature review and the surveillance study, with a slightly higher rate of enterococci and Enterobacterales and a decreased rate of Pseudomonas aeruginosa in the surveillance study.93 These changes in etiology appear to be associated with an important increase in the proportion of resistant pathogens, such as ESBL-producing Enterobacteriaceae, VRE, MDR P. aeruginosa, and the most worrisome, carbapenem-resistant Enterobacterales (CRE). In addition, in leukemic patients, most staphylococci are resistant to methicillin, whereas most gram-negative pathogens are resistant to FQs.93

FIG. 311.1 Etiology of bloodstream infections in cancer patients.Top, Data from recent literature review. Bottom, Data from surveillance study for the Fourth European Conference of Infections in Leukemia (ECIL-4) in 2011. (Modified from Mikulska M, Viscoli C, Orasch C, et al. Fourth European Conference on Infections in Leukemia Group (ECIL-4), a joint venture of EBMT, EORTC,

ICHS, ELN and ESGICH/ESCMID. Aetiology and resistance in bacteraemias among adult and paediatric haematology and cancer patients. J Infect. 2014;68:321–331.)
Follow for extended description
Of note, the rates of resistance were generally higher in southern and eastern than in northern and western Europe, and this trend is also evident in the non–hematologic cancer population.93,94,95 However, the problem of increasing infections due to resistant pathogens is worldwide.

The increase in infections caused by resistant strains usually stems from an increase in colonization with these pathogens. Indeed, colonization with resistant bacteria is one of the most important risk factors for infection with resistant bacteria (Table 311.3). The association between colonization and subsequent infection has been reported for VRE, ESBL-producing Enterobacterales, P. aeruginosa, Stenotrophomonas maltophilia, and carbapenem- or colistin-resistant Klebsiella pneumoniae.96–99 These data may differ in pediatrics. A recent multinational survey on antibiotic-resistant bacteremias in children receiving Page 3654antineoplastic chemotherapy or HSCT showed that colonization was a significant risk factor for MRSA bacteremia, whereas it was not associated with an increase in the risk of antibiotic-resistant gram-negatives, with the exception of resistance to piperacillin-tazobactam.5 This observation can have important management implications because patients carrying MRSA can be successfully decolonized100 and might benefit from the use of a glycopeptide in the initial empirical therapy of febrile neutropenia (see later).

TABLE 311.3

Risk Factors for Infections Caused by Resistant Bacteria in Cancer Patients
1. Previous infection or previous or current colonization by resistant bacteria, in particular:
• Enterobacteriaceae resistant to third-generation cephalosporins or carbapenems
• Multidrug-resistant, nonfermenting, gram-negative rods: Pseudomonas aeruginosa, Acinetobacter baumannii, Stenotrophomonas maltophilia
• Vancomycin-resistant enterococci
• Methicillin-resistant Staphylococcus aureus
2. Previous exposure to broad-spectrum antibiotics, in particular to third-generation cephalosporins
3. Health care–related infection
4. Prolonged hospital stay and/or repeated hospitalizations
5. Presence of urinary catheter
6. Older age
7. Intensive care unit stay

Obviously, not all colonized patients will develop infection due to the colonizing strain, because the risk depends on multiple host- and chemotherapy-related variables. However, colonization-informed choice of empiric treatment of febrile neutropenia targeting the MDR colonial might prevent breakthrough infections. In a study on carbapenem-resistant KPC-producing K. pneumoniae, the introduction of prompt screening and a targeted empirical treatment lowered the rate of BSI due to KPC-producing K. pneumoniae among colonized patients from 67% to 11%, and the mortality of BSI due to the resistant strain was also significantly reduced.101 Although early detection of colonization with MDR gram-negative bacteria is useful, the impact of screening for VRE is less clear. VRE BSI has been associated with lower survival at 3 and 12 months after allogeneic HSCT102; however, there was no benefit of VRE-targeting empirical therapy in colonized patients with VRE BSI, suggesting that VRE BSI might be a marker of mortality risk as opposed to causative.103

In a recent cross-sectional study in adults from the United States, the rate of critical illness (defined as ventilation, vasopressor therapy, or death) within 30 days from BSI was not influenced by FQ prophylaxis, organism type, initial antibiotics, or adequacy of coverage, although in these patients approximately 10% of gram-negatives were susceptible to the standard empirical therapy with cefepime or piperacillin-tazobactam.104 In a multicenter pediatric study in which over 20% of gram-negatives were resistant to cefepime or piperacillin-tazobactam, the risk of ICU admission or death was related to antibiotic resistance and prior antibiotic exposure.5 In both these studies the treatment center played an important role in the distribution of gram-positive versus gram-negative species and development of infections due to resistant pathogens.

Anaerobic bacteria are isolated in less than 1% of positive blood cultures in cancer patients, but the proportion increases to 3% among those undergoing abdominal surgery.79 Anaerobes are usually isolated in polymicrobial bacteremias, especially together with gram-negative rods, with a rate that seems to be higher than that observed in non-oncology patients undergoing similar surgery (0.597 vs. 0.033 per 1000 hospital days, respectively).105 Differences in the etiology of bacterial infection between neutropenic and nonneutropenic cancer patients have been reported.12

CVC-related bacteremias are generally caused by gram-positive cocci (especially coagulase-negative staphylococci), which are isolated in more than 50% of the episodes compared with 25% to 40% for gram-negative rods.7,106–113 The source of infection is likely to be partially different in cases of gram-positive and gram-negative CVC-related bacteremias. Infusate contamination is rare but possible, particularly with Klebsiella, Enterobacter, Citrobacter, Achromobacter, Serratia, Ralstonia, and Pseudomonas (other than P. aeruginosa). Polymicrobial infections are not uncommon, especially with gram-negative bacteria.

Bacterial gastroenteritis caused by classic enteric pathogens (Salmonella and Shigella) is a rare event in patients with acute leukemia, involving less than 1% of acute enteritis after chemotherapy.114 In contrast, C. difficile is common in cancer patients, with an incidence that is twofold higher than in the noncancer population115 and up to sixfold higher in hematology patients.116 Helicobacter pylori has also been described as a possible cause of gastrointestinal disease in this patient population.

Legionellosis and nocardiosis are rare but potentially life-threatening infections, with Legionella pneumonia sometimes presenting with unusual radiologic patterns. Mycoplasma pneumoniae and Chlamydia pneumoniae have been rarely described as a cause of pneumonia in cancer patients, but it is possible that their incidence is underreported.

Finally, TB might be underestimated and underdiagnosed in cancer populations. Older data showed a rate of approximately 90 cases per 100,000 persons (i.e., ninefold higher than in the general population in developed countries).117 However, a recent review found very low rates of TB in hematology settings, with no evidence that universal screening with immunoassays and treatment of latent TB provides clinical benefits in people with low risks in nonendemic areas.118 Patients from high-incidence countries or communities account for most of the cases of TB in the cancer population. Findings of pleuroparenchymal imaging abnormalities (mainly on the upper lobes) suggestive of previous TB and people with reported or predicted exposures should be screened for prior infection using immunoassays, with consideration for anti-TB therapy.118 Routine screening may be warranted in people scheduled to undergo therapy that causes prolonged T-cell depletion. Infections caused by nontuberculous mycobacteria are rare outside of cases of CVC-related BSIs. Local or even disseminated M. bovis infections can occur in patients receiving BCG immunotherapy for bladder cancer.

Fungal Infections
Aspergillus spp. and Candida spp. are the most common fungal pathogens in cancer patients, with the former now seen more frequently than the latter. Other fungal pathogens include P. jirovecii, cryptococci, molds such as Mucorales or Fusarium, and rare yeasts.

Among yeasts, Candida is the most frequently isolated, usually in BSIs, with an increasing proportion of non-albicans strains. This phenomenon, known since 1999, was confirmed by a study from the European Organization for Research and Treatment of Cancer (EORTC), which reported that in cancer patients the overall incidence of fungemia was 2.3%, with C. albicans responsible for 72% of candidemia cases in the whole study population, but only 27% in patients with hematologic malignancies. This is due to fluconazole prophylaxis, which reduced the overall rate of candidiasis but favored breakthrough infections with fluconazole-resistant species.119,120 Similar proportions were observed recently in pediatric patients.121 The majority of candidemias were observed in nonneutropenic patients. In general, yeasts belonging to the Candida parapsilosis complex are usually associated with CVC contamination, whereas other Candida species typically originate from the gastrointestinal tract after selection and translocation. C. glabrata is increasing in frequency and resistance, not only to fluconazole but also to echinocandins.122 Less common is infection with Candida auris, a frequently drug-resistant species causing outbreaks worldwide, mainly due to its ability to form biofilms and to persist on surfaces.

Among molds, Aspergillus represents the most frequently isolated or suspected. The majority of episodes are caused by Aspergillus fumigatus, although some centers report a predominance of infections caused by Aspergillus flavus and Aspergillus terreus.126,127 Invasive aspergillosis in patients with malignancy or HSCT is relatively common and usually community acquired, but outbreaks can occur with environmental exposures, both in and outside the hospital. The incidence of invasive aspergillosis depends mainly on the underlying malignancy and its treatment. It is highest in patients with prolonged neutropenia, followed by those receiving high doses of steroid therapy.128 In a multicenter Italian study, the incidence of aspergillosis among hematologic cancer patients varied from 7.9% in acute nonlymphoblastic leukemia to 4.3% in ALL, 2.3% in chronic myelogenous leukemia, and less than 1% in CLL, Hodgkin and non-Hodgkin lymphoma, and multiple myeloma.126 Additionally, the risk of aspergillosis in adult patients with ALL was reported to be as high as 11.7%.129 Recent pediatric data showed a higher incidence of invasive mycoses in high-risk ALL (up to 15% in the absence of pharmacologic prophylaxis) occurring since the beginning of treatment (which includes the use of high-dose steroids) and in all the most aggressive phases.130,131 Patients with chronic lymphoproliferative disorders represent another at-risk group in which aspergillosis can be seen, probably as a result of more intensive treatment protocols. Despite the high number of patients with these diseases, the rate of IFD is <5%.127,128 The incidence of invasive aspergillosis after autologous HSCT is low (0.3%–2%), and it typically occurs during pre-engraftment neutropenia.132

Increased attention to azole resistance among Aspergillus species, especially members of the A. fumigatus species complex, is necessitated by recognition of organisms harboring mutations in erg11 along with Page 3655recognition of MDR Aspergillus species (e.g., A. calidoustus) and the recently described “cryptic” species (e.g., A. lentulus).133 Rates of infections with these organisms can vary depending on geographic region and agricultural use of azole antifungals. Despite the fact that the phenomenon is so far relatively limited in most settings, knowledge of the frequency of azole resistance at the country and hospital level is fundamental. Secondary azole resistance, with different genetic resistance mutations, has been reported in patients with chronic aspergillosis after prolonged treatment, particularly in the case of suboptimal blood levels or high fungal burden (see Chapter 263).134

Mucormycosis is increasingly reported in some centers, especially in the United States. It is unclear whether this represents a general trend influenced by geography or prior antifungal therapies or if these infections are simply diagnosed more often because of increased awareness or improved survival. Other fungi, such as Fusarium and Scedosporium, are reported sporadically, with some organisms having higher prevalence in certain regions.135

P. jirovecii is a well-known cause of pneumonia in cancer patients not receiving trimethoprim-sulfamethoxazole (TMP-SMX) prophylaxis, especially if treated with high-dose and prolonged steroid therapy or certain antineoplastic drugs such as fludarabine, alemtuzumab, idelalisib, or temozolomide. Attack rates vary from 6.5% to 43% in ALL, 4% to 25% in rhabdomyosarcomas, and nearly 1% in Hodgkin lymphoma and primary or metastatic central nervous system tumors.136,137

Finally, infections with Cryptococcus spp. and geographically influenced “endemic” fungi, such as Histoplasma or Coccidioides, are possible in patients who live or used to live in endemic areas.

Viral Infections

Apart from herpes simplex virus (HSV) reactivation, which occurs in up to 60% of HSV-seropositive patients with acute leukemia, there are few data describing viral infections in febrile neutropenia.3 Cytomegalovirus (CMV) reactivation, as measured by pp65 antigenemia or PCR, occurs in people with severe illness, without necessarily being followed by CMV disease.138 For this reason, routine monitoring of CMV reactivation and preemptive therapy is not considered necessary in cancer patients other than HSCT recipients. A recent study evaluating febrile neutropenia in children and adults receiving intensive chemotherapy showed viremia in 15% of episodes, with the highest frequency in children and the highest viral loads in patients otherwise classifiable as fever of unknown origin (FUO), suggesting that at least some cases of febrile neutropenia in children could be attributable to viral infection/reactivation.139 The risk of viral reactivation (herpes simplex virus [HSV], VZV, CMV) and disease increases significantly with use of T-cell–suppressing agents or drug combinations, such as alemtuzumab or idelalisib or a combination of bendamustine and rituximab.59,140–142

Community-acquired respiratory viruses, such as influenza, parainfluenza, respiratory syncytial virus (RSV), human metapneumovirus, adenoviruses, rhinoviruses, and coronaviruses, are a frequent cause of respiratory disease in cancer patients and are probably underestimated as a cause of fever. Whereas most cancer patients experience self-limited upper respiratory illness, those with severe immune deficits, which can be estimated using dedicated scores that take into consideration lymphopenia, steroid use, hypogammaglobulinemia and recent transplant, and B-cell or T-cell depletion,143 are at increased risk for progression from upper respiratory tract infection to pneumonia, with possible respiratory failure.144 The rate of viral respiratory infections in cancer patients depends on the ongoing rate of infections in the community and on the testing policy. In cancer patients with viral respiratory disease, deferral of chemotherapy may be considered. Specific treatment is warranted for influenza and in some cases for RSV infection (e.g., in leukemic patients with risk factors for RSV-related mortality; see Chapter 165).144

SARS-CoV-2 infection has caused significant morbidity and mortality in cancer patients—much higher than in the general population.145,146 In the cancer setting, the additional negative effect of even mild SARS-CoV-2 infection is caused by interruption of chemotherapy, which can be extended in case of prolonged viral shedding. Patients with non-Hodgkin lymphoma and CLL, particularly if treated with B-cell–depleting therapy, are at the highest risk of prolonged or relapsing infection, and optimal management remains still to be defined. Early treatment is warranted to prevent progression to severe disease and potentially shorten the duration of infection.146

Viral gastroenteritis, mainly caused by rotavirus but also norovirus or sapovirus, may be a frequent complication in pediatric oncology, with a potential to cause outbreaks in cancer centers because of persistent gastrointestinal shedding in immunocompromised hosts. Both adenoviruses and parvovirus B19 have been reported as rare causes of severe gastrointestinal disease in cancer patients.

Finally, the reactivation/exacerbation of infections due to hepatotropic viruses (HBV and hepatitis C virus [HCV]) represents an important problem in areas of high endemicity. HBV reactivation is frequent in cancer patients with chronic inactive HBV infection (HBsAg positive, with negative or low-level serum HBV DNA), but it can also occur in patients with resolved HBV infection (HBsAg negative, HBcAb positive, or with low-level serum HBV DNA), particularly after treatment with rituximab (up to 40% of patients).60 Antiviral prophylaxis, preferably with a high barrier to resistance such as entecavir or tenofovir, is usually given to these patients for at least 6 to 12 months after the end of chemotherapy.

HCV flares might occur in cancer patients undergoing chemotherapy, and HCV eradication could lower the risk of hepatic toxicity and even improve the outcome in certain lymphomas.147 The possibility of chronic hepatitis E virus infection due to genotype 3, which is endemic in certain industrialized regions, has been reported in patients with hematologic malignancies and transplant recipients.148,149 Repeated transfusions might be a risk factor for hepatitis E virus infection.150,151

Other Pathogens
The risk of rare infections or reactivations caused by protozoa (leishmaniasis, South American trypanosomiasis, and malaria), helminths (strongyloidiasis), endemic fungi, and other tropical diseases should be considered in patients who live or lived in endemic areas. Obtaining a detailed history to identify potential exposures is the most important screening tool, with additional serology to document past exposure. However, in the case of strongyloidiasis, for example, the suboptimal performance of stool examination or serologic screening may warrant empirical treatment with ivermectin in patients who present with unexplained eosinophilia and who lived in endemic areas, such as the tropics, the subtropics, or the southeastern United States and Europe.

Prevention of Infections in Cancer Patients

Prevention is desirable, given the remarkable mortality and morbidity associated with infections in cancer patients. In general, to evaluate the cost-effectiveness of a prophylactic protocol, one should know the rate of the target infection in specific patient populations and the number of patients needed to treat to prevent a single infectious episode, severe infection, and attributable death for each center. The rate of severe side effects, reported as the number needed to harm, must also be taken into consideration.

Table 311.4 summarizes different regimens for primary chemoprophylaxis and other approaches appropriate in cancer patients based on clinical trials and guidelines. Advantages and disadvantages of these procedures are discussed here.

Prevention of Bacterial Infections

The use of antibiotics to prevent bacterial infections should be weighed against efficacy, toxicity, and impact on the development of resistance. Chemoprophylaxis for the prevention of bacterial infections was first proposed in clinical practice based on the recognition that 80% of bacterial pathogens causing infection in neutropenic cancer patients originated from the endogenous flora, with approximately half of infections acquired during the hospital stay. The first approach relied on the oral administration of nonabsorbable antibiotics (gentamicin, vancomycin, and nystatin) aimed at totally (including anaerobes) or partially (excluding anaerobes) suppressing the intestinal bacterial flora and preventing the acquisition of exogenous organisms (known also as gut decontamination). Subsequently, absorbable drugs such as TMP-SMX, usually given in combination with oral nystatin and then FQs, were introduced. Large meta-analyses of quinolone prophylaxis in cancer patients have confirmed a statistically significant reduction in febrile Page 3656neutropenia, BSIs, and mortality, with the number needed to treat (NNT) to prevent one death ranging from 24 to 50, with no untoward effect in induction of resistance.152,153 Most guidelines recommended FQ prophylaxis for adult patients with expected neutropenia longer than 7 days. These results were based on studies published between 1973 and 2011, largely before the spread of bacterial resistance.154

Given the alarming increase in bacterial resistance worldwide, two issues need to be addressed: (1) whether the benefit of FQ prophylaxis can still be expected and (2) what is the contemporary impact of routine prolonged administration of broad-spectrum agents, such as FQ, on the selection of resistant strains and local epidemiology. Meta-analyses of recent studies did not confirm the benefit on mortality, although prophylaxis was still associated with lower rates of BSI and febrile episodes.154,155 The reason might be that the sample size was insufficient to show a difference in mortality, because the outcome of febrile neutropenia has significantly improved. Alternatively, infections prevented by FQ may be associated with low mortality, because they can be effectively treated with standard empirical antibiotics.95,156,157 As demonstrated by studies performed in both children and adults, BSI-associated mortality is mainly driven by MDR pathogens, which are unlikely to be prevented by FQs.5,95,158,159 This might explain the fact that FQ prophylaxis may no longer have any effect on overall mortality, despite efficacy in preventing some infections (i.e., those due to organisms that are effectively treated with classic empirical therapy).154,156

Some studies have suggested that FQ prophylaxis may increase rates of MDR infections.154 Finally, FQ can be associated with toxicities, including QT prolongation, tendon rupture, and interaction with drugs such as cyclophosphamide.160 Therefore, some recent guidelines do not support the use of FQ prophylaxis.161–163 In patients with solid tumors, who usually experience low rates of BSI and shorter (<7 days) neutropenia, FQ prophylaxis is not recommended.20

Additional strategies for prevention of MDR bacterial infection include decolonization with orally nonabsorbable antibiotics and fecal microbiota transplantation (FMT), which reduce intestinal burden of MDR strains. Although decolonization with gentamicin or colistin can decrease infections with MDR pathogens, this strategy may also increase resistance among colonizing bacteria.164 FMT has been evaluated in a small prospective trial of MDR-colonized patients with hematologic malignancies; partial-to-complete decolonization was reported.165 The FMT procedure was more effective in the absence of concomitant antibiotic administration (79% vs.

36%), and no severe adverse events were observed.165 Further studies are warranted to determine all the potential benefits and real-life applications of manipulation of gut microbiota.

TABLE 311.4

Suggested Prophylaxis for Infections in Cancer Patients
DRUG SCHEDULE COMMENTS
Antibacterial Ciprofloxacin 500mg twice daily
Use of FQs should be based on local epidemiology and careful evaluation of potential drawbacks of FQ prophylaxis.
Probably not active anymore in many centers. Some studies showed possible effects on increasing resistance.
Traditionally administered to adults receiving chemotherapy for acute leukemia with expected neutropenia >7–10 days; starting with chemotherapy and continuing until resolution of neutropenia or initiation of empirical antibacterial therapy for febrile neutropenia.
Levofloxacin 500mg once daily; children 10 mg/kg twice daily
Antifungal Posaconazole
100-mg tablets: 3 tablets twice daily on the first day, then 3 tablets daily
As alternative, oral solution 200 mg tid orally with a (fatty) meal or acidic drink
IV: 300 mg twice daily on the first day, then 300 mg daily
Patients receiving high-intensity chemotherapy or certain combination therapies with targeted agents for acute myelogenous leukemia or myelodysplastic syndrome.
Therapeutic drug monitoring is warranted in case of oral solution.
Fluconazole 400mg once daily Patients at risk for candidiasis but not aspergillosis.
Other Secondary prophylaxis according to isolated pathogen, clinical presentation, or both.
Anti–Pneumocystis jirovecii Trimethoprim-sulfamethoxazole (TMP-SMX)
One double-strength tablet (160/800mg) three times weekly, or
25 mg/kg TMP-SMX (5 mg/kg of TMP), in 2 divided doses for 3 consecutive days/wk
All patients receiving chemotherapy with steroids, including those with solid tumors (e.g., brain cancer); treatment with agents like fludarabine, alemtuzumab, or idelalisib; or some combination therapies (e.g., bendamustine plus anti-CD20).
Dapsone 100 mg daily in adults, or 2mg/kg/day (max. 100mg) on alternate days three times/wk In patients who cannot tolerate TMP-SMX.
Aerosolized pentamidine 300mg once a month with nebulizer In patients who cannot tolerate TMP-SMX; effective, but it is more difficult to administer.
Atovaquone 750mg twice daily or 1500mg once daily In patients who cannot tolerate TMP-SMX.
Antiviral
Acyclovir
or
40mg/kg in children in 2 divided doses
In adults >40 kg: 800 mg twice daily
Patients with positive anti-HSV antibodies and severe mucositis or receiving treatment for acute leukemia or other drugs which increased the risk for herpes zoster.
VZV-susceptible patients exposed to chickenpox who did not receive prompt administration of specific immunoglobulins.
Valacyclovir
For HSV or VZV prophylaxis: 500mg twice daily or 1000mg daily
For VZV exposure: 1 g three times daily (see text)
Lamivudine 100mg once daily Patients with resolved HBV infection (HBsAg negative and HBcAb positive).
Entecavir 0.5mg daily HBsAg-positive/HBV-DNA <2000IU/mL or high-risk (e.g. receiving anti-CD20 antibodies) patients with resolved HBV infection.
Tenofovir disoproxil fumarate (TDF) 300 mg daily HBsAg-positive/HBV-DNA <2000IU/mL.
Antituberculosis Isoniazid 5mg/kg (max. 300mg) daily in adults, 10mg/kg (max. 300mg) daily in children for 9months
Patients with latent tuberculosis.
Efficacy not specifically evaluated in cancer patients. Add pyridoxine (50 mg/daily for adults).

Rifampicin	600 mg/daily for 4 months20 mg/kg daily (max. 600 mg) in children	Patients with latent TB who do not receive concomitant medications subject to drug interactions. This regimen has been associated with the lowest toxicity and highest compliance rate.
Isoniazid and rifampicin
Dose as noted earlier
For 3 months
Patients with latent TB who do not receive concomitant medications subject to drug interactions of rifampicin.
Anti–CVC-associated infection	None
Good skin preparation and the use of sterile technique at time of device insertion.
Good maintenance procedures.
All patients with indwelling central venous catheter.
Table Continued

DRUG	SCHEDULE	COMMENTS
Others Growth factors	Filgrastim either subcutaneously or as an intravenous infusion over at least 1h, or pegylated filgrastim
For the prevention of febrile neutropenia in patients who have a high risk of this complication based on age, medical history, disease characteristics, and myelotoxicity of the chemotherapy regimen.
Secondary prophylaxis with G-CSFs recommended for patients who experienced a neutropenic complication from a prior cycle of chemotherapy (for which primary prophylaxis was not received), in whom a reduced dose of chemotherapy may compromise disease-free or overall survival or treatment outcome.
Efficacy not fully demonstrated for pegylated filgrastim.
Immunoglobulins	Polyclonal immunoglobulins: 400mg/kg every 21–28 days
Patients with chronic lymphocytic leukemia after the second episode of severe bacterial infection.
Patients with leukemia or lymphoma with hypogammaglobulinemia (<400 g/dL) and severe bacterial infections (reasonable, but not proved).
Specific anti-VZV (VariZIG): 125IU for every 10kg of body weight (max., 625IU); adults 650 IU	In high-risk contact with a negative history of varicella preferably within 96h after exposure to chickenpox.
Vaccines	Influenza
Influenza vaccination of patients, especially during less aggressive treatment phases.
Vaccination of household contact and health care workers.
Pneumococcus	Conjugated (PCV) vaccine recommended; with new 15-valent or 20-valent PCV vaccines available, the role of PPSV23 to be determined.
Varicella	VZV-seronegative household contacts and health care workers.
Herpes zoster Recombinant inactivated 2-dose vaccine in VZV seropositive patients.
COVID-19	Vaccinate with 3-dose primary cycle, and provide boosters following updated recommendations.
Isolation procedures
Perform hand hygiene with an alcohol-based hand rub or by washing hands with soap and water if soiled, before and after all patient contacts or contact with the patients' potentially contaminated equipment or environment.
Use contact precautions (gowns and gloves).
Ensure adherence to standard environmental cleaning with an effective disinfectant.
Patients colonized or infected with multidrug-resistant pathogens (e.g., VRE, CRE) or infected with other pathogens for which contact isolation precautions are advisable (e.g., C. difficile, norovirus); of note, alcohol-based hand rubs are not cidal against C. difficile or norovirus.
CRE, Carbapenem-resistant Enterobacteriaceae; CVC, central venous catheter; FQ, fluoroquinolone; G-CSFs, granulocyte colony-stimulating factors; HBcAb, hepatitis B core antibody; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; HSCT, hematopoietic stem cell transplantation; HSV, herpes simplex virus; IU, international unit; max., maximum; VRE, vancomycin-resistant enterococci; VZV, varicella-zoster virus.

Prevention of Tuberculosis
Risks for reactivation of latent TB are highest among people with sustained defects in T-cell immunity and are not clearly increased in people with short or intermittent episodes of neutropenia. Few data exist, and they depend on geographic prevalence.118 Standard regimens

for latent TB are recommended (rifampin 4 months, rifampin and isoniazid 3 months, isoniazid 9 months), but drug-drug interactions between rifampin and hematologic treatments, combined toxicities (hepatic and neurologic), and the negative impact of delaying hematologic treatment118 must be carefully weighed.

Prevention of Central Venous Catheter–Related Infections (See Chapter 306)

Prevention of CVC-related infections starts with the correct insertion and manipulation of the devices, not only by health care workers but also by caregivers who perform CVC maintenance procedures (and sometimes drug infusions) at home. Aseptic bundles during insertion and postinsertion care, which include assessment of need for CVC, daily inspection of the site of insertion, change of dressing at least weekly or whenever wet or soiled, use of chlorhexidine gluconate–impregnated sponges, application of alcohol scrub for 15 seconds before each access of the line, and hand hygiene, are effective measures to prevent infectious complications.41 Continuous education and regular audits of bundle implementation represent other helpful activities.166 Antibiotic catheter flushing has also been suggested to reduce the risk of infections, but this could represent a risk for selection of resistant bacteria. On the contrary, taurolidine lock has been associated with a significant reduction of CVC-related BSI, even if the susceptibilities of gram-positives and gram-negatives to taurolidine are indeterminate due to limited data.167

Prevention of Fungal Infections

Primary Antifungal Chemoprophylaxis

Although fungal infections usually represent no more than 10% of all infections (see Table 311.2), their associated morbidity and mortality remain high. Two main drugs used for antifungal prophylaxis in patients with acute leukemia include fluconazole (active against yeasts such as Candida but not molds) and posaconazole (mold-active prophylaxis). Fluconazole is effective in preventing Candida infections in people with acute myeloid leukemia and may benefit other populations that have an incidence of invasive candidemia higher than 7% to 10%.128 Mold-active prophylaxis with posaconazole in adults receiving multiple cycles of chemotherapy for acute myeloid leukemia or myelodysplastic syndrome reduces the incidence of invasive mycosis from 8% to 2% (primary end point) compared with standard prophylaxis with fluconazole or itraconazole, with improved survival.168 The NNT to prevent one case of proven/probable invasive mycosis was 16, and the NNT to prevent one fungal infection–related death was 27.169 However, if the incidence of invasive mycoses is lower than 8%, the NNT to prevent one infection would be higher, and this underlines the requirement to tailor needs in different hosts. Specific recommendations on drug dosing and formulations are summarized in Table 311.4 and Table 311.5. Recently introduced targeted agents in the treatment of AML, such as venetoclax, midostaurin, quizartinib, or gilteritinib, have led to challenges in the management of antifungal prophylaxis Page 3658because coadministration of these agents with posaconazole leads to significant increase in their blood levels, which carries the risk of toxicity (see Table 311.2). Posaconazole prophylaxis is still recommended when these agents are used in combination with other chemotherapy drugs, and dose reduction has been proposed for some (venetoclax), whereas careful monitoring was recommended with a low level of evidence of others (midostaurin).66

TABLE 311.5

Antibacterial and Antifungal Agents Usually Used in Cancer Patients

PATIENTS' TYPE OF THERAPY	DRUG	ROUTE OF ADMINISTRATION	DAILY PEDIATRIC DOSAGE	USUAL DAILY ADULT DOSAGE	NO. OF DAILY DIVIDED DOSES
Antibacterial, intravenous	Piperacillin-tazobactam	IV	400 mg/kg (as piperacillin)	13.5–18g (as both drugs)	3–4 in adults, 4 in children; preferably in prolonged or continuous infusion with a loading dose
	Ceftazidime	IV	100 mg/kg	6000mg	3
	Cefepime	IV	100 mg/kg	6000mg	3
	Meropenem	IV	60 mg/kg	3000–4000mg	3, infuse over 3–6 hours
	Ceftolozane-tazobactam	IV	90 mg/kg, up to a maximum dose of 4.5 ga (as both drugs)	4500 mg (as both drugs) 9000 mg for nosocomial pneumonia	3, infuse over 1 hour

Ceftazidime-avibactam IV
6 months to 18 years 187.5 mg/kg, up to a maximum of 7500 mg/daily (as both drugs)
<3 months 150 mg/kga
7500mg (as both drugs) 3, infuse over 2hours
Ceftaroline IV
2months to <2years: 8mg/kg every 8h
≥2 years to <18 years (≤33 kg): 12 mg/kg every 8 hours
≥2 years to <18 years (>33 kg): 400 mg every 8 hours or 600 mg every 12 hours
1200mg 2, infuse over 5–60minutes
Ceftobiprole IV 3 months to 12 years, <33/kg: 60 mg/kg, ≥33 kg 1500 mg; 12–18 years, <50 kg: 30 mg/kg, ≥50 kg 1500 mg 1500mg (as ceftobiprole) (2 g for the first 7 days in case of S. aureus BSI) 4
Imipenem-cilastatin IV 60–100mg/kg (as imipenem) 2000–4000mg (as both drugs) 3–4
Meropenem-vaborbactam IV ND 12 g (as both drugs) 3, infuse over 3hours
Ciprofloxacin IV 15–30mg/kg 800–1200mg 2–3
Ceftriaxone IV 80mg/kg 2000mg 1
Amikacin IV 20mg/kg 1000–1500mg 1
Vancomycin IV 40–60mg/kg 2000mg 2 or as continuous infusion after a loading dose
Teicoplanin IV; not available in United States 10mg/kg (loading dose of 10mg/kg bid on first day of treatment) 600–1200mg (loading dose of 600mg bid on first day of treatment) 1 (2 on first day of treatment)
Daptomycin IV 10 mg/kg 6–12 mg/kgb 1
Linezolid IV (also available oral) 30 mg/kg in three doses if <12years, then 20mg/kg 1200mg 2
Tedizolid Oral/IV ND 200mg 1
Dalbavancin IV
Birth to <6 years 22.5 mg/kg (max. 1500 mg)
6–18 years 18 mg/kg (max. 1500 mg)
1500 mg Single-dose regimen
Tigecycline IV 2.4mg/kg loading dose, then 2.4mg/kg (proposed) 100mg loading dose, then 100mg 2
Colistimethate sodium (colistin) IV 150,000–200,000IU/kg loading dose, then 150,000–200,000IU/kgc 9,000,000IU loading dose, then 9,000,000 IUc 2
Polymyxin B IV
Infants: up to 40,000U/kg
Children: 15,000–25,000 U/kg
15,000–25,000U/kg (= 1.25–2.5mg/kg, 1mg = 10,000U; a loading dose might be indicated)
Fosfomycin IV; not available in United States 300mg/kg 15–24g 3–4
Cefiderocol IV 60 mg/kg 6 g (8 g if augmented renal clearance) 3–4
Aztreonam-avibactam IV ND Loading dose 2670 mg (as both drugs) then 8000 mg daily 4, infuse over 3 hours
Imipenem-cilastin-relebactam IV ND 5000 mg (as all three components, vial 500/500/250) 4, infuse over 30 minutes
Sulbactam-durlobactam IV ND 4000 mg (as of both) 4, infuse over 3 hours
Table Continued
Page 3659
PATIENTS' TYPE OF THERAPY DRUG ROUTE OF ADMINISTRATION DAILY PEDIATRIC DOSAGE USUAL DAILY ADULT DOSAGE NO. OF DAILY DIVIDED DOSES
Antibacterial, oral Amoxicillin-clavulanate Oral 60–80mg/kg (as amoxicillin) 2–3g (as both) 2–3
Ciprofloxacin Oral 30mg/kg 1000–1500mg 2
Cefixime Oral 6–8mg/kg 400mg 2–3
Moxifloxacin Oral, IV Based on age, 4–6mg/kg twice daily285 400mg 1
Antifungal Amphotericin B deoxycholated IV 0.5–1mg/kg 0.5–1mg/kg 1
Liposomal amphotericin B IV 3–5mg/kg 3–5mg/kg 1
Amphotericin B lipid complex IV 5mg/kg 5mg/kg 1
Caspofungin IV 50mg/m2 for age <17years 70mg the first day, then 50mg the following days 1

Micafungin IV 2–4mg/kg 100mg 1
Anidulafungin IV 3mg/kg q24h the first day, then 1.5mg/kg q24h 200mg the first day, then 100mg the following days 1
Rezafungin IV ND 400 mg loading dose, then from day 8 200 mg once/wk
Voriconazolef IV, oral 9mg/kg bid the first day, then 8mg/kg bide 6mg/kg bid the first day, then 4mg/kg bid 2
Fluconazole IV, oral 10mg/kg 400mg 1
Posaconazolef Oral solution (doses are different for tablets and IV) If >12years, dose as adults 800mg (with a fatty meal or acidic carbonated drink) 4
Oral delayed-release tablets Not approved for <18years but PK/PD data available for per-kg dosing286,287 600mg/day divided in 2 doses, then 300mg daily 1
Isavuconazolef IV/oral 5.2 mg/kg/dose as isavuconazole 200mg q8h for 6 doses, then 200mg daily (372mg if ordered as isavuconazonium sulfate) 3 during the first 48hour, 1 thereafter
ND, No established dosing for children; PK/PD, pharmacokinetic/pharmacodynamic.

a Approved for abdominal and urinary tract infection only, not pneumonia.

b High doses necessary for enterococcal infections due to high epidemiological cutoff for E. faecium (4 mg/L).292

c An important and potentially confusing characteristic is colistin's dosage expression. Colistin is available in two salt forms: colistin sulfate and colistimethate sodium. In Europe, colistimethate sodium (salt) is available, and dosing is expressed usually in international units (IU) and sometimes in milligrams of colistimethate sodium, whereas in the United States, the dosage of US Food and Drug Administration–approved colistimethate sodium is defined in milligrams of colistin base activity. Thus particular attention should be paid to avoid dosage errors. Approximate dose conversion: 1,000,000 IU = 80 mg of colistimethate sodium = 30 mg colistin base.288 The optimal dosing of colistin remains to be established. Recent studies in adults with normal renal function reported the use of 6,000,000–9,000,000 IU daily, with pharmacologic analyses supporting the use of a loading dose of 9,000,000 IU, followed by 4,500,000 IU every 12 hours.289–291 See Chapter 31 for more details.

d Contraindicated in the presence of risk factors for renal toxicity (e.g., impaired renal function at baseline; nephrotoxic comedication, including aminoglycoside antibiotics; and history of previous toxicity).

e In patients 2–12 years of age and 12 to 14 years if weighing less than 50 kg; otherwise, dose as for adults if 12 years of age and older. Therapeutic drug monitoring is highly recommended. Children metabolize voriconazole more rapidly than adults.

f Therapeutic drug monitoring might be useful to assess if trough levels are in the range for efficacy in case of oral solution.

Voriconazole prophylaxis in patients with AML has been studied largely in small, single-center and retrospective studies with similar efficacy and toxicity to comparators. Therefore voriconazole has been only moderately recommended, with a lower level of evidence than posaconazole, fluconazole, or itraconazole, in patients with AML or MDS undergoing intensive remission-induction chemotherapy.170 In addition to CYP3A4-mediated drug-drug interactions typical for most of the azoles, voriconazole metabolism is also influenced by the activity of CYP2C19, resulting in significant interpatient variability.171 As certain genetic variations in CYP2C19 may result in increased metabolic activity of voriconazole leading to subtherapeutic blood levels, a strategy based on genetic identification of rapid and ultrarapid metabolizers leading either to increased dosing of voriconazole or alternative prophylaxis reduced the risk of subtherapeutic trough concentrations.172 Isavuconazole prophylaxis has been studied in single-center, mainly retrospective, trials. The three largest studies that included exclusively or mainly AML patients reported the rate of breakthrough proven/probable IFDs ranging from 3%173 to 6% to 7%.174,175

Prophylaxis with nebulized liposomal amphotericin B plus systemic fluconazole, compared with fluconazole only, was effective in reducing proven/probable IFD during repeated periods at risk after chemotherapy for acute leukemia in adults (a 10% reduction in IFD events).176 Systemic antifungal prophylaxis with liposomal amphotericin B at 2.5 mg/kg twice weekly was feasible and safe in high-risk pediatric cancer patients compared with a historical control group.177 On the other hand, a randomized controlled trial that included 355 adult patients undergoing remission-induction chemotherapy for newly diagnosed ALL demonstrated a high incidence of IFD in this population but failed to demonstrate the efficacy of liposomal amphotericin B at 5 mg/kg twice weekly (IFD 7.9% in prophylaxis arm vs. 11.7% in placebo arm).4 In a randomized study in children and young adults with AML, caspofungin prophylaxis during neutropenia resulted in a lower rate Page 3660of proven/probable IFDs and aspergillosis compared with fluconazole (respectively, 3.1% vs. 7.2% and 0.5% vs. 3.1%).178

Prophylaxis Against Pneumocystis jirovecii

In recent years, P. jirovecii pneumonia has been described with increasing frequency in patients without HIV.179 Certain underlying diseases, particularly lymphoproliferative ones, and their treatment confer a high risk of pneumocystosis. These include ALL, non-Hodgkin lymphoma, multiple myeloma, and CLL treated with standard chemotherapy or undergoing autologous transplantation.137 Other patients at risk for this complication are those with central nervous system solid tumors, associated with the prolonged use of high-dose steroids and the alkylating agent temozolomide, or those receiving bendamustine for breast cancer.179 Several other drugs affecting cell-mediated immunity have also been associated with the risk of P. jirovecii pneumonia, including fludarabine, ara-C (cytarabine), methotrexate, D-actinomycin, bleomycin, and L-asparaginase. In addition, pneumocystosis has been associated with alemtuzumab, which causes profound depletion of T lymphocytes; rituximab if used in combination with bendamustine or CHOP (cyclophosphamide, doxorubicin [hydroxydaunomycin], vincristine [Oncovin], and prednisolone) chemotherapy every 14 days; with idelalisib and ibrutinib (rare); and, in two cases, with the tyrosine kinase inhibitor dasatinib.59,137 It remains to be determined whether novel biologic drugs are responsible for an increased infection risk or whether other concomitant or previous treatments (purine analogues, alkylating agents, or steroids) contribute significantly to the risk of pneumocystosis. Finally, the duration of risk after treatment discontinuation is not known. Indications for prophylaxis of pneumocystosis include certain diseases (e.g., ALL) and treatments (e.g., fludarabine, alemtuzumab, or corticosteroid at the dose of >20 mg of prednisone daily for at least 4 weeks).180 An absolute CD4+ lymphocyte count of 200/mm3 or less, or a proportion of 15% of all T cells or less (or similar age-related values for children), has been suggested as an indication for prophylaxis, at least after HSCT, but qualitative defects might be present, leading to pneumocystosis even with higher CD4+ levels. Because the efficacy and tolerability of 160/800 mg of TMP-SMX given three or seven times a week are comparable, one 160/800-mg TMP-SMX tablet three times weekly remains the best prophylactic option, provided the patient is strictly compliant. Other drugs, such as daily oral dapsone or atovaquone or monthly aerosolized pentamidine, have demonstrated efficacy and are alternatives for patients who cannot tolerate TMP-SMX (see Chapter 275).180

Prevention of Viral Infections

For the prevention of HSV or VZV reactivation, antiviral prophylaxis with acyclovir or valacyclovir is recommended for seropositive patients undergoing therapy for acute leukemia or treatment with certain new agents such as bortezomib, alemtuzumab, idelalisib, or daratumumab. In patients not on active prophylaxis but who are exposed to varicella, intravenous administration of anti-VZV immunoglobulins (VariZIG) within 72 to 96 hours of the exposure is recommended, although protection is not guaranteed181 (see Chapter 143). If anti-VZV immunoglobulins are not available or in case of delayed notification of the risk contact, prophylaxis with valacyclovir (1 g three times daily for adults weighing more than 40 kg) or acyclovir (in children orally 20 mg/kg per dose, administered four times per day, with a maximum daily dose of 3200 mg; in adults 800 mg five times a day) is an option, although the efficacy has not been established. Prophylactic antiviral chemotherapy should be started on day 7 after exposure and continued for 7 days thereafter.

Routine CMV prophylaxis is not recommended outside of the transplantation setting. Blood CMV DNA testing should be performed in all cancer patients with signs or symptoms compatible with CMV reactivation (cytopenias, unexplained fever) or disease (gastroenteritis, pneumonia, retinitis, hepatitis, encephalitis) (see Chapter 144).141 In certain settings, such as patients receiving alemtuzumab for hematologic malignancies, idelalisib, rituximab and bendamustine, and possibly also brentuximab, the risk of CMV reactivation and subsequent disease may warrant CMV monitoring.141

All cancer patients should be screened for active or past HBV infection (with at least HBsAg and HBcAb) before starting chemotherapy because viral reactivation and acute hepatitis have consequences ranging from delaying chemotherapy to fatal fulminant infection. Patients with chronic hepatitis should receive antiviral treatment, whereas those with inactive infection (i.e., HBsAg positive, HBV DNA negative) should receive prophylaxis to prevent reactivation if they are scheduled to receive intensive chemotherapy, particularly rituximab. Lamivudine (100 mg/day) used to be the mainstay of antiviral prophylaxis, but antivirals with a higher barrier to resistance are preferred. Limited data on new antivirals are available in this setting, with only one recent randomized, open-label trial showing entecavir (0.5 mg/day) to be superior to lamivudine in preventing clinical hepatitis in patients with chronic inactive HBV infection undergoing chemotherapy, including rituximab with CHOP (R-CHOP) for lymphoma.182 Based on data from immunocompromised patients treated for chronic HBV infection, most guidelines recommend agents with a high genetic barrier (such as entecavir or tenofovir in any of the two formulations) for prophylaxis in HBsAg-positive patients.60 Prophylaxis should be started before chemotherapy if possible, but chemotherapy should not be delayed, and prophylaxis should be continued for at least 6 to 12 months after the end of chemotherapy (18 months for rituximab), depending on the predicted duration and intensity of immunosuppression. Monitoring of reactivation with HBV-DNA and HBsAg is warranted after discontinuing the prophylaxis. In a recent study of 77 noncirrhotic HBsAg-positive patients who received mainly entecavir treatment after rituximab-containing chemotherapy, clinical relapse of HBV occurred in 32.5%, with hepatic decompensation in 14.3% after drug discontinuation, with most cases occurring during the first year after discontinuation.183

HBV reactivation is also possible, although less likely, in patients with resolved HBV infection (HBsAg negative, HBcAb positive). The risk of reactivation can be as high as 42% in patients with lymphoma treated with rituximab and 20% in CLL, 13% in T-cell leukemia, and 5% in multiple myeloma.60 The options are either to provide lamivudine prophylaxis in those with high risk (i.e., >10%) of reactivation (e.g., patients receiving anti-CD20 agents) or to monitor them closely for HBV DNA and treat at the first sign of reactivation.

For influenza, vaccination is the most effective preventive measure globally. During community or nosocomial outbreaks of influenza infection, prophylaxis with oseltamivir 75 mg once daily for 10 days or postexposure prophylaxis with 75 mg twice daily for 5 days should be considered, especially in severely immunocompromised patients, because of the risk of severe infection and its complications.

Vaccination is also the most effective preventive measure for SARS-CoV-2 infection. All cancer patients (and their caregivers) should receive a full primary immunization schedule (three doses for the immunocompromised) and all recommended booster doses.146 In addition to vaccination, passive immunization with antispike monoclonal antibodies (mAbs) has been successfully studied, either as preexposure or postexposure prophylaxis. However, the neutralizing activity and consequent efficacy of approved antispike mAbs have been significantly reduced or abolished by mutations in spike proteins of novel viral variants. There are currently no antibodies for preexposure prophylaxis in high-risk patients or early treatment, but this may change as new viruses emerge.

Currently no antiviral has been approved for prophylaxis of SARS-CoV-2 infection. Oral nirmatrelvir/ritonavir, intravenous remdesivir, and oral molnupiravir are approved for treatment of mild-to-moderate SRS-CoV-2 infection in cancer patients, with the aim to prevent progression to severe COVID-19146 (see Chapter 136).

Other Prophylactic Strategies

Role of Colony-Stimulating Factors in Prophylaxis

Colony-stimulating factors (CSFs) are used as primary or secondary prophylaxis with the aim of facilitating dose-intense treatments and decreasing infections by preventing the development of neutropenia or reducing its duration. Granulocyte colony–stimulating factor (G-CSF) is the most widely used CSF, and many compounds with this activity are now available.184 A systematic review and meta-analysis on the efficacy and safety of 11 different CSFs showed a higher risk of febrile neutropenia for short-acting G-CSF filgrastim compared Page 3661with prolonged-release molecules of pegfilgrastim.185 Among the new molecules and biosimilars, mecapegfilgrastim, lipegfilgrastim, and balugrastim significantly reduced the incidence of febrile neutropenia (cumulative probabilities: 58%, 15%, and 11%, respectively), but empegfilgrastim (a short-acting G-CSF) and a long-acting G-CSF biosimilar were best in reducing severe neutropenia (cumulative probabilities: 21%, 20%, 15%, respectively) after cytotoxic chemotherapy. Unfortunately, results from different studies in different underlying conditions are not reported uniformly or with similar end points, and therefore comparison is difficult.186 Moreover, the use of G-CSF is frequently performed outside of guidelines,186,187 with reports of efficacy in reducing febrile neutropenia incidence and duration of hospitalization in people with solid tumors.186,187 A systematic review188 demonstrated improved survival in patients receiving intensified chemotherapy with primary G-CSF prophylaxis, but this was also associated with an increased risk of secondary neoplasia, including AML and myelodysplastic syndrome.

Finally, the effectiveness of G-CSF versus FQ for preventing febrile neutropenia was assessed in two studies that reported a significant difference in favor of antibiotics.189 In a recent study of patients with breast cancer randomized to receive ciprofloxacin or G-CSF, a nonsignificant difference was observed in favor of G-CSF in reducing febrile neutropenia and non-febrile neutropenia–related hospitalization.190

Role of Immunoglobulins in Prophylaxis

Based on the experience in patients with primary immunodeficiency syndromes, the administration of immunoglobulins has been recommended in those with secondary iatrogenic immunoglobulin deficiencies, such as in patients with acute and chronic lymphocytic leukemia or non-Hodgkin lymphoma, and in those receiving rituximab. Keeping the immunoglobulin G (IgG) level higher than 600 mg/dL has been associated with a reduction of infectious episodes due to bacteria, provided the treatment could be given for at least 6 months, but with no effect on viral and fungal diseases.191 Similarly, a meta-analysis of studies in patients with CLL or multiple myeloma receiving immunoglobulins reported a reduction in the occurrence of infections but without any impact on mortality.192

Among 4479 rituximab-treated patients, 4.5% of 3478 patients with cancer and 9.7% of 340 patients with a hematologic disorder received replacement therapy with immunoglobulins, and a higher cumulative dose of administered immunoglobulins was associated with a reduced risk of serious infectious complications (hazard ratio [HR], 0.98; 95% CI, 0.96–0.99; $p = 0.002$).193 Patients who benefit from regular IgG replacement therapy are those with CLL and recurrent infections (mainly pneumonia or sinusitis) and an IgG level less than 500 mg/dL.194 Subcutaneous administration of immunoglobulins has been demonstrated to be safe and effective.195

Infection Control: Isolation and Antimicrobial Stewardship

In consideration of growing antimicrobial resistance and a shortage of new antibiotics, infection control and antimicrobial stewardship programs should be implemented in all cancer centers.196 Infection control practices include surveillance, promoting and auditing the use of standard and transmission mode–based precautions, appropriate screening for colonization with resistant pathogens, and preventive measures, such as proper hand hygiene and contact precautions for patients colonized or infected with resistant bacteria (see Chapter 304). Active surveillance—for example, with rectal swabs for colonization with carbapenemase-producing Enterobacterales or VRE and nasal and skin colonization by MRSA—should be performed in institutions where these pathogens are regularly encountered. Additional transmission mode–based precautions include the use of high-level masks/respirators for health care personnel, such as N95 or higher level or free-flow pressure 3 (FFP3) masks. These airborne precautions should be applied, for example, in cases of active TB or measles. Contact precautions (hand hygiene, use of disposable gowns and gloves) are indispensable when caring for patients colonized or infected with MDR bacteria. In these cases, patients should be isolated in single rooms, possibly with staff cohorting. If single rooms are not available, then patient cohorting is recommended. In addition, in cases of patient transfer to another ward or hospital, it is of utmost importance to notify the receiving unit about carrier status and need for precautions. Droplet precautions, including disposable gown and gloves and a surgical mask, should be applied in cases of infection with respiratory viruses such as influenza. For all precaution measures, regular monitoring of adherence is warranted, and the facilities should ensure access to adequate hand hygiene stations (e.g., sinks, alcohol-based rubs) and personal protective equipment.

For neutropenic patients, it is not clear whether staying at home or in a hospital may have an impact on the development of chemotherapy-related complications. In the hospital, keeping the patient in a reverse isolation/protective environment is believed to reduce colonization or infection with various pathogens. Unlike for transmission-based precautions, there is no universal consensus on the appropriate protective environment for neutropenic patients, and standards may vary between institutions. However, they usually include single rooms and, for people entering the room, surgical masks (mainly to prevent respiratory viruses) and in some institutions also disposable gloves and gowns (to prevent contamination with contact-transmitted pathogens). General guidelines for preventing the diffusion of infectious diseases in health care facilities should be followed in any ward where cancer patients are admitted.197 Additionally, because aspergillosis and other mold infections are acquired via the respiratory route, the use of high-efficiency particulate air (HEPA) filters in rooms or wards where leukemic or transplant patients are hospitalized is recommended.198 The use of masks with adequate filtration power (FFP3 or free-flow pressure 2 [FFP2] or N95 respirators) could reduce Aspergillus colonization and infection when the patient is moved from the protective environment. This approach has been demonstrated effective, together with other physical barriers, during building renovations.199 These precautions should also be recommended at home when even small masonry works are performed. The use of laminar airflow rooms is not deemed necessary and does not have a substantial impact on the rate of infection. On the contrary, it negatively affects patient quality of life and the possibility for the health care providers to care for them properly.

Antimicrobial stewardship (see Chapter 54), including antifungal stewardship, should include four main aspects: (1) local surveillance of antibiotic and antifungal resistance, antibiotic and antifungal consumption, and patient outcomes in selected infections; (2) development and regular update of protocols and algorithms for the diagnosis, prevention, and treatment of infections; (3) updated diagnostic methods and prompt reporting of microbiologic results by the laboratory, allowing timely de-escalation of broad-spectrum empirical regimens and shortening of antibiotic therapy; and (4) optimization of dosing regimens, including therapeutic drug monitoring for voriconazole and posaconazole. These call for a multidisciplinary approach and close collaboration between the treating physicians, the microbiology laboratory, the infectious diseases consultation service, the infection control unit, the hospital pharmacy, and, if possible, a medical pharmacologist.200

Food and Lifestyle

A low-bacterial-count diet seems not to offer any benefit compared with a normal diet.201,202 A recent systematic review and meta-analysis showed that a neutropenic diet and a control diet had no effect on infection and mortality rates in neutropenic oncology patients.203 However, some precautions should be recommended, such as avoiding unpasteurized milk; unpasteurized or moldy cheese products; raw or undercooked (including insufficiently reheated) meat, fish, shellfish, tofu, or eggs; and unpeeled fruits and salad ingredients, unless properly washed at home. Listeria colonization and infection have been described in association with unpasteurized dairy products and ready-to-eat deli meats, whereas raw vegetable sprouts have been associated with outbreaks of E. coli, Salmonella, and Listeria. Transmission of toxoplasmosis can also occur with raw meat and vegetables contaminated with cat feces. Although probiotics (foods with live yeast or bacteria cultures) are advertised as useful in reducing the risk of antibiotic-associated diarrhea, BSIs from probiotic administration have been reported. Safety practices for food handling should be followed, and specific information for cancer patients is available online.204 Diet recommendations that are too restrictive may have a negative impact on the patient's nutritional status or quality of life, or both.

Cancer patients, especially during less intensive treatment, may seek information about the safety of traveling or other recreational Page 3662activities. Although few data exist that quantify the risk of travel or recreational activities, some considerations can be made. First, the assessment of the underlying conditions should be made, with particular attention to the stability of the patient's clinical condition and their potential need for rapid access to health care facilities, in which case, remote destinations or cruises are not recommended. Evaluation of any ongoing treatment that might constitute a contraindication to the disease prevention measures recommended for the proposed destination, such as vaccines or antimalaria prophylaxis, is necessary. In immunocompromised hosts, live-attenuated vaccines (such as those against yellow fever or Salmonella typhi) are contraindicated, whereas effectiveness of other vaccines, such as that against hepatitis A virus (HAV), might be reduced.

Cancer patients should not be advised to part with their pets, although some precautions are necessary. For example, a different household member should be assigned to scoop cat litter because of potential Toxoplasma cyst exposure. Aquaria should not be touched or maintained by patients because water in fish tanks may be contaminated with nontuberculous mycobacteria, and Salmonella can be acquired directly from reptiles or from fomites. Finally, large pet birds should be avoided because they may transmit Chlamydia psittaci. In general, the potential benefit of recommendations on the safety of food, pet care, travel, and other lifestyle measures should be weighed against the unclear value of such recommendations and their potential to have a negative impact on patient nutritional intake and quality of life.

Vaccination

Inactivated vaccines have been demonstrated to be safe and effective in cancer patients, especially during nonaggressive/maintenance treatment phases. The only pitfall is that the immune response can be poor, especially with certain drugs. Vaccinations against easily transmitted diseases (e.g., viral respiratory diseases such as influenza, SARS-CoV-2, or measles and varicella) should also be recommended for caregivers. International recommendations are available for vaccination of patients with hematologic malignancies and HSCT recipients.194,205–207

Influenza vaccination with inactivated vaccines is strongly recommended for all immunocompromised patients, with the possible exception/postponing of those treated with intensive chemotherapy (e.g., a remission induction regimen for acute leukemia) and those who received rituximab within the previous 6 months. This is not because of safety concerns, but because the probability of responding is very low. Both unvaccinated and vaccinated but severely immunocompromised patients should be offered empirical antiviral treatment in cases of clinical signs or symptoms of influenza during influenza season. Influenza vaccination is also recommended for household contacts and health care personnel caring for cancer patients.

Pneumococcal vaccination is strongly recommended for immunocompromised patients, even though response to vaccination may be suboptimal. The pneumococcal conjugate vaccine (PCV) is thought to elicit better responses than the pneumococcal polysaccharide vaccine (PPSV23). Vaccination at the onset of disease and before chemotherapy is recommended for those with chronic proliferative diseases. The new PCV20, containing antigens from 20 pneumococcal strains, may not require a subsequent PSV23 dose.208 See Chapter 136 for SARS-CoV-2 immunization.

Treatment of Complications in Neutropenic Cancer Patients

Empirical Antibacterial Therapy for Fever During Neutropenia

Fever during neutropenia has always been considered a medical emergency and should always be considered as due to an infection, unless proven otherwise. Febrile episodes during the course of neutropenia are classified according to infection status as (1) microbiologically documented infections with bacteremia (isolation of a significant pathogen from one or more blood cultures); (2) microbiologically documented infections without bacteremia (isolation of a significant pathogen from a well-defined site of infection, usually urine or respiratory secretions obtained with sterile procedures or fluid aspiration); (3) clinically documented infections in the presence of signs and symptoms clearly and objectively due to infection but without microbiologic proof; and (4) unexplained fever when both clinical and microbiologic proof are lacking but the clinical course is compatible with infection. For the purpose of starting empirical antibiotic therapy, fever is now defined as one oral or axillary temperature measurement ≥38.5°C or two oral or axillary temperature measurements ≥38.0°C separated by at least 1 hour. An ear temperature of 39°C is equivalent to 38.4°C and can be used to decide to start empirical antibiotics in a neutropenic pediatric patient, except for patients with acute myeloid leukemia or HSCT.209 Additionally, in neutropenic cancer patients, the development of other signs of infection (e.g., altered mental status, hypotension, skin or mucosal lesions, respiratory failure) without fever or even with hypothermia (temperature <35.5°C) should always raise the suspicion of an infection and must receive prompt attention, including blood cultures and empirical antibacterial treatment, in addition to appropriate diagnostic procedures.

The effectiveness of the empirical therapy approach has been clearly demonstrated, and empirical antibiotic therapy has contributed substantially to the impressive reduction in mortality from infectious complications over the last decades.210 Unfortunately, some recent studies on infection in patients with acute leukemia and HSCT show that the mortality rate associated with infections due to gram-negative rods, and possibly also VRE, may be increasing. This is driven by MDR isolates with peaks as high as 70% of BSI due to carbapenem-resistant K. pneumoniae in allogeneic HSCT recipients due to the absence of effective empirical threatment.5,157,164,211,212 Indeed, in the era of bacterial resistance and the shortage of new antibiotics, especially those active against gram-negative rods, infectious complications may once again significantly compromise the success of cancer treatment. Bacterial epidemiology and patterns of resistance may vary from patient to patient (in cases of individual colonization with resistant pathogens), from center to center (in cases of environmental colonization), and from country to country (different endemicity of resistant strains).5,95 Therefore the risk of a severe and complicated clinical course and the risk of infection caused by resistant pathogens should be evaluated individually and the treatment chosen accordingly.213 Fig. 311.2 summarizes a possible initial approach to a patient with febrile neutropenia. Table 311.5 summarizes the major antibiotics used for empirical or targeted therapy in febrile neutropenia.

In recent years attention has been paid to the pharmacokinetics of antibiotics in neutropenic cancer patients to improve effectiveness and reduce the risk of resistance. Continuous infusion of β-lactams (ceftazidime, cefepime, piperacillin-tazobactam, and meropenem) or vancomycin (at least in children) achieves the highest likelihood of adequate concentrations for treatment of BSIs.214–222 Neutropenic cancer patients can have “augmented renal clearance” (generally defined as creatinine clearance >130 mL/min/1.73 m2), which can increase drug elimination and therefore jeopardize therapeutic concentrations in plasma and tissues.223 The association of hypoalbuminemia with augmented renal clearance can reduce plasma concentrations of antibiotics, especially those with high protein binding, with further risk of therapeutic failure.224

Patients at Low Risk

Increased numbers of patients with solid tumors and

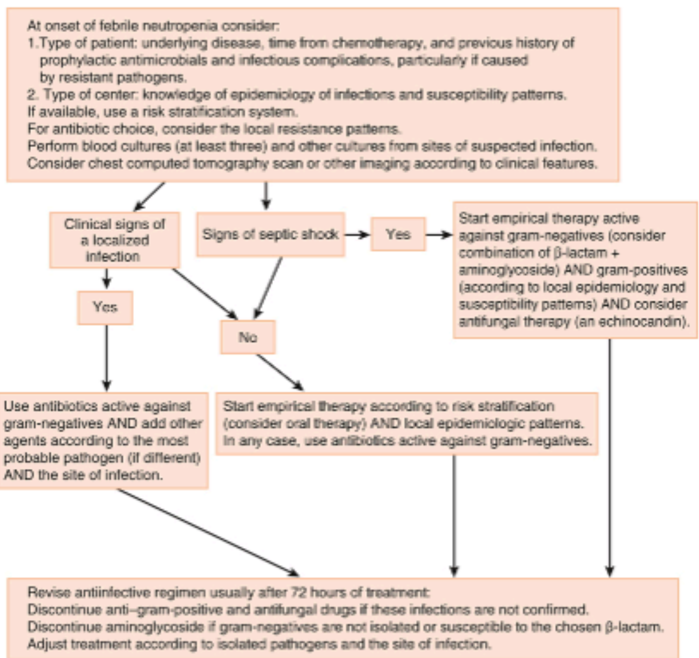

FIG. 311.2 Possible initial approach to a patient with febrile neutropenia.

nonleukemic hematologic diseases are treated with high-dose chemotherapy as outpatients. In some patients, especially those with solid tumors, neutropenia is rarely severe or protracted. In most cases, these patients are clinically stable within 48 hours after the appearance of fever and are without fever within 3 to 4 days. The Multinational Association for Supportive Care in Cancer

FIG. 311.2 Possible initial approach to a patient with febrile neutropenia.

Follow for extended description

TABLE 311.6

Clinical Scores to Predict Uncomplicated Clinical Course in Febrile Neutropenic Cancer Patients

	MASCC SCORE		CISNE SCORE	
	Clinical Parameters	**Score** [a]	**Clinical Parameters**	**Score**
Clinical data available at onset of febrile neutropenia or soon after admission	1. Burden of illness: no or mild symptoms [b]	0, 3, or 5	ECOG performance status ≥2	2
	2. No hypotension	5	Stress-induced hyperglycemia	2
	3. No chronic obstructive pulmonary disease	4	Chronic obstructive pulmonary disease	1
	4. Solid tumor or hematologic malignancy with no prior fungal infection	4	Cardiovascular disease	1
	5. No dehydration	3	Mucositis of grade ≥2 (National Cancer Institute Common Toxicity Criteria)	1
	6. Outpatient status	3	Monocytes <200/μL	1
	7. Patient's age <60 years	2		

[a] The maximal theoretical score is 26. Low-risk patient: score ≥21.

[b] No or mild symptoms = 5, moderate = 3, severe or morbid = 0.*ECOG*, European Cooperative Oncology Group; *MASCC*, Multinational Association for Supportive Care in Cancer.

(MASCC) score has helped identify patients with a low probability of complicated febrile neutropenia. A risk-index score >21 identified low-risk patients, with positive and negative predictive values of 91% and 36%, respectively. At this threshold, sensitivity and specificity were 71% and 68%, respectively, with a 30% misclassification rate. In recent years this scoring system has been considered inadequate for decision making.225 The Clinical Index of Stable Febrile Neutropenia (CISNE), ranging from 0 to 8, has been developed and Page 3663validated for low risk febrile neutropenic patients with different solid tumors.226,225,227 This system classifies patients into 3 classes: 0-2 points - low, 1-2 points - intermediate, while a score ≥ 3 identifies a high-risk patient.226 A systematic review and meta-analysis found that CISNE had a higher sensitivity.228 The combined use of both indices has been suggested in the Emergency Department to improve identification of low risk patients.229,230 Table 311.6 presents the variables included in MASCC and CISNE scores.

Even if validated clinical prediction rules for identification of children with neutropenic cancer at risk of severe complications are recommended in the most recent international guidelines,163 the pediatric clinical decision rules do not represent an efficient predictive tool when used outside the groups and the contexts that were used to generate them.231 A recent Australian study compared nine different clinical decision rules (CDRs) using prospectively collected data in children with cancer. They adopted a pragmatic “real-life” approach that included repeated episodes of febrile neutropenia occurring in the same patient.232 Eight of the nine CDRs were reproducible, with similar sensitivity or specificity, but none of them was able to accurately differentiate high- from low-risk episodes. The performance of CDRs Page 3664improved when the same parameters were re-evaluated on day 2, confirming the difficulties of using “a priori” predictive indicators. This also suggests potential utility for selecting patients suitable for short in-hospital evaluation (12–48 hours) before discharge. Early discharge or home therapy of low-risk patients is the next logical step, both in children and adults, and recent guidelines from the American Society of Clinical Oncology and Infectious Diseases Society of America (ASCO and IDSA) have been dedicated to outpatient management of fever and neutropenia in adult cancer patients.233 The mandatory conditions for home or outpatient treatment include the presence of a reliable caregiver at home, stable intravenous access, hospital proximity and adequate transportation, and necessary home facilities (telephone, running water, heating, and refrigeration). Antibiotic choices for oral therapy in the absence of risk factors for resistant pathogens include combined ciprofloxacin plus amoxicillin-clavulanate, moxifloxacin monotherapy, or intravenous antibiotics, such as ceftriaxone plus amikacin once daily.234,235 In neutropenic patients with FUO, switching therapy from intravenous to oral (e.g., ciprofloxacin or cefixime) has been demonstrated to be safe. Skin and soft tissue infections due to gram-positive cocci may be safely treated with a single administration of 1.5 g of dalbavancin, allowing outpatient treatment of infections due to resistant gram-positive bacteria, although there are limited data in neutropenic patients.236

These findings should be considered in light of contemporary practices. First, targeted therapies and biologic response modifiers have partially changed the spectrum of infecting pathogens and put patients at higher and different risk than previously realized. Second, all these studies were performed in times when antibiotic resistance was not a major problem and the risk for a bacterial infection caused by ESBL-producing and/or quinolone-resistant gram-negative rods was low or absent. Every center should consider patient risks in the context of local epidemiology in developing the best approach to fever during neutropenia.

Patients at High Risk
Administering empirical antimicrobial therapy very early after the onset of fever during neutropenia has been a major driver of progress in this field. However, because of the decreasing activity of the classic antibiotics used for fever during neutropenia and the shortage of new drugs, strategies have been rediscussed. Physicians have been forced to diversify empirical regimens based on colonization, local epidemiology, antibiotic policies, and patient-related factors such as clinical presentation, organ failure, and status of the underlying disease. Currently, two strategies for empirical therapy are envisaged: the escalation approach and the de-escalation approach.163,213,237 In both strategies, there are two key time points: the day of onset of fever or suspected infection (day 0), when the patient is initially evaluated, cultures are drawn, and antibiotic therapy is started, and the day in which everything should be reevaluated, based on clinical conditions and availability of microbiologic results (days 2–4, depending on the performance of local laboratories). These are appropriate times to consider broadening or restricting the antibiotic coverage or focus on a specific pathogen. Table 311.7 summarizes the approaches, as discussed next.

Etiology
Surveillance studies on pathogens causing infections in cancer patients are of the utmost importance for the implementation of management strategies. Large-scale studies are obviously crucial because they can provide information about worldwide trends, but also single-center surveillance reports may be useful, because every geographic region, country, or even single center may have peculiarities related to the type of patient, type of care, and previous local antibiotic policies. Most of the available information concerns bacterial and fungal pathogens isolated in the bloodstream, whereas the role of deep-seated infections, in addition to the impact of viral infections, is less well described.

Bacterial Infections
Over the last 30 years, gram-positive bacteria were the most frequent pathogens causing BSIs in cancer patients. However, an increase in the frequency of bacteremias caused by gram-negative rods has been reported, with gram-negative pathogens becoming either predominant or at least as frequently isolated as the gram-positives.12,91,92 This trend has also been reported in a retrospective literature review and Page 3653contemporary surveillance study performed in 2011 in 39 European hematology centers from 18 countries belonging to the network of the European Conference of Infections in Leukemia (ECIL).93 As shown in Fig. 311.1, this study found that gram-negative pathogens were almost as frequently isolated as the gram-positives, with gram-positive/gram-negative ratios in BSIs of 60%/40% and 55%/45% in the literature review and the ECIL-4 surveillance, respectively. The detailed etiology was similar (see Fig. 311.1) in the literature review and the surveillance study, with a slightly higher rate of enterococci and Enterobacterales and a decreased rate of Pseudomonas aeruginosa in the surveillance study.93 These changes in etiology appear to be associated with an important increase in the proportion of resistant pathogens, such as ESBL-producing Enterobacteriaceae, VRE, MDR P. aeruginosa, and the most worrisome, carbapenem-resistant Enterobacterales (CRE). In addition, in leukemic patients, most staphylococci are resistant to methicillin, whereas most gram-negative pathogens are resistant to FQs.93

FIG. 311.1 Etiology of bloodstream infections in cancer patients.Top, Data from recent literature review. Bottom, Data from surveillance study for the Fourth European Conference of Infections in Leukemia (ECIL-4) in 2011. (Modified from Mikulska M, Viscoli C, Orasch C, et al. Fourth

European Conference on Infections in Leukemia Group (ECIL-4), a joint venture of EBMT, EORTC, ICHS, ELN and ESGICH/ESCMID. Aetiology and resistance in bacteraemias among adult and paediatric haematology and cancer patients. J Infect. 2014;68:321–331.)
Follow for extended description
Of note, the rates of resistance were generally higher in southern and eastern than in northern and western Europe, and this trend is also evident in the non–hematologic cancer population.93,94,95 However, the problem of increasing infections due to resistant pathogens is worldwide.

The increase in infections caused by resistant strains usually stems from an increase in colonization with these pathogens. Indeed, colonization with resistant bacteria is one of the most important risk factors for infection with resistant bacteria (Table 311.3). The association between colonization and subsequent infection has been reported for VRE, ESBL-producing Enterobacterales, P. aeruginosa, Stenotrophomonas maltophilia, and carbapenem- or colistin-resistant Klebsiella pneumoniae.96–99 These data may differ in pediatrics. A recent multinational survey on antibiotic-resistant bacteremias in children receiving Page 3654antineoplastic chemotherapy or HSCT showed that colonization was a significant risk factor for MRSA bacteremia, whereas it was not associated with an increase in the risk of antibiotic-resistant gram-negatives, with the exception of resistance to piperacillin-tazobactam.5 This observation can have important management implications because patients carrying MRSA can be successfully decolonized100 and might benefit from the use of a glycopeptide in the initial empirical therapy of febrile neutropenia (see later).

TABLE 311.3

Risk Factors for Infections Caused by Resistant Bacteria in Cancer Patients
1. Previous infection or previous or current colonization by resistant bacteria, in particular:
• Enterobacteriaceae resistant to third-generation cephalosporins or carbapenems
• Multidrug-resistant, nonfermenting, gram-negative rods: Pseudomonas aeruginosa, Acinetobacter baumannii, Stenotrophomonas maltophilia
• Vancomycin-resistant enterococci
• Methicillin-resistant Staphylococcus aureus
2. Previous exposure to broad-spectrum antibiotics, in particular to third-generation cephalosporins
3. Health care–related infection
4. Prolonged hospital stay and/or repeated hospitalizations
5. Presence of urinary catheter
6. Older age
7. Intensive care unit stay

Obviously, not all colonized patients will develop infection due to the colonizing strain, because the risk depends on multiple host- and chemotherapy-related variables. However, colonization-informed choice of empiric treatment of febrile neutropenia targeting the MDR colonial might prevent breakthrough infections. In a study on carbapenem-resistant KPC-producing K. pneumoniae, the introduction of prompt screening and a targeted empirical treatment lowered the rate of BSI due to KPC-producing K. pneumoniae among colonized patients from 67% to 11%, and the mortality of BSI due to the resistant strain was also significantly reduced.101 Although early detection of colonization with MDR gram-negative bacteria is useful, the impact of screening for VRE is less clear. VRE BSI has been associated with lower survival at 3 and 12 months after allogeneic HSCT102; however, there was no benefit of VRE-targeting empirical therapy in colonized patients with VRE BSI, suggesting that VRE BSI might be a marker of mortality risk as opposed to causative.103

In a recent cross-sectional study in adults from the United States, the rate of critical illness (defined as ventilation, vasopressor therapy, or death) within 30 days from BSI was not influenced by FQ prophylaxis, organism type, initial antibiotics, or adequacy of coverage, although in these patients approximately 10% of gram-negatives were susceptible to the standard empirical therapy with cefepime or piperacillin-tazobactam.104 In a multicenter pediatric study in which over 20% of gram-negatives were resistant to cefepime or piperacillin-tazobactam, the risk of ICU admission or death was related to antibiotic resistance and prior antibiotic exposure.5 In both these studies the treatment center played an important role in the distribution of gram-positive versus gram-negative species and development of infections due to resistant pathogens.

Anaerobic bacteria are isolated in less than 1% of positive blood cultures in cancer patients, but the proportion increases to 3% among those undergoing abdominal surgery.79 Anaerobes are usually isolated in polymicrobial bacteremias, especially together with gram-negative rods, with a rate that seems to be higher than that observed in non-oncology patients undergoing similar surgery (0.597 vs. 0.033 per 1000 hospital days, respectively).105 Differences in the etiology of bacterial infection between neutropenic and nonneutropenic cancer patients have been reported.12

CVC-related bacteremias are generally caused by gram-positive cocci (especially coagulase-negative staphylococci), which are isolated in more than 50% of the episodes compared with 25% to 40% for gram-negative rods.7,106–113 The source of infection is likely to be partially different in cases of gram-positive and gram-negative CVC-related bacteremias. Infusate contamination is rare but possible, particularly with Klebsiella, Enterobacter, Citrobacter, Achromobacter, Serratia, Ralstonia, and Pseudomonas (other than P. aeruginosa). Polymicrobial infections are not uncommon, especially with gram-negative bacteria.

Bacterial gastroenteritis caused by classic enteric pathogens (Salmonella and Shigella) is a rare event in patients with acute leukemia, involving less than 1% of acute enteritis after chemotherapy.114 In contrast, C. difficile is common in cancer patients, with an incidence that is twofold higher than in the noncancer population115 and up to sixfold higher in hematology patients.116 Helicobacter pylori has also been described as a possible cause of gastrointestinal disease in this patient population.

Legionellosis and nocardiosis are rare but potentially life-threatening infections, with Legionella pneumonia sometimes presenting with unusual radiologic patterns. Mycoplasma pneumoniae and Chlamydia pneumoniae have been rarely described as a cause of pneumonia in cancer patients, but it is possible that their incidence is underreported.

Finally, TB might be underestimated and underdiagnosed in cancer populations. Older data showed a rate of approximately 90 cases per 100,000 persons (i.e., ninefold higher than in the general population in developed countries).117 However, a recent review found very low rates of TB in hematology settings, with no evidence that universal screening with immunoassays and treatment of latent TB provides clinical benefits in people with low risks in nonendemic areas.118 Patients from high-incidence countries or communities account for most of the cases of TB in the cancer population. Findings of pleuroparenchymal imaging abnormalities (mainly on the upper lobes) suggestive of previous TB and people with reported or predicted exposures should be screened for prior infection using immunoassays, with consideration for anti-TB therapy.118 Routine screening may be warranted in people scheduled to undergo therapy that causes prolonged T-cell depletion. Infections caused by nontuberculous mycobacteria are rare outside of cases of CVC-related BSIs. Local or even disseminated M. bovis infections can occur in patients receiving BCG immunotherapy for bladder cancer.

Fungal Infections

Aspergillus spp. and Candida spp. are the most common fungal pathogens in cancer patients, with the former now seen more frequently than the latter. Other fungal pathogens include P. jirovecii, cryptococci, molds such as Mucorales or Fusarium, and rare yeasts.

Among yeasts, Candida is the most frequently isolated, usually in BSIs, with an increasing proportion of non-albicans strains. This phenomenon, known since 1999, was confirmed by a study from the European Organization for Research and Treatment of Cancer (EORTC), which reported that in cancer patients the overall incidence of fungemia was 2.3%, with C. albicans responsible for 72% of candidemia cases in the whole study population, but only 27% in patients with hematologic malignancies. This is due to fluconazole prophylaxis, which reduced the overall rate of candidiasis but favored breakthrough infections with fluconazole-resistant species.119,120 Similar proportions were observed recently in pediatric patients.121 The majority of candidemias were observed in nonneutropenic patients. In general, yeasts belonging to the Candida parapsilosis complex are usually associated with CVC contamination, whereas other Candida species typically originate from the gastrointestinal tract after selection and translocation. C. glabrata is increasing in frequency and resistance, not only to fluconazole but also to echinocandins.122 Less common is infection with Candida auris, a frequently drug-resistant species causing outbreaks worldwide, mainly due to its ability to form biofilms and to persist on surfaces.

Among molds, Aspergillus represents the most frequently isolated or suspected. The majority of episodes are caused by Aspergillus fumigatus, although some centers report a predominance of infections caused by Aspergillus flavus and Aspergillus terreus.126,127 Invasive aspergillosis in patients with malignancy or HSCT is relatively common and usually community acquired, but outbreaks can occur with environmental exposures, both in and outside the hospital. The incidence of invasive aspergillosis depends mainly on the underlying malignancy and its treatment. It is highest in patients with prolonged neutropenia, followed by those receiving high doses of steroid therapy.128 In a multicenter Italian study, the incidence of aspergillosis among hematologic cancer patients varied from 7.9% in acute nonlymphoblastic leukemia to 4.3% in ALL, 2.3% in chronic myelogenous leukemia, and less than 1% in CLL, Hodgkin and non-Hodgkin lymphoma, and multiple myeloma.126 Additionally, the risk of aspergillosis in adult patients with ALL was reported to be as high as 11.7%.129 Recent pediatric data showed a higher incidence of invasive mycoses in high-risk ALL (up to 15% in the absence of pharmacologic prophylaxis) occurring since the beginning of treatment (which includes the use of high-dose steroids) and in all the most aggressive phases.130,131 Patients with chronic lymphoproliferative disorders represent another at-risk group in which aspergillosis can be seen, probably as a result of more intensive treatment protocols. Despite the high number of patients with these diseases, the rate of IFD is <5%.127,128 The incidence of invasive aspergillosis after autologous HSCT is low (0.3%–2%), and it typically occurs during pre-engraftment neutropenia.132

Increased attention to azole resistance among Aspergillus species, especially members of the A. fumigatus species complex, is necessitated by recognition of organisms harboring mutations in erg11 along with Page 3655recognition of MDR Aspergillus species (e.g., A. calidoustus) and the recently described “cryptic” species (e.g., A. lentulus).133 Rates of infections with these organisms can vary depending on geographic region and agricultural use of azole antifungals. Despite the fact that the phenomenon is so far relatively limited in most settings, knowledge of the frequency of azole resistance at the country and hospital level is fundamental. Secondary azole resistance, with different genetic resistance mutations, has been reported in patients with chronic aspergillosis after prolonged treatment, particularly in the case of suboptimal blood levels or high fungal burden (see Chapter 263).134

Mucormycosis is increasingly reported in some centers, especially in the United States. It is unclear whether this represents a general trend influenced by geography or prior antifungal therapies or if these infections are simply diagnosed more often because of increased awareness or improved survival. Other fungi, such as Fusarium and Scedosporium, are reported sporadically, with some organisms having higher prevalence in certain regions.135

P. jirovecii is a well-known cause of pneumonia in cancer patients not receiving trimethoprim-sulfamethoxazole (TMP-SMX) prophylaxis, especially if treated with high-dose and prolonged steroid therapy or certain antineoplastic drugs such as fludarabine, alemtuzumab, idelalisib, or temozolomide. Attack rates vary from 6.5% to 43% in ALL, 4% to 25% in rhabdomyosarcomas, and nearly 1% in Hodgkin lymphoma and primary or metastatic central nervous system tumors.136,137

Finally, infections with Cryptococcus spp. and geographically influenced “endemic” fungi, such as Histoplasma or Coccidioides, are possible in patients who live or used to live in endemic areas.

Viral Infections
Apart from herpes simplex virus (HSV) reactivation, which occurs in up to 60% of HSV-seropositive patients with acute leukemia, there are few data describing viral infections in febrile neutropenia.3 Cytomegalovirus (CMV) reactivation, as measured by pp65 antigenemia or PCR, occurs in people with severe illness, without necessarily being followed by CMV disease.138 For this reason, routine monitoring of CMV reactivation and preemptive therapy is not considered necessary in cancer patients other than HSCT recipients. A recent study evaluating febrile

neutropenia in children and adults receiving intensive chemotherapy showed viremia in 15% of episodes, with the highest frequency in children and the highest viral loads in patients otherwise classifiable as fever of unknown origin (FUO), suggesting that at least some cases of febrile neutropenia in children could be attributable to viral infection/reactivation.139 The risk of viral reactivation (herpes simplex virus [HSV], VZV, CMV) and disease increases significantly with use of T-cell–suppressing agents or drug combinations, such as alemtuzumab or idelalisib or a combination of bendamustine and rituximab.59,140–142

Community-acquired respiratory viruses, such as influenza, parainfluenza, respiratory syncytial virus (RSV), human metapneumovirus, adenoviruses, rhinoviruses, and coronaviruses, are a frequent cause of respiratory disease in cancer patients and are probably underestimated as a cause of fever. Whereas most cancer patients experience self-limited upper respiratory illness, those with severe immune deficits, which can be estimated using dedicated scores that take into consideration lymphopenia, steroid use, hypogammaglobulinemia and recent transplant, and B-cell or T-cell depletion,143 are at increased risk for progression from upper respiratory tract infection to pneumonia, with possible respiratory failure.144 The rate of viral respiratory infections in cancer patients depends on the ongoing rate of infections in the community and on the testing policy. In cancer patients with viral respiratory disease, deferral of chemotherapy may be considered. Specific treatment is warranted for influenza and in some cases for RSV infection (e.g., in leukemic patients with risk factors for RSV-related mortality; see Chapter 165).144

SARS-CoV-2 infection has caused significant morbidity and mortality in cancer patients—much higher than in the general population.145,146 In the cancer setting, the additional negative effect of even mild SARS-CoV-2 infection is caused by interruption of chemotherapy, which can be extended in case of prolonged viral shedding. Patients with non-Hodgkin lymphoma and CLL, particularly if treated with B-cell–depleting therapy, are at the highest risk of prolonged or relapsing infection, and optimal management remains still to be defined. Early treatment is warranted to prevent progression to severe disease and potentially shorten the duration of infection.146

Viral gastroenteritis, mainly caused by rotavirus but also norovirus or sapovirus, may be a frequent complication in pediatric oncology, with a potential to cause outbreaks in cancer centers because of persistent gastrointestinal shedding in immunocompromised hosts. Both adenoviruses and parvovirus B19 have been reported as rare causes of severe gastrointestinal disease in cancer patients.

Finally, the reactivation/exacerbation of infections due to hepatotropic viruses (HBV and hepatitis C virus [HCV]) represents an important problem in areas of high endemicity. HBV reactivation is frequent in cancer patients with chronic inactive HBV infection (HBsAg positive, with negative or low-level serum HBV DNA), but it can also occur in patients with resolved HBV infection (HBsAg negative, HBcAb positive, or with low-level serum HBV DNA), particularly after treatment with rituximab (up to 40% of patients).60 Antiviral prophylaxis, preferably with a high barrier to resistance such as entecavir or tenofovir, is usually given to these patients for at least 6 to 12 months after the end of chemotherapy.

HCV flares might occur in cancer patients undergoing chemotherapy, and HCV eradication could lower the risk of hepatic toxicity and even improve the outcome in certain lymphomas.147 The possibility of chronic hepatitis E virus infection due to genotype 3, which is endemic in certain industrialized regions, has been reported in patients with hematologic malignancies and transplant recipients.148,149 Repeated transfusions might be a risk factor for hepatitis E virus infection.150,151

Other Pathogens

The risk of rare infections or reactivations caused by protozoa (leishmaniasis, South American trypanosomiasis, and malaria), helminths (strongyloidiasis), endemic fungi, and other tropical diseases should be considered in patients who live or lived in endemic areas. Obtaining a detailed history to identify potential exposures is the most important screening tool, with additional serology to document past exposure. However, in the case of strongyloidiasis, for example, the suboptimal performance of stool examination or serologic screening may warrant empirical treatment with ivermectin in patients who present with unexplained eosinophilia and who lived in endemic areas, such as the tropics, the subtropics, or the southeastern United States and Europe.

Prevention of Infections in Cancer Patients

Prevention is desirable, given the remarkable mortality and morbidity associated with infections in cancer patients. In general, to evaluate the cost-effectiveness of a prophylactic protocol, one should know the rate of the target infection in specific patient populations and the number of patients needed to treat to prevent a single infectious episode, severe infection, and attributable death for each center. The rate of severe side effects, reported as the number needed to harm, must also be taken into consideration.

Table 311.4 summarizes different regimens for primary chemoprophylaxis and other approaches appropriate in cancer patients based on clinical trials and guidelines. Advantages and disadvantages of these procedures are discussed here.

Prevention of Bacterial Infections

The use of antibiotics to prevent bacterial infections should be weighed against efficacy, toxicity, and impact on the development of resistance. Chemoprophylaxis for the prevention of bacterial infections was first proposed in clinical practice based on the recognition that 80% of bacterial pathogens causing infection in neutropenic cancer patients originated from the endogenous flora, with approximately half of infections acquired during the hospital stay. The first approach relied on the oral administration of nonabsorbable antibiotics (gentamicin, vancomycin, and nystatin) aimed at totally (including anaerobes) or partially (excluding anaerobes) suppressing the intestinal bacterial flora and preventing the acquisition of exogenous organisms (known also as gut decontamination). Subsequently, absorbable drugs such as TMP-SMX, usually given in combination with oral nystatin and then FQs, were introduced. Large meta-analyses of quinolone prophylaxis in cancer patients have confirmed a statistically significant reduction in febrile Page 3656neutropenia, BSIs, and mortality, with the number needed to treat (NNT) to prevent one death ranging from 24 to 50, with no untoward effect in induction of resistance.152,153 Most guidelines recommended FQ prophylaxis for adult patients with expected neutropenia longer than 7 days. These results were based on studies published between 1973 and 2011, largely before the spread of bacterial resistance.154

Given the alarming increase in bacterial resistance worldwide, two issues need to be addressed: (1) whether the benefit of FQ prophylaxis can still be expected and (2) what is the contemporary impact of routine prolonged administration of broad-spectrum agents, such as FQ, on the selection of resistant strains and local epidemiology. Meta-analyses of recent studies did not confirm the benefit on mortality, although prophylaxis was still associated with lower rates of BSI and febrile episodes.154,155 The reason might be that the sample size was insufficient to show a difference in mortality, because the outcome of febrile neutropenia has significantly improved. Alternatively, infections prevented by FQ may be associated with low mortality, because they can be effectively treated with standard empirical antibiotics.95,156,157 As demonstrated by studies performed in both children and adults, BSI-associated mortality is mainly driven by MDR pathogens, which are unlikely to be prevented by FQs.5,95,158,159 This might explain the fact that FQ prophylaxis may no longer have any effect on overall mortality, despite efficacy in preventing some infections (i.e., those due to organisms that are effectively treated with classic empirical therapy).154,156

Some studies have suggested that FQ prophylaxis may increase rates of MDR infections.154 Finally, FQ can be associated with toxicities, including QT prolongation, tendon rupture, and interaction with drugs such as cyclophosphamide.160 Therefore, some recent guidelines do not support the use of FQ prophylaxis.161–163 In patients with solid tumors, who usually experience low rates of BSI and shorter (<7 days) neutropenia, FQ prophylaxis is not recommended.20

Additional strategies for prevention of MDR bacterial infection include decolonization with orally nonabsorbable antibiotics and fecal microbiota transplantation (FMT), which reduce intestinal burden of MDR strains. Although decolonization with gentamicin or colistin can decrease infections with MDR pathogens, this strategy may also increase resistance among colonizing bacteria.164 FMT has been evaluated in a small prospective trial of MDR-colonized patients with hematologic malignancies; partial-to-complete decolonization was reported.165 The FMT procedure was more effective in the absence of concomitant antibiotic administration (79% vs. 36%), and no severe adverse events were observed.165 Further studies are warranted to determine all the potential benefits and real-life applications of manipulation of gut microbiota.

TABLE 311.4

Suggested Prophylaxis for Infections in Cancer Patients
DRUG SCHEDULE COMMENTS
Antibacterial Ciprofloxacin 500mg twice daily
Use of FQs should be based on local epidemiology and careful evaluation of potential drawbacks of FQ prophylaxis.
Probably not active anymore in many centers. Some studies showed possible effects on increasing resistance.
Traditionally administered to adults receiving chemotherapy for acute leukemia with expected neutropenia >7–10 days; starting with chemotherapy and continuing until resolution of neutropenia or initiation of empirical antibacterial therapy for febrile neutropenia.
Levofloxacin 500mg once daily; children 10 mg/kg twice daily
Antifungal Posaconazole
100-mg tablets: 3 tablets twice daily on the first day, then 3 tablets daily
As alternative, oral solution 200 mg tid orally with a (fatty) meal or acidic drink
IV: 300 mg twice daily on the first day, then 300 mg daily
Patients receiving high-intensity chemotherapy or certain combination therapies with targeted agents for acute myelogenous leukemia or myelodysplastic syndrome.
Therapeutic drug monitoring is warranted in case of oral solution.
Fluconazole 400mg once daily Patients at risk for candidiasis but not aspergillosis.
Other Secondary prophylaxis according to isolated pathogen, clinical presentation, or both.
Anti–Pneumocystis jirovecii Trimethoprim-sulfamethoxazole (TMP-SMX)
One double-strength tablet (160/800mg) three times weekly, or
25 mg/kg TMP-SMX (5 mg/kg of TMP), in 2 divided doses for 3 consecutive days/wk
All patients receiving chemotherapy with steroids, including those with solid tumors (e.g., brain cancer); treatment with agents like fludarabine, alemtuzumab, or idelalisib; or some combination therapies (e.g., bendamustine plus anti-CD20).
Dapsone 100 mg daily in adults, or 2mg/kg/day (max. 100mg) on alternate days three times/wk In patients who cannot tolerate TMP-SMX.
Aerosolized pentamidine 300mg once a month with nebulizer In patients who cannot tolerate TMP-SMX; effective, but it is more difficult to administer.
Atovaquone 750mg twice daily or 1500mg once daily In patients who cannot tolerate TMP-SMX.
Antiviral
Acyclovir
or
40mg/kg in children in 2 divided doses
In adults >40 kg: 800 mg twice daily
Patients with positive anti-HSV antibodies and severe mucositis or receiving treatment for acute leukemia or other drugs which increased the risk for herpes zoster.
VZV-susceptible patients exposed to chickenpox who did not receive prompt administration of specific immunoglobulins.
Valacyclovir
For HSV or VZV prophylaxis: 500mg twice daily or 1000mg daily
For VZV exposure: 1 g three times daily (see text)
Lamivudine 100mg once daily Patients with resolved HBV infection (HBsAg negative and HBcAb positive).
Entecavir 0.5mg daily HBsAg-positive/HBV-DNA <2000IU/mL or high-risk (e.g. receiving anti-CD20 antibodies) patients with resolved HBV infection.
Tenofovir disoproxil fumarate (TDF) 300 mg daily HBsAg-positive/HBV-DNA <2000IU/mL.
Antituberculosis Isoniazid 5mg/kg (max. 300mg) daily in adults, 10mg/kg (max. 300mg) daily in children for 9months
Patients with latent tuberculosis.
Efficacy not specifically evaluated in cancer patients. Add pyridoxine (50 mg/daily for adults).

Rifampicin 600 mg/daily for 4 months20 mg/kg daily (max. 600 mg) in children Patients with latent TB who do not receive concomitant medications subject to drug interactions. This regimen has been associated with the lowest toxicity and highest compliance rate.
Isoniazid and rifampicin
Dose as noted earlier
For 3 months
Patients with latent TB who do not receive concomitant medications subject to drug interactions of rifampicin.
Anti–CVC-associated infection None
Good skin preparation and the use of sterile technique at time of device insertion.
Good maintenance procedures.
All patients with indwelling central venous catheter.
Table Continued
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DRUG SCHEDULE COMMENTS
Others Growth factors Filgrastim either subcutaneously or as an intravenous infusion over at least 1h, or pegylated filgrastim
For the prevention of febrile neutropenia in patients who have a high risk of this complication based on age, medical history, disease characteristics, and myelotoxicity of the chemotherapy regimen.
Secondary prophylaxis with G-CSFs recommended for patients who experienced a neutropenic complication from a prior cycle of chemotherapy (for which primary prophylaxis was not received), in whom a reduced dose of chemotherapy may compromise disease-free or overall survival or treatment outcome.
Efficacy not fully demonstrated for pegylated filgrastim.
Immunoglobulins Polyclonal immunoglobulins: 400mg/kg every 21–28 days
Patients with chronic lymphocytic leukemia after the second episode of severe bacterial infection.
Patients with leukemia or lymphoma with hypogammaglobulinemia (<400 g/dL) and severe bacterial infections (reasonable, but not proved).
Specific anti-VZV (VariZIG): 125IU for every 10kg of body weight (max., 625IU); adults 650 IU In high-risk contact with a negative history of varicella preferably within 96h after exposure to chickenpox.
Vaccines Influenza
Influenza vaccination of patients, especially during less aggressive treatment phases.
Vaccination of household contact and health care workers.
Pneumococcus Conjugated (PCV) vaccine recommended; with new 15-valent or 20-valent PCV vaccines available, the role of PPSV23 to be determined.
Varicella VZV-seronegative household contacts and health care workers.
Herpes zoster Recombinant inactivated 2-dose vaccine in VZV seropositive patients.
COVID-19 Vaccinate with 3-dose primary cycle, and provide boosters following updated recommendations.
Isolation procedures
Perform hand hygiene with an alcohol-based hand rub or by washing hands with soap and water if soiled, before and after all patient contacts or contact with the patients' potentially contaminated equipment or environment.
Use contact precautions (gowns and gloves).
Ensure adherence to standard environmental cleaning with an effective disinfectant.
Patients colonized or infected with multidrug-resistant pathogens (e.g., VRE, CRE) or infected with other pathogens for which contact isolation precautions are advisable (e.g., C. difficile, norovirus); of note, alcohol-based hand rubs are not cidal against C. difficile or norovirus.
CRE, Carbapenem-resistant Enterobacteriaceae; CVC, central venous catheter; FQ, fluoroquinolone; G-CSFs, granulocyte colony-stimulating factors; HBcAb, hepatitis B core antibody; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus; HSCT, hematopoietic stem cell transplantation; HSV, herpes simplex virus; IU, international unit; max., maximum; VRE, vancomycin-resistant enterococci; VZV, varicella-zoster virus.

Prevention of Tuberculosis
Risks for reactivation of latent TB are highest among people with sustained defects in T-cell immunity and are not clearly increased in people with short or intermittent episodes of neutropenia. Few data exist, and they depend on geographic prevalence.118 Standard regimens

for latent TB are recommended (rifampin 4 months, rifampin and isoniazid 3 months, isoniazid 9 months), but drug-drug interactions between rifampin and hematologic treatments, combined toxicities (hepatic and neurologic), and the negative impact of delaying hematologic treatment118 must be carefully weighed.

Prevention of Central Venous Catheter–Related Infections (See Chapter 306)

Prevention of CVC-related infections starts with the correct insertion and manipulation of the devices, not only by health care workers but also by caregivers who perform CVC maintenance procedures (and sometimes drug infusions) at home. Aseptic bundles during insertion and postinsertion care, which include assessment of need for CVC, daily inspection of the site of insertion, change of dressing at least weekly or whenever wet or soiled, use of chlorhexidine gluconate–impregnated sponges, application of alcohol scrub for 15 seconds before each access of the line, and hand hygiene, are effective measures to prevent infectious complications.41 Continuous education and regular audits of bundle implementation represent other helpful activities.166 Antibiotic catheter flushing has also been suggested to reduce the risk of infections, but this could represent a risk for selection of resistant bacteria. On the contrary, taurolidine lock has been associated with a significant reduction of CVC-related BSI, even if the susceptibilities of gram-positives and gram-negatives to taurolidine are indeterminate due to limited data.167

Prevention of Fungal Infections

Primary Antifungal Chemoprophylaxis

Although fungal infections usually represent no more than 10% of all infections (see Table 311.2), their associated morbidity and mortality remain high. Two main drugs used for antifungal prophylaxis in patients with acute leukemia include fluconazole (active against yeasts such as Candida but not molds) and posaconazole (mold-active prophylaxis). Fluconazole is effective in preventing Candida infections in people with acute myeloid leukemia and may benefit other populations that have an incidence of invasive candidemia higher than 7% to 10%.128 Mold-active prophylaxis with posaconazole in adults receiving multiple cycles of chemotherapy for acute myeloid leukemia or myelodysplastic syndrome reduces the incidence of invasive mycosis from 8% to 2% (primary end point) compared with standard prophylaxis with fluconazole or itraconazole, with improved survival.168 The NNT to prevent one case of proven/probable invasive mycosis was 16, and the NNT to prevent one fungal infection–related death was 27.169 However, if the incidence of invasive mycoses is lower than 8%, the NNT to prevent one infection would be higher, and this underlines the requirement to tailor needs in different hosts. Specific recommendations on drug dosing and formulations are summarized in Table 311.4 and Table 311.5. Recently introduced targeted agents in the treatment of AML, such as venetoclax, midostaurin, quizartinib, or gilteritinib, have led to challenges in the management of antifungal prophylaxis Page 3658because coadministration of these agents with posaconazole leads to significant increase in their blood levels, which carries the risk of toxicity (see Table 311.2). Posaconazole prophylaxis is still recommended when these agents are used in combination with other chemotherapy drugs, and dose reduction has been proposed for some (venetoclax), whereas careful monitoring was recommended with a low level of evidence of others (midostaurin).66

TABLE 311.5

Antibacterial and Antifungal Agents Usually Used in Cancer Patients

PATIENTS' TYPE OF THERAPY	DRUG	ROUTE OF ADMINISTRATION	DAILY PEDIATRIC DOSAGE	USUAL DAILY ADULT DOSAGE	NO. OF DAILY DIVIDED DOSES
Antibacterial, intravenous	Piperacillin-tazobactam	IV	400 mg/kg (as piperacillin)	13.5–18g (as both drugs)	3–4 in adults, 4 in children; preferably in prolonged or continuous infusion with a loading dose
	Ceftazidime	IV	100 mg/kg	6000mg	3
	Cefepime	IV	100 mg/kg	6000mg	3
	Meropenem	IV	60 mg/kg	3000–4000mg	3, infuse over 3–6 hours
	Ceftolozane-tazobactam	IV	90 mg/kg, up to a maximum dose of 4.5 ga (as both drugs)	4500 mg (as both drugs) 9000 mg for nosocomial pneumonia	3, infuse over 1 hour

Ceftazidime-avibactam IV
6 months to 18 years 187.5 mg/kg, up to a maximum of 7500 mg/daily (as both drugs)
<3 months 150 mg/kga
7500mg (as both drugs) 3, infuse over 2hours
Ceftaroline IV
2months to <2years: 8mg/kg every 8h
≥2 years to <18 years (≤33 kg): 12 mg/kg every 8 hours
≥2 years to <18 years (>33 kg): 400 mg every 8 hours or 600 mg every 12 hours
1200mg 2, infuse over 5–60minutes
Ceftobiprole IV 3 months to 12 years, <33/kg: 60 mg/kg, ≥33 kg 1500 mg; 12–18 years, <50 kg: 30 mg/kg, ≥50 kg 1500 mg 1500mg (as ceftobiprole) (2 g for the first 7 days in case of S. aureus BSI) 4
Imipenem-cilastatin IV 60–100mg/kg (as imipenem) 2000–4000mg (as both drugs) 3–4
Meropenem-vaborbactam IV ND 12 g (as both drugs) 3, infuse over 3hours
Ciprofloxacin IV 15–30mg/kg 800–1200mg 2–3
Ceftriaxone IV 80mg/kg 2000mg 1
Amikacin IV 20mg/kg 1000–1500mg 1
Vancomycin IV 40–60mg/kg 2000mg 2 or as continuous infusion after a loading dose
Teicoplanin IV; not available in United States 10mg/kg (loading dose of 10mg/kg bid on first day of treatment) 600–1200mg (loading dose of 600mg bid on first day of treatment) 1 (2 on first day of treatment)
Daptomycin IV 10 mg/kg 6–12 mg/kgb 1
Linezolid IV (also available oral) 30 mg/kg in three doses if <12years, then 20mg/kg 1200mg 2
Tedizolid Oral/IV ND 200mg 1
Dalbavancin IV
Birth to <6 years 22.5 mg/kg (max. 1500 mg)
6–18 years 18 mg/kg (max. 1500 mg)
1500 mg Single-dose regimen
Tigecycline IV 2.4mg/kg loading dose, then 2.4mg/kg (proposed) 100mg loading dose, then 100mg 2
Colistimethate sodium (colistin) IV 150,000–200,000IU/kg loading dose, then 150,000–200,000IU/kgc 9,000,000IU loading dose, then 9,000,000 IUc 2
Polymyxin B IV
Infants: up to 40,000U/kg
Children: 15,000–25,000 U/kg
15,000–25,000U/kg (= 1.25–2.5mg/kg, 1mg = 10,000U; a loading dose might be indicated)
Fosfomycin IV; not available in United States 300mg/kg 15–24g 3–4
Cefiderocol IV 60 mg/kg 6 g (8 g if augmented renal clearance) 3–4
Aztreonam-avibactam IV ND Loading dose 2670 mg (as both drugs) then 8000 mg daily 4, infuse over 3 hours
Imipenem-cilastin-relebactam IV ND 5000 mg (as all three components, vial 500/500/250) 4, infuse over 30 minutes
Sulbactam-durlobactam IV ND 4000 mg (as of both) 4, infuse over 3 hours
Table Continued
Page 3659
PATIENTS' TYPE OF THERAPY DRUG ROUTE OF ADMINISTRATION DAILY PEDIATRIC DOSAGE USUAL DAILY ADULT DOSAGE NO. OF DAILY DIVIDED DOSES
Antibacterial, oral Amoxicillin-clavulanate Oral 60–80mg/kg (as amoxicillin) 2–3g (as both) 2–3
Ciprofloxacin Oral 30mg/kg 1000–1500mg 2
Cefixime Oral 6–8mg/kg 400mg 2–3
Moxifloxacin Oral, IV Based on age, 4–6mg/kg twice daily285 400mg 1
Antifungal Amphotericin B deoxycholated IV 0.5–1mg/kg 0.5–1mg/kg 1
Liposomal amphotericin B IV 3–5mg/kg 3–5mg/kg 1
Amphotericin B lipid complex IV 5mg/kg 5mg/kg 1
Caspofungin IV 50mg/m2 for age <17years 70mg the first day, then 50mg the following days 1

Micafungin IV 2–4mg/kg 100mg 1
Anidulafungin IV 3mg/kg q24h the first day, then 1.5mg/kg q24h 200mg the first day, then 100mg the following days 1
Rezafungin IV ND 400 mg loading dose, then from day 8 200 mg once/wk
Voriconazolef IV, oral 9mg/kg bid the first day, then 8mg/kg bide 6mg/kg bid the first day, then 4mg/kg bid 2
Fluconazole IV, oral 10mg/kg 400mg 1
Posaconazolef Oral solution (doses are different for tablets and IV) If >12years, dose as adults 800mg (with a fatty meal or acidic carbonated drink) 4
Oral delayed-release tablets Not approved for <18years but PK/PD data available for per-kg dosing286,287 600mg/day divided in 2 doses, then 300mg daily 1
Isavuconazolef IV/oral 5.2 mg/kg/dose as isavuconazole 200mg q8h for 6 doses, then 200mg daily (372mg if ordered as isavuconazonium sulfate) 3 during the first 48hour, 1 thereafter
ND, No established dosing for children; PK/PD, pharmacokinetic/pharmacodynamic.

a Approved for abdominal and urinary tract infection only, not pneumonia.

b High doses necessary for enterococcal infections due to high epidemiological cutoff for E. faecium (4 mg/L).292

c An important and potentially confusing characteristic is colistin's dosage expression. Colistin is available in two salt forms: colistin sulfate and colistimethate sodium. In Europe, colistimethate sodium (salt) is available, and dosing is expressed usually in international units (IU) and sometimes in milligrams of colistimethate sodium, whereas in the United States, the dosage of US Food and Drug Administration–approved colistimethate sodium is defined in milligrams of colistin base activity. Thus particular attention should be paid to avoid dosage errors. Approximate dose conversion: 1,000,000 IU = 80 mg of colistimethate sodium = 30 mg colistin base.288 The optimal dosing of colistin remains to be established. Recent studies in adults with normal renal function reported the use of 6,000,000–9,000,000 IU daily, with pharmacologic analyses supporting the use of a loading dose of 9,000,000 IU, followed by 4,500,000 IU every 12 hours.289–291 See Chapter 31 for more details.

d Contraindicated in the presence of risk factors for renal toxicity (e.g., impaired renal function at baseline; nephrotoxic comedication, including aminoglycoside antibiotics; and history of previous toxicity).

e In patients 2–12 years of age and 12 to 14 years if weighing less than 50 kg; otherwise, dose as for adults if 12 years of age and older. Therapeutic drug monitoring is highly recommended. Children metabolize voriconazole more rapidly than adults.

f Therapeutic drug monitoring might be useful to assess if trough levels are in the range for efficacy in case of oral solution.

Voriconazole prophylaxis in patients with AML has been studied largely in small, single-center and retrospective studies with similar efficacy and toxicity to comparators. Therefore voriconazole has been only moderately recommended, with a lower level of evidence than posaconazole, fluconazole, or itraconazole, in patients with AML or MDS undergoing intensive remission-induction chemotherapy.170 In addition to CYP3A4-mediated drug-drug interactions typical for most of the azoles, voriconazole metabolism is also influenced by the activity of CYP2C19, resulting in significant interpatient variability.171 As certain genetic variations in CYP2C19 may result in increased metabolic activity of voriconazole leading to subtherapeutic blood levels, a strategy based on genetic identification of rapid and ultrarapid metabolizers leading either to increased dosing of voriconazole or alternative prophylaxis reduced the risk of subtherapeutic trough concentrations.172 Isavuconazole prophylaxis has been studied in single-center, mainly retrospective, trials. The three largest studies that included exclusively or mainly AML patients reported the rate of breakthrough proven/probable IFDs ranging from 3%173 to 6% to 7%.174,175

Prophylaxis with nebulized liposomal amphotericin B plus systemic fluconazole, compared with fluconazole only, was effective in reducing proven/probable IFD during repeated periods at risk after chemotherapy for acute leukemia in adults (a 10% reduction in IFD events).176 Systemic antifungal prophylaxis with liposomal amphotericin B at 2.5 mg/kg twice weekly was feasible and safe in high-risk pediatric cancer patients compared with a historical control group.177 On the other hand, a randomized controlled trial that included 355 adult patients undergoing remission-induction chemotherapy for newly diagnosed ALL demonstrated a high incidence of IFD in this population but failed to demonstrate the efficacy of liposomal amphotericin B at 5 mg/kg twice weekly (IFD 7.9% in prophylaxis arm vs. 11.7% in placebo arm).4 In a randomized study in children and young adults with AML, caspofungin prophylaxis during neutropenia resulted in a lower rate Page 3660of proven/probable IFDs and aspergillosis compared with fluconazole (respectively, 3.1% vs. 7.2% and 0.5% vs. 3.1%).178

Prophylaxis Against Pneumocystis jirovecii
In recent years, P. jirovecii pneumonia has been described with increasing frequency in patients without HIV.179 Certain underlying diseases, particularly lymphoproliferative ones, and their treatment confer a high risk of pneumocystosis. These include ALL, non-Hodgkin lymphoma, multiple myeloma, and CLL treated with standard chemotherapy or undergoing autologous transplantation.137 Other patients at risk for this complication are those with central nervous system solid tumors, associated with the prolonged use of high-dose steroids and the alkylating agent temozolomide, or those receiving bendamustine for breast cancer.179 Several other drugs affecting cell-mediated immunity have also been associated with the risk of P. jirovecii pneumonia, including fludarabine, ara-C (cytarabine), methotrexate, D-actinomycin, bleomycin, and L-asparaginase. In addition, pneumocystosis has been associated with alemtuzumab, which causes profound depletion of T lymphocytes; rituximab if used in combination with bendamustine or CHOP (cyclophosphamide, doxorubicin [hydroxydaunomycin], vincristine [Oncovin], and prednisolone) chemotherapy every 14 days; with idelalisib and ibrutinib (rare); and, in two cases, with the tyrosine kinase inhibitor dasatinib.59,137 It remains to be determined whether novel biologic drugs are responsible for an increased infection risk or whether other concomitant or previous treatments (purine analogues, alkylating agents, or steroids) contribute significantly to the risk of pneumocystosis. Finally, the duration of risk after treatment discontinuation is not known. Indications for prophylaxis of pneumocystosis include certain diseases (e.g., ALL) and treatments (e.g., fludarabine, alemtuzumab, or corticosteroid at the dose of >20 mg of prednisone daily for at least 4 weeks).180 An absolute CD4+ lymphocyte count of 200/mm3 or less, or a proportion of 15% of all T cells or less (or similar age-related values for children), has been suggested as an indication for prophylaxis, at least after HSCT, but qualitative defects might be present, leading to pneumocystosis even with higher CD4+ levels. Because the efficacy and tolerability of 160/800 mg of TMP-SMX given three or seven times a week are comparable, one 160/800-mg TMP-SMX tablet three times weekly remains the best prophylactic option, provided the patient is strictly compliant. Other drugs, such as daily oral dapsone or atovaquone or monthly aerosolized pentamidine, have demonstrated efficacy and are alternatives for patients who cannot tolerate TMP-SMX (see Chapter 275).180

Prevention of Viral Infections
For the prevention of HSV or VZV reactivation, antiviral prophylaxis with acyclovir or valacyclovir is recommended for seropositive patients undergoing therapy for acute leukemia or treatment with certain new agents such as bortezomib, alemtuzumab, idelalisib, or daratumumab. In patients not on active prophylaxis but who are exposed to varicella, intravenous administration of anti-VZV immunoglobulins (VariZIG) within 72 to 96 hours of the exposure is recommended, although protection is not guaranteed181 (see Chapter 143). If anti-VZV immunoglobulins are not available or in case of delayed notification of the risk contact, prophylaxis with valacyclovir (1 g three times daily for adults weighing more than 40 kg) or acyclovir (in children orally 20 mg/kg per dose, administered four times per day, with a maximum daily dose of 3200 mg; in adults 800 mg five times a day) is an option, although the efficacy has not been established. Prophylactic antiviral chemotherapy should be started on day 7 after exposure and continued for 7 days thereafter.

Routine CMV prophylaxis is not recommended outside of the transplantation setting. Blood CMV DNA testing should be performed in all cancer patients with signs or symptoms compatible with CMV reactivation (cytopenias, unexplained fever) or disease (gastroenteritis, pneumonia, retinitis, hepatitis, encephalitis) (see Chapter 144).141 In certain settings, such as patients receiving alemtuzumab for hematologic malignancies, idelalisib, rituximab and bendamustine, and possibly also brentuximab, the risk of CMV reactivation and subsequent disease may warrant CMV monitoring.141

All cancer patients should be screened for active or past HBV infection (with at least HBsAg and HBcAb) before starting chemotherapy because viral reactivation and acute hepatitis have consequences ranging from delaying chemotherapy to fatal fulminant infection. Patients with chronic hepatitis should receive antiviral treatment, whereas those with inactive infection (i.e., HBsAg positive, HBV DNA negative) should receive prophylaxis to prevent reactivation if they are scheduled to receive intensive chemotherapy, particularly rituximab. Lamivudine (100 mg/day) used to be the mainstay of antiviral prophylaxis, but antivirals with a higher barrier to resistance are preferred. Limited data on new antivirals are available in this setting, with only one recent randomized, open-label trial showing entecavir (0.5 mg/day) to be superior to lamivudine in preventing clinical hepatitis in patients with chronic inactive HBV infection undergoing chemotherapy, including rituximab with CHOP (R-CHOP) for lymphoma.182 Based on data from immunocompromised patients treated for chronic HBV infection, most guidelines recommend agents with a high genetic barrier (such as entecavir or tenofovir in any of the two formulations) for prophylaxis in HBsAg-positive patients.60 Prophylaxis should be started before chemotherapy if possible, but chemotherapy should not be delayed, and prophylaxis should be continued for at least 6 to 12 months after the end of chemotherapy (18 months for rituximab), depending on the predicted duration and intensity of immunosuppression. Monitoring of reactivation with HBV-DNA and HBsAg is warranted after discontinuing the prophylaxis. In a recent study of 77 noncirrhotic HBsAg-positive patients who received mainly entecavir treatment after rituximab-containing chemotherapy, clinical relapse of HBV occurred in 32.5%, with hepatic decompensation in 14.3% after drug discontinuation, with most cases occurring during the first year after discontinuation.183

HBV reactivation is also possible, although less likely, in patients with resolved HBV infection (HBsAg negative, HBcAb positive). The risk of reactivation can be as high as 42% in patients with lymphoma treated with rituximab and 20% in CLL, 13% in T-cell leukemia, and 5% in multiple myeloma.60 The options are either to provide lamivudine prophylaxis in those with high risk (i.e., >10%) of reactivation (e.g., patients receiving anti-CD20 agents) or to monitor them closely for HBV DNA and treat at the first sign of reactivation.

For influenza, vaccination is the most effective preventive measure globally. During community or nosocomial outbreaks of influenza infection, prophylaxis with oseltamivir 75 mg once daily for 10 days or postexposure prophylaxis with 75 mg twice daily for 5 days should be considered, especially in severely immunocompromised patients, because of the risk of severe infection and its complications.

Vaccination is also the most effective preventive measure for SARS-CoV-2 infection. All cancer patients (and their caregivers) should receive a full primary immunization schedule (three doses for the immunocompromised) and all recommended booster doses.146 In addition to vaccination, passive immunization with antispike monoclonal antibodies (mAbs) has been successfully studied, either as preexposure or postexposure prophylaxis. However, the neutralizing activity and consequent efficacy of approved antispike mAbs have been significantly reduced or abolished by mutations in spike proteins of novel viral variants. There are currently no antibodies for preexposure prophylaxis in high-risk patients or early treatment, but this may change as new viruses emerge.

Currently no antiviral has been approved for prophylaxis of SARS-CoV-2 infection. Oral nirmatrelvir/ritonavir, intravenous remdesivir, and oral molnupiravir are approved for treatment of mild-to-moderate SRS-CoV-2 infection in cancer patients, with the aim to prevent progression to severe COVID-19146 (see Chapter 136).

Other Prophylactic Strategies
Role of Colony-Stimulating Factors in Prophylaxis

Colony-stimulating factors (CSFs) are used as primary or secondary prophylaxis with the aim of facilitating dose-intense treatments and decreasing infections by preventing the development of neutropenia or reducing its duration. Granulocyte colony–stimulating factor (G-CSF) is the most widely used CSF, and many compounds with this activity are now available.184 A systematic review and meta-analysis on the efficacy and safety of 11 different CSFs showed a higher risk of febrile neutropenia for short-acting G-CSF filgrastim compared Page 3661with prolonged-release molecules of pegfilgrastim.185 Among the new molecules and biosimilars, mecapegfilgrastim, lipegfilgrastim, and balugrastim significantly reduced the incidence of febrile neutropenia (cumulative probabilities: 58%, 15%, and 11%, respectively), but empegfilgrastim (a short-acting G-CSF) and a long-acting G-CSF biosimilar were best in reducing severe neutropenia (cumulative probabilities: 21%, 20%, 15%, respectively) after cytotoxic chemotherapy. Unfortunately, results from different studies in different underlying conditions are not reported uniformly or with similar end points, and therefore comparison is difficult.186 Moreover, the use of G-CSF is frequently performed outside of guidelines,186,187 with reports of efficacy in reducing febrile neutropenia incidence and duration of hospitalization in people with solid tumors.186,187 A systematic review188 demonstrated improved survival in patients receiving intensified chemotherapy with primary G-CSF prophylaxis, but this was also associated with an increased risk of secondary neoplasia, including AML and myelodysplastic syndrome.

Finally, the effectiveness of G-CSF versus FQ for preventing febrile neutropenia was assessed in two studies that reported a significant difference in favor of antibiotics.189 In a recent study of patients with breast cancer randomized to receive ciprofloxacin or G-CSF, a nonsignificant difference was observed in favor of G-CSF in reducing febrile neutropenia and non-febrile neutropenia–related hospitalization.190

Role of Immunoglobulins in Prophylaxis

Based on the experience in patients with primary immunodeficiency syndromes, the administration of immunoglobulins has been recommended in those with secondary iatrogenic immunoglobulin deficiencies, such as in patients with acute and chronic lymphocytic leukemia or non-Hodgkin lymphoma, and in those receiving rituximab. Keeping the immunoglobulin G (IgG) level higher than 600 mg/dL has been associated with a reduction of infectious episodes due to bacteria, provided the treatment could be given for at least 6 months, but with no effect on viral and fungal diseases.191 Similarly, a meta-analysis of studies in patients with CLL or multiple myeloma receiving immunoglobulins reported a reduction in the occurrence of infections but without any impact on mortality.192

Among 4479 rituximab-treated patients, 4.5% of 3478 patients with cancer and 9.7% of 340 patients with a hematologic disorder received replacement therapy with immunoglobulins, and a higher cumulative dose of administered immunoglobulins was associated with a reduced risk of serious infectious complications (hazard ratio [HR], 0.98; 95% CI, 0.96–0.99; p = 0.002).193 Patients who benefit from regular IgG replacement therapy are those with CLL and recurrent infections (mainly pneumonia or sinusitis) and an IgG level less than 500 mg/dL.194 Subcutaneous administration of immunoglobulins has been demonstrated to be safe and effective.195

Infection Control: Isolation and Antimicrobial Stewardship

In consideration of growing antimicrobial resistance and a shortage of new antibiotics, infection control and antimicrobial stewardship programs should be implemented in all cancer centers.196 Infection control practices include surveillance, promoting and auditing the use of standard and transmission mode–based precautions, appropriate screening for colonization with resistant pathogens, and preventive measures, such as proper hand hygiene and contact precautions for patients colonized or infected with resistant bacteria (see Chapter 304). Active surveillance—for example, with rectal swabs for colonization with carbapenemase-producing Enterobacterales or VRE and nasal and skin colonization by MRSA—should be performed in institutions where these pathogens are regularly encountered. Additional transmission mode–based precautions include the use of high-level masks/respirators for health care personnel, such as N95 or higher level or free-flow pressure 3 (FFP3) masks. These airborne precautions should be applied, for example, in cases of active TB or measles. Contact precautions (hand hygiene, use of disposable gowns and gloves) are indispensable when caring for patients colonized or infected with MDR bacteria. In these cases, patients should be isolated in single rooms, possibly with staff cohorting. If single rooms are not available, then patient cohorting is recommended. In addition, in cases of patient transfer to another ward or hospital, it is of utmost importance to notify the receiving unit about carrier status and need for precautions. Droplet precautions, including disposable gown and gloves and a surgical mask, should be applied in cases of infection with respiratory viruses such as influenza. For all precaution measures, regular monitoring of adherence is warranted, and the facilities should ensure access to adequate hand hygiene stations (e.g., sinks, alcohol-based rubs) and personal protective equipment.

For neutropenic patients, it is not clear whether staying at home or in a hospital may have an impact on the development of chemotherapy-related complications. In the hospital, keeping the patient in a reverse isolation/protective environment is believed to reduce colonization or infection with various pathogens. Unlike for transmission-based precautions, there is no universal consensus on the appropriate protective environment for neutropenic patients, and standards may vary between institutions. However, they usually include single rooms and, for people entering the room, surgical masks (mainly to prevent respiratory viruses) and in some institutions also disposable gloves and gowns (to prevent contamination with contact-transmitted pathogens). General guidelines for preventing the diffusion of infectious diseases in health care facilities should be followed in any ward where cancer patients are admitted.197 Additionally, because aspergillosis and other mold infections are acquired via the respiratory route, the use of high-efficiency particulate air (HEPA) filters in rooms or wards where leukemic or transplant patients are hospitalized is recommended.198 The use of masks with adequate filtration power (FFP3 or free-flow pressure 2 [FFP2] or N95 respirators) could reduce Aspergillus colonization and infection when the patient is moved from the protective environment. This approach has been demonstrated effective, together with other physical barriers, during building renovations.199 These precautions should also be recommended at home when even small masonry works are performed. The use of laminar airflow rooms is not deemed necessary and does not have a substantial impact on the rate of infection. On the contrary, it negatively affects patient quality of life and the possibility for the health care providers to care for them properly.

Antimicrobial stewardship (see Chapter 54), including antifungal stewardship, should include four main aspects: (1) local surveillance of antibiotic and antifungal resistance, antibiotic and antifungal consumption, and patient outcomes in selected infections; (2) development and regular update of protocols and algorithms for the diagnosis, prevention, and treatment of infections; (3) updated diagnostic methods and prompt reporting of microbiologic results by the laboratory, allowing timely de-escalation of broad-spectrum empirical regimens and shortening of antibiotic therapy; and (4) optimization of dosing regimens, including therapeutic drug monitoring for voriconazole and posaconazole. These call for a multidisciplinary approach and close collaboration between the treating physicians, the microbiology laboratory, the infectious diseases consultation service, the infection control unit, the hospital pharmacy, and, if possible, a medical pharmacologist.200

Food and Lifestyle
A low-bacterial-count diet seems not to offer any benefit compared with a normal diet.201,202 A recent systematic review and meta-analysis showed that a neutropenic diet and a control diet had no effect on infection and mortality rates in neutropenic oncology patients.203 However, some precautions should be recommended, such as avoiding unpasteurized milk; unpasteurized or moldy cheese products; raw or undercooked (including insufficiently reheated) meat, fish, shellfish, tofu, or eggs; and unpeeled fruits and salad ingredients, unless properly washed at home. Listeria colonization and infection have been described in association with unpasteurized dairy products and ready-to-eat deli meats, whereas raw vegetable sprouts have been associated with outbreaks of E. coli, Salmonella, and Listeria. Transmission of toxoplasmosis can also occur with raw meat and vegetables contaminated with cat feces. Although probiotics (foods with live yeast or bacteria cultures) are advertised as useful in reducing the risk of antibiotic-associated diarrhea, BSIs from probiotic administration have been reported. Safety practices for food handling should be followed, and specific information for cancer patients is available online.204 Diet recommendations that are too restrictive may have a negative impact on the patient's nutritional status or quality of life, or both.

Cancer patients, especially during less intensive treatment, may seek information about the safety of traveling or other recreational Page 3662activities. Although few data exist that quantify the risk of travel or recreational activities, some considerations can be made. First, the assessment of the underlying conditions should be made, with particular attention to the stability of the patient's clinical condition and their potential need for rapid access to health care facilities, in which case, remote destinations or cruises are not recommended. Evaluation of any ongoing treatment that might constitute a contraindication to the disease prevention measures recommended for the proposed destination, such as vaccines or antimalaria prophylaxis, is necessary. In immunocompromised hosts, live-attenuated vaccines (such as those against yellow fever or Salmonella typhi) are contraindicated, whereas effectiveness of other vaccines, such as that against hepatitis A virus (HAV), might be reduced.

Cancer patients should not be advised to part with their pets, although some precautions are necessary. For example, a different household member should be assigned to scoop cat litter because of potential Toxoplasma cyst exposure. Aquaria should not be touched or maintained by patients because water in fish tanks may be contaminated with nontuberculous mycobacteria, and Salmonella can be acquired directly from reptiles or from fomites. Finally, large pet birds should be avoided because they may transmit Chlamydia psittaci. In general, the potential benefit of recommendations on the safety of food, pet care, travel, and other lifestyle measures should be weighed against the unclear value of such recommendations and their potential to have a negative impact on patient nutritional intake and quality of life.

Vaccination

Inactivated vaccines have been demonstrated to be safe and effective in cancer patients, especially during nonaggressive/maintenance treatment phases. The only pitfall is that the immune response can be poor, especially with certain drugs. Vaccinations against easily transmitted diseases (e.g., viral respiratory diseases such as influenza, SARS-CoV-2, or measles and varicella) should also be recommended for caregivers. International recommendations are available for vaccination of patients with hematologic malignancies and HSCT recipients.194,205–207

Influenza vaccination with inactivated vaccines is strongly recommended for all immunocompromised patients, with the possible exception/postponing of those treated with intensive chemotherapy (e.g., a remission induction regimen for acute leukemia) and those who received rituximab within the previous 6 months. This is not because of safety concerns, but because the probability of responding is very low. Both unvaccinated and vaccinated but severely immunocompromised patients should be offered empirical antiviral treatment in cases of clinical signs or symptoms of influenza during influenza season. Influenza vaccination is also recommended for household contacts and health care personnel caring for cancer patients.

Pneumococcal vaccination is strongly recommended for immunocompromised patients, even though response to vaccination may be suboptimal. The pneumococcal conjugate vaccine (PCV) is thought to elicit better responses than the pneumococcal polysaccharide vaccine (PPSV23). Vaccination at the onset of disease and before chemotherapy is recommended for those with chronic proliferative diseases. The new PCV20, containing antigens from 20 pneumococcal strains, may not require a subsequent PSV23 dose.208 See Chapter 136 for SARS-CoV-2 immunization.

Treatment of Complications in Neutropenic Cancer Patients

Empirical Antibacterial Therapy for Fever During Neutropenia

Fever during neutropenia has always been considered a medical emergency and should always be considered as due to an infection, unless proven otherwise. Febrile episodes during the course of neutropenia are classified according to infection status as (1) microbiologically documented infections with bacteremia (isolation of a significant pathogen from one or more blood cultures); (2) microbiologically documented infections without bacteremia (isolation of a significant pathogen from a well-defined site of infection, usually urine or respiratory secretions obtained with sterile procedures or fluid aspiration); (3) clinically documented infections in the presence of signs and symptoms clearly and objectively due to infection but without microbiologic proof; and (4) unexplained fever when both clinical and microbiologic proof are lacking but the clinical course is compatible with infection. For the purpose of starting empirical antibiotic therapy, fever is now defined as one oral or axillary temperature measurement ≥38.5°C or two oral or axillary temperature measurements ≥38.0°C separated by at least 1 hour. An ear temperature of 39°C is equivalent to 38.4°C and can be used to decide to start empirical antibiotics in a neutropenic pediatric patient, except for patients with acute myeloid leukemia or HSCT.209 Additionally, in neutropenic cancer patients, the development of other signs of infection (e.g., altered mental status, hypotension, skin or mucosal lesions, respiratory failure) without fever or even with hypothermia (temperature <35.5°C) should always raise the suspicion of an infection and must receive prompt attention, including blood cultures and empirical antibacterial treatment, in addition to appropriate diagnostic procedures.

The effectiveness of the empirical therapy approach has been clearly demonstrated, and empirical antibiotic therapy has contributed substantially to the impressive reduction in mortality from infectious complications over the last decades.210 Unfortunately, some recent studies on infection in patients with acute leukemia and HSCT show that the mortality rate associated with infections due to gram-negative rods, and possibly also VRE, may be increasing. This is driven by MDR isolates with peaks as high as 70% of BSI due to carbapenem-resistant K. pneumoniae in allogeneic HSCT recipients due to the absence of effective empirical threatment.5,157,164,211,212 Indeed, in the era of bacterial resistance and the shortage of new antibiotics, especially those active against gram-negative rods, infectious complications may once again significantly compromise the success of cancer treatment. Bacterial epidemiology and patterns of resistance may vary from patient to patient (in cases of individual colonization with resistant pathogens), from center to center (in cases of environmental colonization), and from country to country (different endemicity of resistant strains).5,95 Therefore the risk of a severe and complicated clinical course and the risk of infection caused by resistant pathogens should be evaluated individually and the treatment chosen accordingly.213 Fig. 311.2 summarizes a possible initial approach to a patient with febrile neutropenia. Table 311.5 summarizes the major antibiotics used for empirical or targeted therapy in febrile neutropenia.

In recent years attention has been paid to the pharmacokinetics of antibiotics in neutropenic cancer patients to improve effectiveness and reduce the risk of resistance. Continuous infusion of β-lactams (ceftazidime, cefepime, piperacillin-tazobactam, and meropenem) or vancomycin (at least in children) achieves the highest likelihood of adequate concentrations for treatment of BSIs.214–222 Neutropenic cancer patients can have “augmented renal clearance” (generally defined as creatinine clearance >130 mL/min/1.73 m2), which can increase drug elimination and therefore jeopardize therapeutic concentrations in plasma and tissues.223 The association of hypoalbuminemia with augmented renal clearance can reduce plasma concentrations of antibiotics, especially those with high protein binding, with further risk of therapeutic failure.224

Patients at Low Risk

Increased numbers of patients with solid tumors and nonleukemic hematologic diseases are treated with high-dose chemotherapy as outpatients. In some patients, especially those with solid tumors, neutropenia is rarely severe or protracted. In most cases, these patients are clinically stable within 48 hours after the appearance of fever and are without fever within 3 to 4 days. The Multinational Association for Supportive Care in Cancer (MASCC) score has helped identify patients with a low probability of complicated febrile neutropenia. A risk-index score >21 identified low-risk patients, with positive and negative predictive values of 91% and 36%, respectively. At this threshold, sensitivity and specificity were 71% and 68%, respectively, with a 30% misclassification rate. In recent years this scoring system has been considered inadequate for decision making.225 The Clinical Index of Stable Febrile Neutropenia (CISNE), ranging from 0 to 8, has been developed and Page 3663validated for low risk febrile neutropenic patients with different solid tumors.226,225,227 This system classifies patients into 3 classes: 0-2 points - low, 1-2 points - intermediate, while a score ≥ 3 identifies a high-risk patient.226 A systematic review and meta-analysis found that CISNE had a higher sensitivity.228 The combined use of both indices has been suggested in the Emergency Department to improve identification of low risk patients.229,230 Table 311.6 presents the variables included in MASCC and CISNE scores.

FIG. 311.2 Possible initial approach to a patient with febrile neutropenia.
Follow for extended description

TABLE 311.6

Clinical Scores to Predict Uncomplicated Clinical Course in Febrile Neutropenic Cancer Patients

MASCC SCORE		CISNE SCORE	
Clinical Parameters	Scorea	Clinical Parameters	Score

Clinical data available at onset of febrile neutropenia or soon after admission
1. Burden of illness: no or mild symptomsb
0, 3, or 5 ECOG performance status ≥2 2
2. No hypotension
5 Stress-induced hyperglycemia 2
3. No chronic obstructive pulmonary disease
4 Chronic obstructive pulmonary disease 1
4. Solid tumor or hematologic malignancy with no prior fungal infection
4 Cardiovascular disease 1
5. No dehydration
3 Mucositis of grade ≥2 (National Cancer Institute Common Toxicity Criteria) 1
6. Outpatient status
3 Monocytes <200/μL 1
7. Patient's age <60years
2
a The maximal theoretical score is 26. Low-risk patient: score ≥21.

b No or mild symptoms = 5, moderate = 3, severe or morbid = 0.ECOG, European Cooperative Oncology Group; MASCC, Multinational Association for Supportive Care in Cancer.

Even if validated clinical prediction rules for identification of children with neutropenic cancer at risk of severe complications are recommended in the most recent international guidelines,163 the pediatric clinical decision rules do not represent an efficient predictive tool when used outside the groups and the contexts that were used to generate them.231 A recent Australian study compared nine different clinical decision rules (CDRs) using prospectively collected data in children with cancer. They adopted a pragmatic "real-life" approach that included repeated episodes of febrile neutropenia occurring in the same patient.232 Eight of the nine CDRs were reproducible, with similar sensitivity or specificity, but none of them was able to accurately differentiate high- from low-risk episodes. The performance of CDRs Page 3664improved when the same parameters were re-evaluated on day 2, confirming the difficulties of using "a priori" predictive indicators. This also suggests potential utility for selecting patients suitable for short in-hospital evaluation (12–48 hours) before discharge. Early discharge or home therapy of low-risk patients is the next logical step, both in children and adults, and recent guidelines from the American Society of Clinical Oncology and Infectious Diseases Society of America (ASCO and IDSA) have been dedicated to outpatient management of fever and neutropenia in adult cancer patients.233 The mandatory conditions for home or outpatient treatment include the presence of a reliable caregiver at home, stable intravenous access, hospital proximity and adequate transportation, and necessary home facilities (telephone, running water, heating, and refrigeration). Antibiotic choices for oral therapy in the absence of risk factors for resistant pathogens include combined ciprofloxacin plus amoxicillin-clavulanate, moxifloxacin monotherapy, or intravenous antibiotics, such as ceftriaxone plus amikacin once daily.234,235 In neutropenic patients with FUO, switching therapy from intravenous to oral (e.g., ciprofloxacin or cefixime) has been demonstrated to be safe. Skin and soft tissue infections due to gram-positive cocci may be safely treated with a single administration of 1.5 g of dalbavancin, allowing outpatient treatment of infections due to resistant gram-positive bacteria, although there are limited data in neutropenic patients.236

These findings should be considered in light of contemporary practices. First, targeted therapies and biologic response modifiers have partially changed the spectrum of infecting pathogens and put patients at higher and different risk than previously realized. Second, all these studies were performed in times when antibiotic resistance was not a major problem and the risk for a bacterial infection caused by ESBL-producing and/or quinolone-resistant gram-negative rods was low or absent. Every center should consider patient risks in the context of local epidemiology in developing the best approach to fever during neutropenia.

Patients at High Risk
Administering empirical antimicrobial therapy very early after the onset of fever during neutropenia has been a major driver of progress in this field. However, because of the decreasing activity of the classic antibiotics used for fever during neutropenia and the shortage of new drugs, strategies have been rediscussed. Physicians have been forced to diversify empirical regimens based on

colonization, local epidemiology, antibiotic policies, and patient-related factors such as clinical presentation, organ failure, and status of the underlying disease. Currently, two strategies for empirical therapy are envisaged: the escalation approach and the de-escalation approach.163,213,237 In both strategies, there are two key time points: the day of onset of fever or suspected infection (day 0), when the patient is initially evaluated, cultures are drawn, and antibiotic therapy is started, and the day in which everything should be reevaluated, based on clinical conditions and availability of microbiologic results (days 2–4, depending on the performance of local laboratories). These are appropriate times to consider broadening or restricting the antibiotic coverage or focus on a specific pathogen. Table 311.7 summarizes the approaches, as discussed next.

The escalation approach consists of starting with monotherapy with piperacillin-tazobactam or a third-generation cephalosporin (e.g., cefepime or ceftazidime) and then adjusting therapy if necessary. This approach is still appropriate in many countries and centers, especially when resistance rates are low among pathogens commonly causing infections in neutropenia.104 With this approach, the carbapenems are used as second-line therapy in patients failing the initial therapy and in case of a documented infection.213,238 In this clinical situation, combining an aminoglycoside with a β-lactam is not deemed necessary because of increased toxicity and no clinical advantage. Clinical trials and meta-analyses confirmed the reliability of this approach.239 A survey from 14 major US cancer centers documented high susceptibility rates to ceftazidime, cefepime, and piperacillin/tazobactam, showing that the monotherapy regimen (mainly cefepime) was effective "since the beginning," with only severity of illness predicting poor outcomes.104

The empirical use of an anti–gram-positive antibiotic such as vancomycin, teicoplanin, linezolid, or daptomycin is not recommended by any guideline, either as initial therapy or in persistently febrile patients, unless MRSA or a penicillin-resistant pneumococcus or enterococcus is isolated or the patient has signs and symptoms suggesting a gram-positive etiology (e.g., CVC involvement or

TABLE 311.7

New Approach to Empirical Therapy of Febrile Neutropenia Divided Into Escalation and De-escalation Strategy

Strategy	Escalation	De-escalation
Definition	Empirical treatment active against susceptible *Enterobacterales* and *P. aeruginosa*	Starting up-front an empirical coverage of MDR bacteria, particularly gram-negatives or MRSA in colonized patients, which is later (72–96 hours) reduced (= de-escalated) if an MDR pathogen is NOT isolated; that is: • Susceptible strain isolated • No microbiologic results
Antibiotics usually used	• Antipseudomonal cephalosporin (cefepime, ceftazidime) • Piperacillin/tazobactam	• Carbapenem • New β-lactam + β-lactamase inhibitor combination (e.g., ceftazidime/avibactam) Combinations, examples: • β-lactam + aminoglycoside • Colistin-based combination regimen • β-lactam + anti-MRSA coverage
Main advantages	Less selection of resistant strains (including carbapenem-sparing strategy) Less toxicity (if aminoglycoside or glycopeptide are not routinely used)	Appropriate therapy before culture results are available and hopefully lower mortality
Main limitations	In case of infection due to a resistant gram-negative, therapy might be inappropriate and the prognosis might be significantly worsened	Overuse of broad-spectrum antibiotics/combinations > high antibiotic pressure, particularly in case of failure to de-escalate Need to have individualized protocols for empirical therapy Need to establish individually if a patient is at risk for particular resistant bacteria and provide clear indications for the antibiotic(s) to be used in empirical therapy
Who	All patients, unless risk factors for resistant bacteria present or severe clinical presentation	Patients at risk for infections due to resistant bacteria, particularly if presenting with severe clinical conditions: • Past or current MDR colonization • Previous MDR infections • Local epidemiology

pneumonia).234 The recent finding that MRSA colonization in pediatric patients is significantly associated with the development of bacteremia5 could change this scenario, with the introduction of an initial short period of glycopeptide administration (see the discussion on the de-escalation approach). Page 3665The escalation approach avoids universal, and usually unnecessary, up-front use of antibiotics with the broadest spectrum, such as a carbapenem or combinations with an aminoglycoside or glycopeptide, and consequently minimizes potential disadvantages, such as selection of resistant pathogens or toxicity. However, there is growing concern with this approach because of increasing antibiotic resistance. If the initial regimen fails to cover the pathogen responsible for infection, the outcome can be fatal.240 For this reason, a new approach has been suggested, which has been called de-escalation empirical therapy.

The de-escalation approach consists of starting with a very broad initial empirical regimen, chosen based on the severity of the patient's clinical presentation, the presence of colonization with MDR bacteria, risk factors for harboring MDR bacteria, and the local bacterial epidemiology. Examples of de-escalation empirical therapy include starting with a carbapenem or new β-lactam plus β-lactamase inhibitor active against ESBL-producing bacteria and some more resistant gram-negatives.241–243

Up-front use of vancomycin or daptomycin in patients colonized with MRSA, methicillin-resistant Staphylococcus epidermidis, or penicillin-resistant gram-positive cocci, or daptomycin or linezolid in the case of VRE colonization or in centers with high incidence of infections due to these pathogens, is more controversial, with no benefit of empirical therapy with linezolid in patients colonized with VRE.103

In the de-escalation approach, the key issues are (1) to safeguard the patient and to avoid the risk of undertreating their infection, which has the potential to be overwhelming, and (2) to reduce as much the unnecessary use of important but frequently overused drugs (e.g., carbapenems or vancomycin) by reassessing the antibiotic treatment after 48 to 96 hours and streamlining therapy, based on microbiology, whenever possible. For example, in our center, where an increase in the incidence of BSIs caused by ampicillin-resistant enterococci and ciprofloxacin-resistant/ESBL-producing gram-negative rods (around 60% of gram-negative pathogens) was noted, piperacillin-tazobactam was the standard therapy when the clinical presentation allowed watchful waiting. However, patients presenting with severe sepsis, septic shock, or suspected bacterial pneumonia and those known to be colonized by resistant pathogens were started on meropenem plus amikacin or vancomycin and then de-escalated to piperacillin-tazobactam or ceftazidime monotherapy if a susceptible pathogen was isolated. In patients with colonization or previous infection with an MDR gram-negative pathogen, empirical antibiotic therapy targeting this strain was decided on at admission and indicated in the chart, to be started at the onset of fever. Currently, it frequently contains a combination regimen of ceftazidime-avibactam or ceftolozane-tazobactam, plus amikacin or daptomycin, or other targeted molecule.92 Recently, we observed that the combination of piperacillin-tazobactam and amikacin (for 3 days in the case of FUO) was effective and safe in children in a center with strict control of local epidemiology,244 confirming that knowledge of local epidemiologic data represents pivotal information for deciding whether to start with an escalation or a de-escalation approach and deciding on the type of drugs to be administered.245 The current use of a β-lactam and aminoglycoside in empirical therapy of febrile neutropenia is different from the way this combination therapy was prescribed decades ago and studied in randomized controlled trials. Only the aim remained the same: to reduce mortality. In the 1980s the addition of an aminoglycoside to an active β-lactam was based on the hope of potential synergy and rapid bactericidal activity, and the combined treatment was continued until the resolution of neutropenia. Randomized trials and meta-analyses reported no improved efficacy but higher renal toxicities in the combination arm.246,247 In the current era of widespread antimicrobial resistance, the short (3 days) combination therapy serves to broaden the antibiotic spectrum, particularly as a carbapenem-sparing strategy, and to improve empirical treatment adequacy within the de-escalation approach. Combination treatment is rapidly followed by streamlining of an antibiotic regimen based on susceptibility testing or discontinuation if no pathogen is isolated.248 Data from neutropenic cancer patients in an ICU and from neutropenic HSCT recipients showed that the de-escalation approach was safe and feasible.249–251 De-escalation is much easier to perform in the case of isolation of a susceptible pathogen (streamlining of treatment) than in cases of negative microbiologic results (discontinuation).252

Duration of Antibacterial Treatment

Another traditional practice that has been challenged by the ECIL guidelines is the need for empirical antibiotics to be continued until neutrophil recovery, with the idea that this is the way to avoid infection relapse and superinfections and to reduce mortality. This recommendation was based on a very small, open-label, randomized trial published in 1979.253 The ECIL position was that antibiotics can be safely discontinued after 72 or more hours of intravenous therapy in patients who are and have been hemodynamically stable since the onset of fever and are without fever for 48 or more hours, irrespective of the granulocyte count and the expected duration of neutropenia. Careful clinical monitoring was recommended to immediately restart antibiotics at the first signs of infection relapse or novel infection. In microbiologically documented infection, the recommendation was to treat for at least 7 days. The rationale for this revolutionary recommendation was based on acknowledgment that alteration of the patient microbiota leads to an increased risk of intestinal colonization by resistant pathogens, which, in turn, increases the risk of infection.254,255 Thus antibiotics, which are the main drivers of dysbiosis, may in fact be harmful if used unnecessarily. Decreasing the antibiotic selective pressure is crucial for future generations and should be a goal of modern medicine. Controversy around the dogma that antibiotic therapy should be continued until the end of neutropenia can be traced back to 1988.

The 2008 IDSA guidelines left some room for the possibility of stopping empirical therapy before recovery of neutropenia in a clinically stable patient, while providing the possibility of resuming quinolone prophylaxis.234,256 Although initial data came from two prospective randomized studies in low-risk children,257,258 subsequent observational studies in high-risk patients with prolonged neutropenia confirmed that discontinuation of antibiotics was associated with relapse of fever in a few patients but without an increase in mortality, providing that antibacterial treatment was restarted immediately.259,260,261

An open-label, multicenter, randomized trial addressed the question of whether a clinically oriented approach was superior to a standard end-of-neutropenia approach for discontinuing empirical antibacterial therapy in patients with fever and neutropenia.262 The study showed that the clinically oriented approach significantly increased the number of antibiotic-free days from 13.6 to 16.1, without affecting mortality or days of fever. The incidence of fever recurrences and fungal or bacterial superinfections in the two groups were basically the same, likely due to timely and adequate interventions.

In conclusion, in the face of the antibiotic crisis and the increasing recognition of the importance of maintaining as much as possible the integrity and diversity of the human microbiota, we must rationalize antibiotic intervention in hematologic patients. Prioritization involves four main aspects:

1. To review antibiotic prophylaxis measures
2. To adopt stringent criteria for driving empirical choices at the onset of fever and neutropenia and on day 3 to 4 after starting treatment
3. To be acquainted with the concept that a planned progressive succession of antibiotics in fever and neutropenia does not exist, and every patient and every episode should be evaluated separately
4. To accept the idea that in neutropenic patients with FUO, empirical antibiotics can be discontinued even before neutropenia recovery, provided close clinical observation as an inpatient is maintained with a high level of suspicion for restarting treatment.

Empirical and Diagnostic-Driven Antifungal Therapy
The empirical antifungal therapy approach consists of administering an antifungal drug in a persistently febrile neutropenic cancer patient after a variable period of empirical antibacterial therapy (usually 4–7 days), in the absence of any clinical, microbiologic, or radiologic documentation of a fungal infection. This practice is based on autopsy Page 3666studies and two small, randomized trials performed many years ago , when periods of neutropenia were prolonged, antifungal prophylaxis was not administered, and the bulk of fungal infections were caused by Candida species.263–266 Despite nondefinitive data, empirical antifungal therapy in persistently febrile neutropenic patients without a documented infection has become common practice in many cancer centers worldwide, and numerous drugs have been tested for, and provided regulatory indications for empirical antifungal therapy. All of these studies have used a composite end point that included five criteria: resolution of fever, no discontinuation for toxicity, treatment of any baseline fungal infections, prevention of breakthrough fungal infections, and survival. In general, no drug has been demonstrated to be significantly more effective than others, with differences driven by toxicities.

Over the most recent decades, treatments have changed such that durations of neutropenia are reduced compared to historic cytotoxic regimens, and fluconazole is commonly administered prophylactically, reducing the threat of candidiasis. An increased burden of IFD is now associated with pulmonary pathogens, especially Aspergillus spp., which can be difficult to diagnose and treat early. This, with availability of in vitro biomarkers, increased use of highly sensitive CT scans and aggressive performance of bronchoscopy, has compelled exploration of “diagnostic-driven,” or “preemptive” antifungal therapy approaches as alternatives to routine treatment of fever during neutropenia (empiricism). Diagnostic-driven and preemptive strategies are not necessarily the same, as truly preemptive strategies rely on biomarker screening to guide antifungal therapy without mandating other evidence of disease, akin to preemptive treatment of CMV viremia. In the diagnostic-driven strategy discussed here, clinical considerations (fever, thoracic pain, cough), biologic markers (e.g., Aspergillus galactomannan or PCR in serum), and/or imaging data (e.g., chest computed tomography [CT], with whole-volume scanning with thin-slice reconstruction preferable to high-resolution CT) are combined together to start therapy.

This approach has been studied in both children and adults. The first "proof of concept" study concluded that the diagnostic-driven approach was feasible, associated with less use of antifungal therapy, and without increased mortality when compared with historical controls. Of particular interest, 10 patients who were diagnosed with positive galactomannan and CT scanning would not have received any antifungal therapy with the classic empirical approach because of absence of fever.267 In the first randomized noninferiority trial that compared the empirical and diagnostic-driven approaches (defined differently from the previous study), no difference was found in the primary end point (survival). As expected, in the arm in which an active diagnostic workup was performed, there were more fungal infections than in the other arm. Other studies compared the two strategies in adults and children, with results consistently in favor of the diagnostic-driven approach.269–271 However, the diagnostic-driven approach requires the rapid availability of CT scanning and noninvasive assays (galactomannan, PCR) able to provide results in short turnaround times, which should be no more than 2 to 3 days to allow timely intervention. Finally, the diagnostic methods used should be chosen thoughtfully for specific populations. Diagnostic-driven screening cannot be reliably performed in people receiving mold-active antifungals, both because prophylaxis affects the sensitivity of the diagnostic tests and prevalence of infection, thus affecting predictive value.272 Currently the use of biomarkers in children is limited to "aid to diagnosis," as opposed to screening.273 Rapid access to the bronchoalveolar lavage (BAL) procedure is also fundamental to confirm the diagnosis and start optimal targeted antifungal treatment.

In our opinion, the empirical and diagnostic-driven strategies are not mutually exclusive. Empirical antifungal treatment can be provided while awaiting the results of diagnostic tests but then discontinued if the results are not confirmatory. Screening with serum galactomannan or PCR in those not receiving mold active prophylaxis is useful to trigger further diagnostic workup, such as CT scanning, before persistent fever occurs. Fig. 311.3 summarizes the possible approaches to a patient with persistent febrile neutropenia. Table 311.6 lists drugs indicated for empirical or targeted antifungal therapy. Drugs approved for empirical therapy include liposomal amphotericin B, caspofungin, and itraconazole. Without regulatory approvals for preemptive therapy, current treatments mimic those typically used as primary therapy of aspergillosis: voriconazole, liposomal amphotericin B, or isavuconazole. Isavuconazole, in addition to aspergillosis, has been also approved for treatment of mucormycosis.274,275 In centers that have a high incidence of infections caused by antifungal-resistant Aspergillus species, such as azole-resistant A. fumigatus and MDR species (A. calidoustus, A. lentulus), consideration of first-line therapy consisting of a combination of an azole and echinocandin or liposomal amphotericin B is reasonable.276

Early Empirical Management of a Neutropenic Patient With a Localized Infection

In consideration of the high risk of severe complications of infections in neutropenic cancer patients, an "empirical" approach can be recommended in the presence of clinical and radiologic signs of localized infection, pending the culture results. The diagnostic approach, and especially imaging (CT also with contrast, ultrasound, MRI), can provide a pivotal tool for the diagnosis of localized infections in the neutropenic cancer patient. Recently, also 18F-fluorodeoxyglucose positron emission tomography (PET) combined with low-dose CT (commonly known as FDG-PET/CT) has been applied for diagnosis of localized infections in neutropenic patients with persistent or recurrent fever with an increased identification (compared with CT) of abdominal foci, perineal infections, and septic thrombophlebitis, with possible important changes in therapeutic strategies.277

Catheter-Related Infection

It is likely that the role of indwelling catheters in causing fever and infection in neutropenic patients may have been overestimated. The suspicion that the catheter is actually involved should be raised in case of septic shock, endocarditis, rapidly progressive bacterial infection, fever with concomitant signs of infection at the catheter site (including the subcutaneous tunnel), blood culture persistence, and with fever developing concomitantly with catheter flushing. In addition to clinical criteria, some microbiologic criteria could be used, although some of them require quantitative evaluation systems: growth of the same pathogen (with the same susceptibility testing results) from blood culture of the peripheral vein and from culture of the CVC tip; growth of the same pathogen from blood culture of the CVC and from blood culture of the peripheral vein and differential time to positivity ≥2 hours (faster growth in CVC blood culture) or more than threefold greater colony count of pathogens grown from blood culture of CVC relative to colony count from a peripheral vein.278 When a catheter-related infection is proven or suspected, the first thing to do is to establish whether the catheter is dispensable. If so, the catheter should be removed and treatment administered, if possible, through a peripheral line.

The choice of the antibiotic regimen should be based on the epidemiology of CVC-related infections in every individual center and on the pharmacokinetic/pharmacodynamic characteristics of the available antibiotics. As a rule, an anti–gram-positive drug should always be included in the initial regimen, although the choice should not necessarily be vancomycin, except for centers with a high rate of MRSA and MRSE. On the other hand, in centers where staphylococci with high MIC values for vancomycin (≥2 µg/mL) have been isolated, daptomycin or linezolid might be considered. Moreover, because gram-negative organisms are not infrequent in single-agent or polymicrobial CVC bacteremias, an anti–gram-negative coverage is recommended in all cases. In contrast, the empirical inclusion of an antifungal drug seems

[Page 3667] therapy of underlying disease) and risks for bleeding, especially that related to thrombocytopenia.

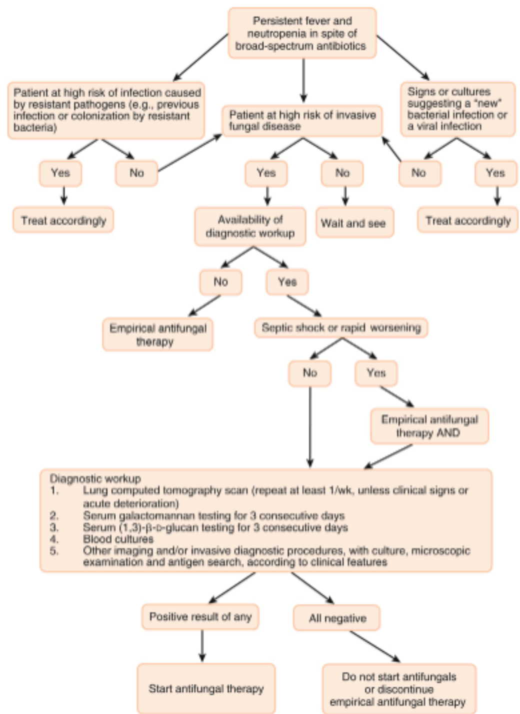

FIG. 311.3 Management of a persistently febrile neutropenic patient.

Follow for extended description

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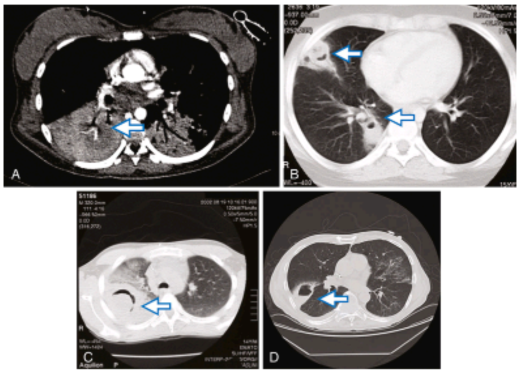

FIG. 311.4 Pulmonary cavitary lesions resulting from infections due to different etiologies in the presence and absence of neutropenia.(A) Neutropenic patient with hemoptysis and pulmonary cavitation *(arrow)* in the presence of *Klebsiella pneumoniae* bacteremia. (B) Neutropenic patient with pulmonary cavitation *(arrows)* in the presence of methicillin-susceptible *Staphylococcus aureus* bacteremia. (C) Air crescent *(arrow)* in a no-longer-neutropenic patient with pulmonary aspergillosis. (D) Cavitary lesion *(arrow)* in a no-longer-neutropenic patient with *Pseudomonas aeruginosa* bacteremia.

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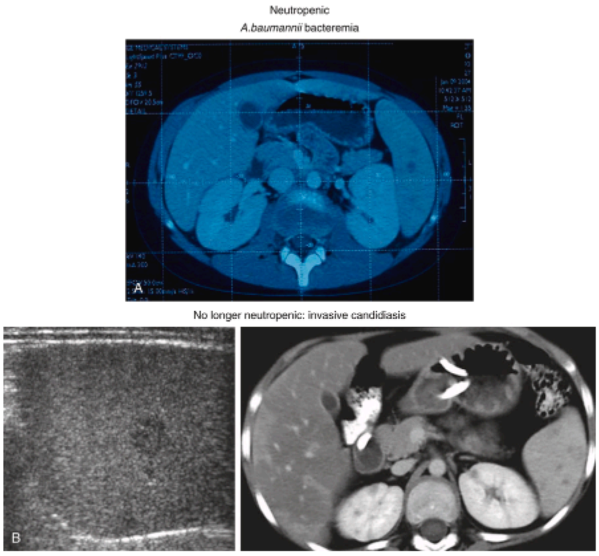
Abdominal Infections

Febrile neutropenic patients may present with gastrointestinal signs and symptoms, such as abdominal pain, nausea, vomiting, and diarrhea, in addition to fever. In these patients, initial conservative management with bowel rest, intravenous fluids, total parenteral nutrition, and broad-spectrum antibiotics with antianaerobic activity should be immediately implemented. In some cases (3%–6%), especially in patients receiving aggressive treatment for acute leukemia, full-blown neutropenic enterocolitis may develop, with high fever, severe abdominal pain, and even hemorrhagic diarrhea evolving into acute abdomen and septic shock. Clostridioides or CMV colitis should be considered in the diagnostic workup and treated accordingly (see Chapters 144 and 249). The clinical picture of neutropenic enterocolitis together with C. difficile infection also can be observed. Surgical intervention is usually not indicated but may be recommended in the setting of obstruction, perforation, persistent gastrointestinal bleeding despite correction of thrombocytopenia and coagulopathy, and clinical deterioration. Acute appendicitis must also be considered in the differential diagnosis, especially in the youngest patients, because in these



patients

FIG. 311.5 Abdominal lesions resulting from infections due to different etiologies in the presence and absence of neutropenia.(A) Lesions in a neutropenic patient with *Acinetobacter baumannii* bacteremia. (B) Lesions due to hepatosplenic candidiasis after resolution of neutropenia.

appendectomy is associated with a lower rate of complications.283,284 Finally, persistent fever in a neutropenic patient can be due to chronic invasive candidiasis (also called hepatosplenic candidiasis); however, in this case as well, the presence of neutrophils is pivotal for imaging detection of the fungal localizations. Therefore, the presence of focal liver or spleen lesions (e.g., abscesses) in a persistently neutropenic patient is generally due to bacterial localization from a (concomitant) bacteremia (Fig. 311.5).

Utility of therapies involving immune-modifiers, such as colony stimulating factors and granulocyte transfusions is discussed extensively in Chapter 52.

Conclusions

The management of fever and infections during neutropenia and in non-neutropenic cancer patients remains a challenge for clinicians. The limited availability of new antibiotics and the emergence of various degrees of resistance to older antibiotics can have detrimental effects and increased mortality rates. The importance of this problem varies among countries and regions, probably due to local epidemiology and antibiotic policies. At the same time, antineoplastic treatments are changing, with consequent need to maintain a high level of attention for unexpected infections and unusual clinical presentations, particularly in case of new agents that modify T-cell immunity and/or require additional immunosuppressive therapy for inflammatory syndromes. Diagnostic microbiologic methods are also changing, with the introduction of new biomarkers and rapid pathogen identification and susceptibility assessments enabling earlier antibacterial and antifungal therapies. In the future, better identification of risk factors for infections, both related to treatment and patient characteristics, will help introduce more individualized approaches to prevention and early antimicrobial therapies.

Acknowledgement

In memory of extraordinary mentor and friend, Professor Claudio Viscoli, who enthusiastically pursued the research leading to better understanding and to improved daily management of infectious complications in the immunocompromised.Page 3669

Figure 1: Utility of therapies involving immune-modifiers, such as colony stimulating factors and granulocyte transfusions is discussed extensively in Chapter 52.

Conclusions

The management of fever and infections during neutropenia and in non-neutropenic cancer patients remains a challenge for clinicians. The limited availability of new antibiotics and the emergence of various degrees of resistance to older antibiotics can have detrimental effects and increased mortality rates. The importance of this problem varies among countries and regions, probably due to local epidemiology and antibiotic policies. At the same time, antineoplastic treatments are changing, with consequent need to maintain a high level of attention for unexpected infections and unusual clinical presentations, particularly in case of new agents that modify T-cell immunity and/or require additional immunosuppressive therapy for inflammatory syndromes. Diagnostic microbiologic methods are also changing, with the introduction of new biomarkers and rapid pathogen identification and susceptibility assessments enabling earlier antibacterial and antifungal therapies. In the future, better identification of risk factors for

infections, both related to treatment and patient characteristics, will help introduce more individualized approaches to prevention and early antimicrobial therapies.

Figure 1: The two-panel pie chart shows the etiology of bloodstream infections in cancer patients. The top panel presents data from a recent literature review from 2005 to 2011, highlighting the distribution of pathogens responsible for bloodstream infections. The pathogens include Staphylococcus aureus, coagulase-negative staphylococci, Viridans streptococci, Enterococcus, other gram positives, Enterobacteriaceae, Pseudomonas aeruginosa, Acinetobacter, and other gram-negative bacteria. The bottom panel presents data from the Fourth European Conference of Infections in Leukemia E C I L-4 surveillance study in 2011, which also shows the percentage distribution of the same pathogens, with additional emphasis on the prevalence of Staphylococcus aureus, coagulase-negative staphylococci, Viridans streptococci, Enterococcus, and Pseudomonas aeruginosa in bloodstream infections.

The flowchart outlines a possible initial approach to managing a patient with febrile neutropenia. The process begins with assessing the type of patient at the onset of febrile neutropenia, several factors should be considered to guide the appropriate treatment. These include type of patient: Consider underlying diseases, time since chemotherapy, and prior history of prophylactic antimicrobials and infectious complications, particularly if caused by resistant pathogens. Type of center: Knowledge of the local epidemiology of infections and susceptibility patterns is crucial. If available, use a risk stratification system to inform the antibiotic choice and consider local resistance patterns. Blood cultures of at least three and cultures from other sites of suspected infection should be performed. A chest computed tomography scan or other imaging should be considered based on the clinical presentation. If clinical signs of localized infection are present, antibiotics should be administered that are active against gram-negative bacteria. Additional agents should be included according to the most probable pathogen if different and the specific site of infection. After 72 hours of treatment, the anti-infective regimen should be revised. Discontinue anti-gram-positive and antifungal drugs if these infections are not confirmed. Aminoglycosides should be discontinued if gram-negative bacteria are not isolated or if they are susceptible to the chosen beta-lactam. Adjust treatment based on isolated pathogens and the site of infection. If there are signs of septic shock, empirical therapy should be started with antibiotics active against gram-negative bacteria consider a combination of beta-lactam and aminoglycoside and gram-positive bacteria based on local epidemiology and susceptibility patterns. Antifungal therapy like echinocandin should also be considered. Again, after 72 hours, the regimen should be revised in the same manner that is previously described: discontinuing drugs for infections not confirmed, adjusting based on isolated pathogens, and the site of infection. If no signs of septic shock are observed, empirical therapy should be started according to risk stratification considering oral therapy and local epidemiologic patterns. In all cases, antibiotics active against gram-negative bacteria should be included. After 72 hours, the same process for revising the anti-infective regimen should be followed: discontinuing unnecessary drugs and adjusting treatment based on isolated pathogens and the site of infection.
Figure 3: The antifungal therapy decision flowchart outlines a clinical approach for managing patients who experience persistent fever and neutropenia despite receiving broad-spectrum antibiotics. The process begins at the top with the identification of such a patient, branching into three initial assessments. The first branch determines if the patient is at high risk of infection from resistant pathogens, including previous infection or colonization. If confirmed, the clinician should treat accordingly; otherwise, proceed to diagnostic considerations for a patient at high risk of invasive fungal disease. The second branch asks whether the patient is at high risk for invasive fungal disease. If not, a wait-and-see approach is recommended. If yes, the next decision point assesses the availability of diagnostic workup. If unavailable, empirical antifungal therapy is initiated. If diagnostics are available, a further assessment is made for septic shock or rapid clinical worsening. In such a case, empirical antifungal therapy is started alongside the workup. If there is no rapid worsening, the process proceeds with diagnostic workup steps, which include lung computed tomography scans, serum galactomannan testing, serum beta-D-glucan testing, blood cultures, and other imaging or invasive diagnostics based on clinical presentation. Following testing, if any result is positive, antifungal therapy should begin. If all tests are negative, antifungal treatment should either be withheld or discontinued. The third branch from the top addresses whether signs or cultures indicate a new bacterial or viral infection. If confirmed, the patient should be treated accordingly. If not, the clinician proceeds to diagnostic considerations

for a patient at high risk of invasive fungal disease. The chart organizes the decision-making process by connecting each step with directional arrows and ensures a logical progression through clinical assessment and treatment choices based on patient risk and diagnostic availability.

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Figure 4: The four-panel C T scan shows pulmonary cavitary lesions resulting from infections due to different etiologies in neutropenic and non-neutropenic patients. Panel A shows a computed tomography scan of a neutropenic patient with hemoptysis and a pulmonary cavitation, indicated by the arrow, caused by Klebsiella pneumoniae bacteremia. Panel B shows a computed tomography scan of a neutropenic patient with pulmonary cavitation, marked by the arrows, due to methicillin-susceptible Staphylococcus aureus bacteremia. In panel C, an air crescent, marked by the arrow, is seen in a non-neutropenic patient with pulmonary aspergillosis. Panel D shows a cavitary lesion, indicated by the arrow, in a non-neutropenic patient with Pseudomonas aeruginosa bacteremia.

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Figure 5: The four-panel C T scan shows pulmonary cavitary lesions resulting from infections due to different etiologies in neutropenic and non-neutropenic patients. Panel A shows a computed tomography scan of a neutropenic patient with hemoptysis and a pulmonary cavitation, indicated by the arrow, caused by Klebsiella pneumoniae bacteremia. Panel B shows a computed tomography scan of a neutropenic patient with pulmonary cavitation, marked by the arrows, due to methicillin-susceptible Staphylococcus aureus bacteremia. In panel C, an air crescent, marked by the arrow, is seen in a non-neutropenic patient with pulmonary aspergillosis. Panel D shows a cavitary lesion, indicated by the arrow, in a non-neutropenic patient with Pseudomonas aeruginosa bacteremia.

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